



GE Nuclear Energy

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ABWR Design Control Document

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B

A

TABLE 1 LPRM TO APRM ASSIGNMENTS

DET AND LPRM LOCATION	APRM CHANNEL				MUX UNIT NO.
	A	B	C	D	
20-61	B	C	D	A	1
28-61	C	D	A	B	2
36-61	B	C	D	A	3
44-61	C	D	A	B	4
52-61	B	C	D	A	1
12-53	D	A	B	C	1
20-53	A	B	C	D	4
28-53	D	A	B	C	3
36-53	A	B	C	D	2
44-53	D	A	B	C	1
52-53	A	B	C	D	4
60-53	D	A	B	C	3
04-45	B	C	D	A	4
12-45	C	D	A	B	3
20-45	B	C	D	A	2
28-45	C	D	A	B	1
36-45	B	C	D	A	4
44-45	C	D	A	B	3
52-45	B	C	D	A	2
60-45	C	D	A	B	1
04-37	A	B	C	D	2
12-37	D	A	B	C	3
20-37	A	B	C	D	3
28-37	D	A	B	C	4
36-37	A	B	C	D	1
44-37	D	A	B	C	2
52-37	A	B	C	D	3
60-37	D	A	B	C	4
04-29	B	C	D	A	3
12-29	C	D	A	B	4
20-29	B	C	D	A	1
28-29	C	D	A	B	2
36-29	B	C	D	A	3
44-29	C	D	A	B	4
52-29	B	C	D	A	1
60-29	C	D	A	B	2

TABLE 1 LPRM TO APRM ASSIGNMENTS (CONT'D)

DET AND LPRM LOCATION	APRM CHANNEL				MUX UNIT NO.
	A	B	C	D	
12-21	D	A	B	C	1
20-21	A	B	C	D	4
28-21	D	A	B	C	3
36-21	A	B	C	D	2
44-21	D	A	B	C	1
52-21	A	B	C	D	4
60-21	D	A	B	C	3
12-13	C	D	A	B	3
20-13	B	C	D	A	2
28-13	C	D	A	B	1
36-13	B	C	D	A	4
44-13	C	D	A	B	3
52-13	B	C	D	A	2
60-05	D	A	B	C	4
36-05	A	B	C	D	1
44-05	D	A	B	C	2

(NOTE 9)

TABLE 3 POWER DISTRIBUTION

DIVISION	120V AC DIVISIONAL UPS BUS			
	A	B	C	D
SRNM CH'S	A,E,J	B,F	C,G,L	D,H
APRM CH'S	A	B	C	D
DIVISION	120V AC NON-DIVISIONAL UPS			
	A	B	C	D
MRBM CH'S	A	B		
SRNM RECORDERS	A,E,J	B,F	C,G,L	D,H
APRM RECORDERS	A	B	C	D
MRBM RECORDERS	A	B	A	B
460V, 3 Ø 3KVA: ATIP DRIVE MECHANISMS				
120V AC: ATIP INSTR. PANEL				

TABLE 2 LPRM STRING ASSIGNMENT TO ATIP MACHINES

ATIP A		ATIP B		ATIP C	
POSITION #	LPRM LOC.	POSITION #	LPRM LOC.	POSITION #	LPRM LOC.
1	04-29	1	28-05	1	44-13
2	04-37	2	28-13	2	44-21
3	04-45	3	28-21	3	44-29
4	12-13	4	28-29	4	44-37
5	12-21	5	28-37	5	44-45
6	12-29	6	28-45	6	44-53
7	12-37	7	28-53	7	44-61
8	12-45	8	28-61	8	52-13
9	12-53	9	36-05	9	52-21
10	20-13	10	36-13	10	52-29
11	20-21	11	36-21	11	52-37
12	20-29	12	36-29	12	52-45
13	20-37	13	36-37	13	52-53
14	20-45	14	36-45	14	52-61
15	20-53	15	36-53	15	60-21
16	20-61	16	44-05	16	60-29
17	-	17	-	17	60-37
18	-	18	-	18	60-45
19	-	19	-	19	60-53
20	36-37	20	36-37	20	36-37

SUPPLEMENTAL DOCUMENTS UNDER THE FOLLOWING IDENTITIES ARE TO BE USED IN CONJUNCTION WITH THIS DRAWING:

	MPL NO.
1. RECIRCULATION FLOW CONTROL SYS, IED	C81-1040
2. PMCS INPUT/OUTPUT LIST	C90-1060
3. PIPING AND INSTRUMENT SYMBOLS DIAGRAM	A10-3030
4. REACTOR PROTECTION SYS, IED	C71-1040
5. ROD CONTROL AND INFORMATION SYS, IED	C11-1040
6. PMCS IED	C90-1040
7. MCRP SYSTEM IED	H11-1040
8. AUTOMATIC POWER REGULATOR SYS IED	C82-1040
9. NEUTRON MONITORING SYSTEM, IBD	C51-1030
10. NEUTRON MONITORING SYS ARGMT	C51-1070

NOTES:

- THIS DOCUMENT PRESENTS THE BASIC REQUIREMENTS OF FUNCTIONAL UNITS ALLOCATION OF THE NMS INSTRUMENTS, THEIR BASIC FUNCTIONS, AND THEIR INPUT AND OUTPUT REQUIREMENTS TO OTHER SYSTEMS. IT ALSO INCLUDES REQUIREMENTS OF DETECTOR LOCATION ASSIGNMENTS IN CORE AS WELL AS THEIR ASSIGNMENTS TO DIFFERENT DIVISIONS.
- ALL EQUIPMENT, INSTRUMENTS AND DISPLAY CONTROL SYSTEM INPUT SIGNALS ARE PREFIXED BY C51- UNLESS OTHERWISE NOTED. THE PART NUMBERS ARE LOCATED ADJACENT TO OR ON THE SIGNAL CONDITIONING EQUIPMENT PERFORMING THE FUNCTION INDICATED.
- AN AREA RADIATION MONITOR IS RECOMMENDED TO BE INSTALLED IN THE TIP DRIVE MECHANISM ROOM, AND THE OUTSIDE OF THE PERSONNEL HATCH DOOR (TO THE DRYWELL).
- REMOTE DISPLAY MEANS DISPLAY ON MAIN CONTROL ROOM PANEL. FOR DISPLAY DEVICE TYPE, REFER TO C51-5030.
- REMOTE RECORDING DEVICE SHALL BE BASED ON THAT SPECIFIED IN H11-4010.
- SRNM NON-COINCIDENT UPSCALE LEVEL TRIP IS ONLY ACTIVATED BY MANUAL SWITCH IN RPS.
- THESE ARE SAFETY RELATED CORE PLATE DIFFERENTIAL PRESSURE (C81-DPT301A-D) SIGNALS OF THE RFCS ESSENTIAL CORE PLATE DIFFERENTIAL PRESSURE SIGNALS.
- COMMAND AND CONTROL SIGNALS BETWEEN THE PROCESS COMPUTER AND THE ATIP OTHER THAN ATIP MEASURED DATA ARE TRANSMITTED THROUGH THIS DATA ENTRY POINT.
- IN CASE OF USING MULTIPLEXING UNITS OR SUB-APRM UNITS, DETECTORS SIGNALS ARE DISTRIBUTED TO FOUR MULTIPLEXER UNITS, IN EACH DIVISION ACCORDING TO TABLE 1 ARRANGEMENT. THE MULTIPLEXER UNIT ASSIGNMENT ONLY DEPENDS ON LPRM STRING LOCATION AND ARE IDENTICAL FOR ALL FOUR DIVISIONS.
- FOR DETAIL TRIP SIGNAL FUNCTIONAL INTERFACES WITH RCIS & RPS, REFER TO SUPPL DOC 9. THERE SHOULD BE A SEPARATE ROSS AND RPS TRIP LOGIC FUNCTION INDEPENDENT OF EACH INDIVIDUAL SRNM CHANNEL IN THE DIVISION. THE SSLO INTERLOCK SIGNAL IS THE "ATWS PERMISSIVE" SIGNAL.
- THIS IED INCLUDES ALL MAIN INPUT/OUTPUT SIGNALS REQUIRED TO BE INCLUDED IN THE INSTRUMENT DESIGN. FOR DETAIL BINARY TRIP SIGNALS, REFER TO NMS IBD, SUPPLEMENTAL DOCUMENT 9.
- THE SIGNAL TRANSMISSION PATH CAN BE EITHER THROUGH MULTIPLEXING OR THROUGH DEDICATED CONDUCTORS.
- TRIP SIGNALS FROM THE NMS TO THE PMCS SHALL BE TRANSMITTED VIA DEDICATED CONDUCTORS (E.G. METAL CABLE WITH PROPER ISOLATION) TO AVOID UNNECESSARY TIME DELAY FOR TRIP SEQUENCE RECORDING.
- THE OPRM IS A FUNCTIONAL SUBSYSTEM OF THE APRM. IT RECEIVES THE SAME LPRM SIGNALS THE APRM OF THE SAME DIVISION RECEIVES AS INPUT, AS SHOWN IN TABLE 1. THE OPRM LOGIC IS DESCRIBED IN SUPPL DOC 9.

LEGEND:

- - LPRM DETECTOR ASM (52)
- + - CONTROL RODS (205)
- - SRNM DETECTORS (10)
- △ - NEUTRON SOURCES (5)
- ▢ - SIGNAL ISOLATION
- △ - COMPUTER INPUTS
- △ - TRANSIENT TEST PANEL INPUTS
- - MULTIPLEXING TRANSMISSION (VIA OPTIC FIBER OR METALLIC CONDUCTOR)
- - SINGLE TRANSMISSION

ABBREVIATIONS

- RPS --- REACTOR PROTECTION SYSTEM
- SRNM --- STARTUP RANGE NEUTRON MONITOR
- LPRM --- LOCAL POWER RANGE MONITOR
- APRM --- AVERAGE POWER RANGE MONITOR
- ATIP --- AUTOMATED TRAVELING INCORE PROBE
- MRBM --- MULTI-CHANNEL ROD BLOCK MONITOR
- UPS --- UNINTERRUPTIBLE POWER SUPPLY
- NMS --- NEUTRON MONITORING SYSTEM
- RC&IS --- ROD CONTROL AND INFORMATION SYSTEM
- MCRP --- MAIN CONTROL ROOM PANEL
- PMCS --- PERFORMANCE MONITORING CONTROL SYSTEM
- APRS --- AUTOMATIC POWER REGULATOR SYSTEM
- MSV --- MEAN SQUARE VOLTAGE
- ATLM --- AUTOMATED THERMAL LIMIT MONITOR
- OPRM --- OSCILLATION POWER RANGE MONITOR

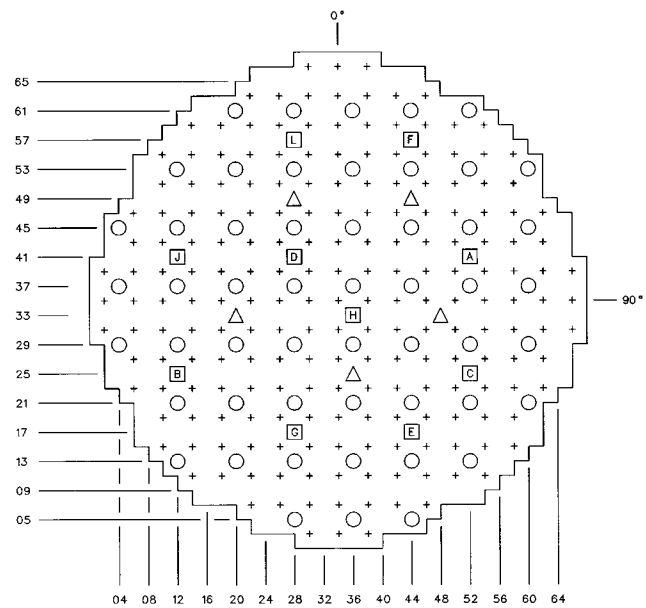


FIGURE 1: DETECTOR AND CONTROL ELEMENT ARRANGEMENT (TOP VIEW OF CORE)

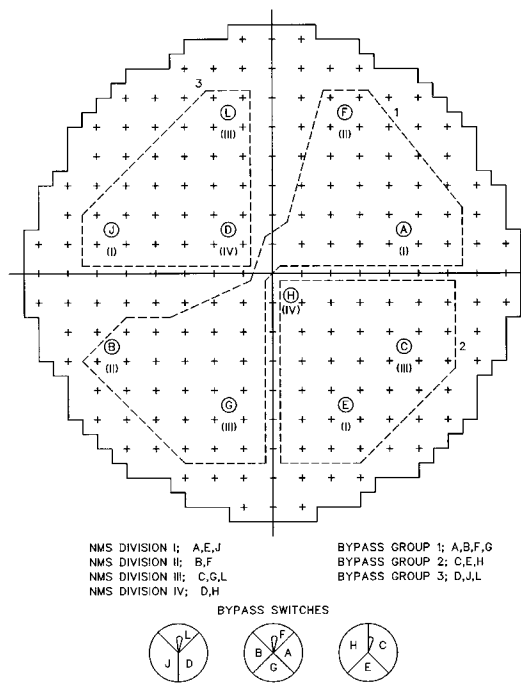


FIGURE 2: SRNM DIVISION & BYPASS GROUP ASSIGNMENTS

MPL NO. C51-1040

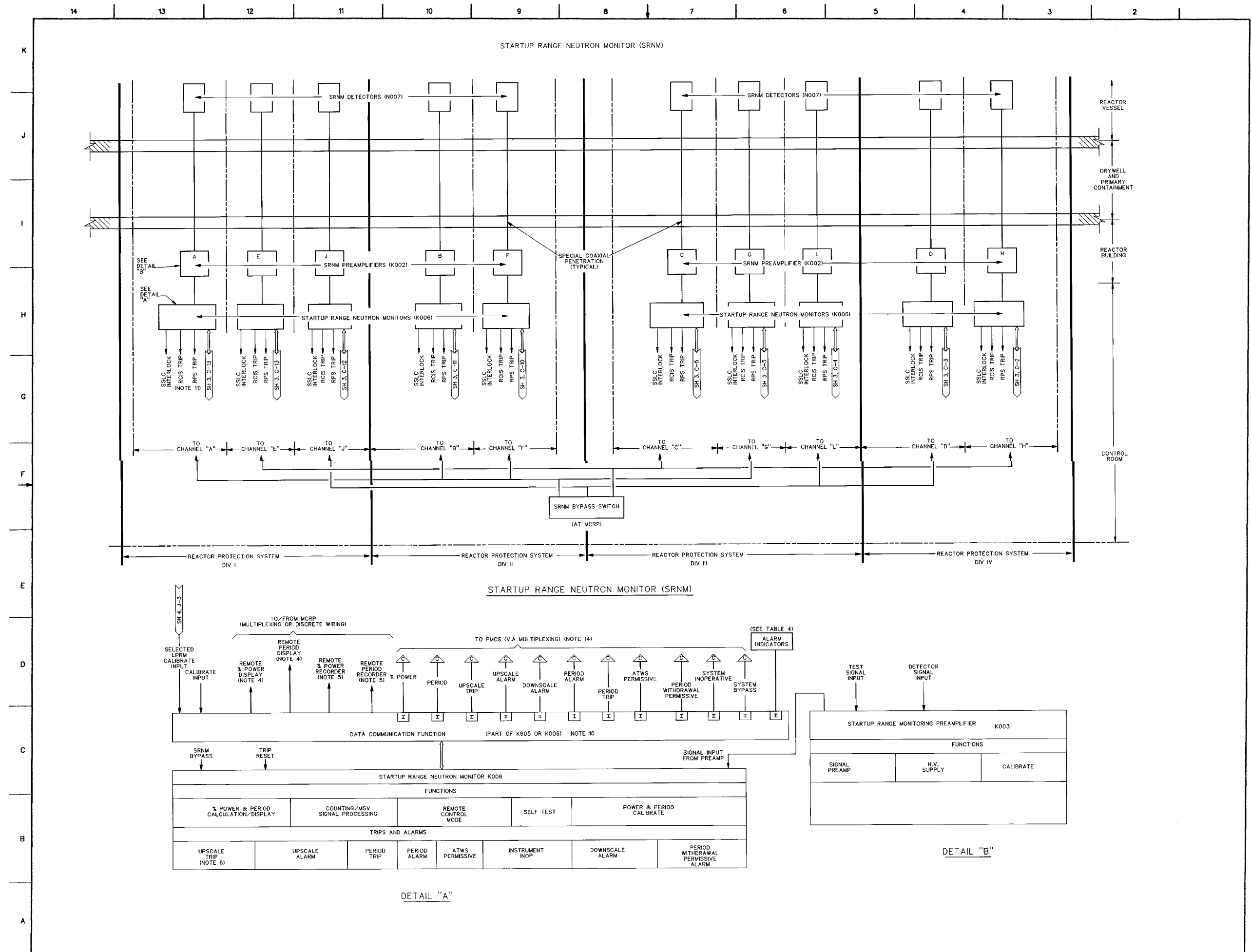


FIGURE 7.6-1 NEUTRON MONITORING SYSTEM (SRNM) SH 2 OF 4

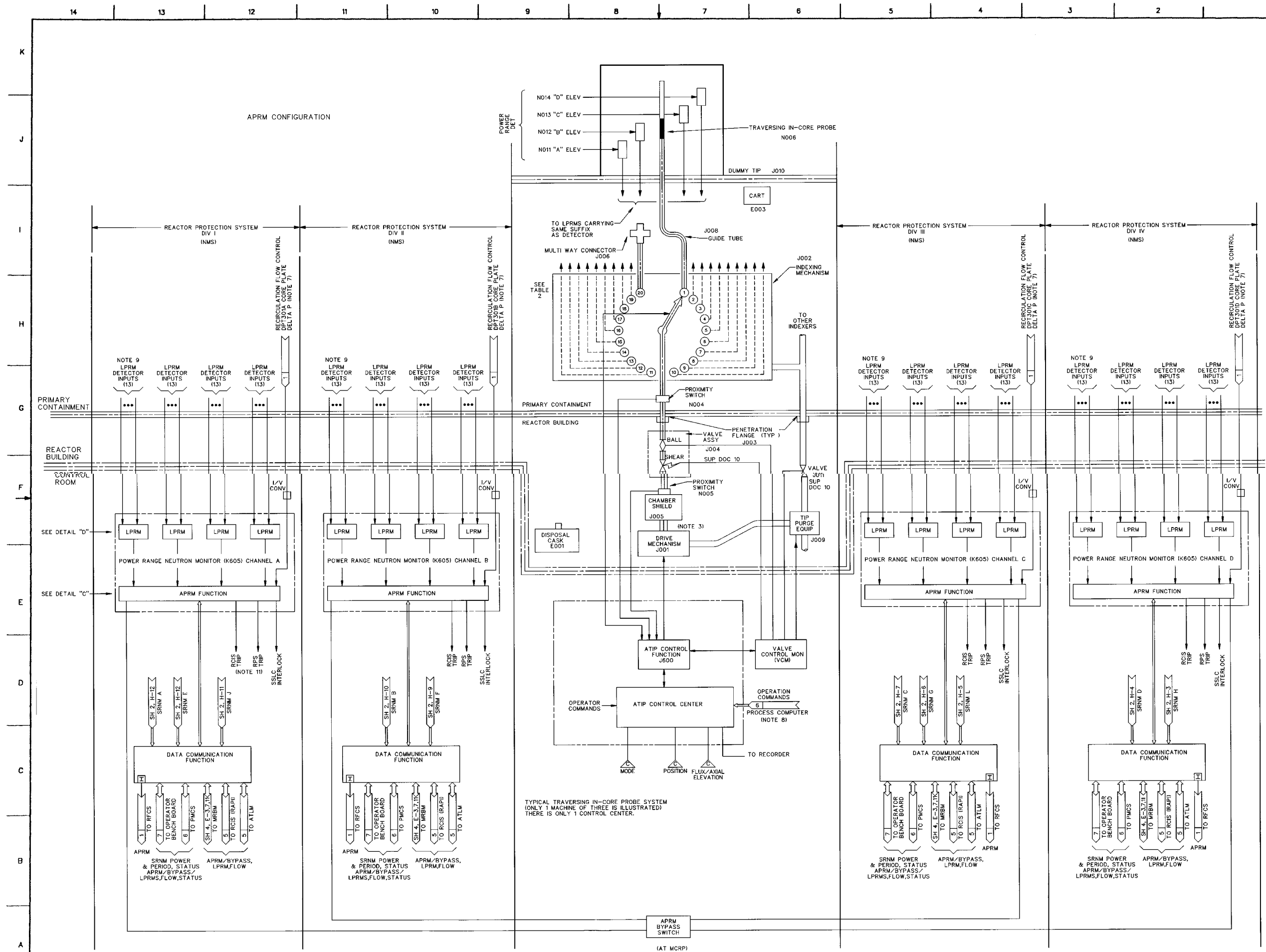


FIGURE 7.6-1 NEUTRON MONITORING SYSTEM IED SH 3 OF 4

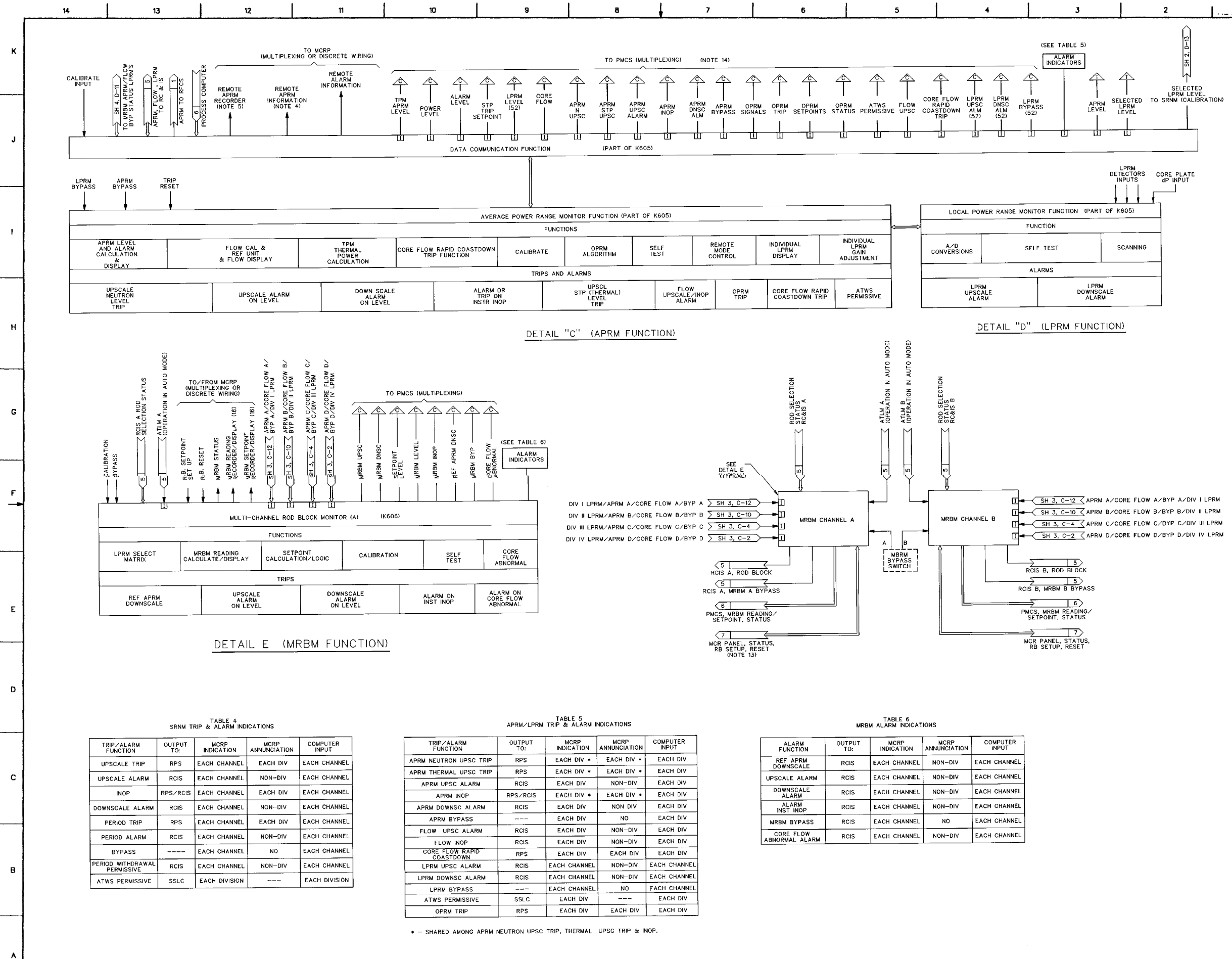
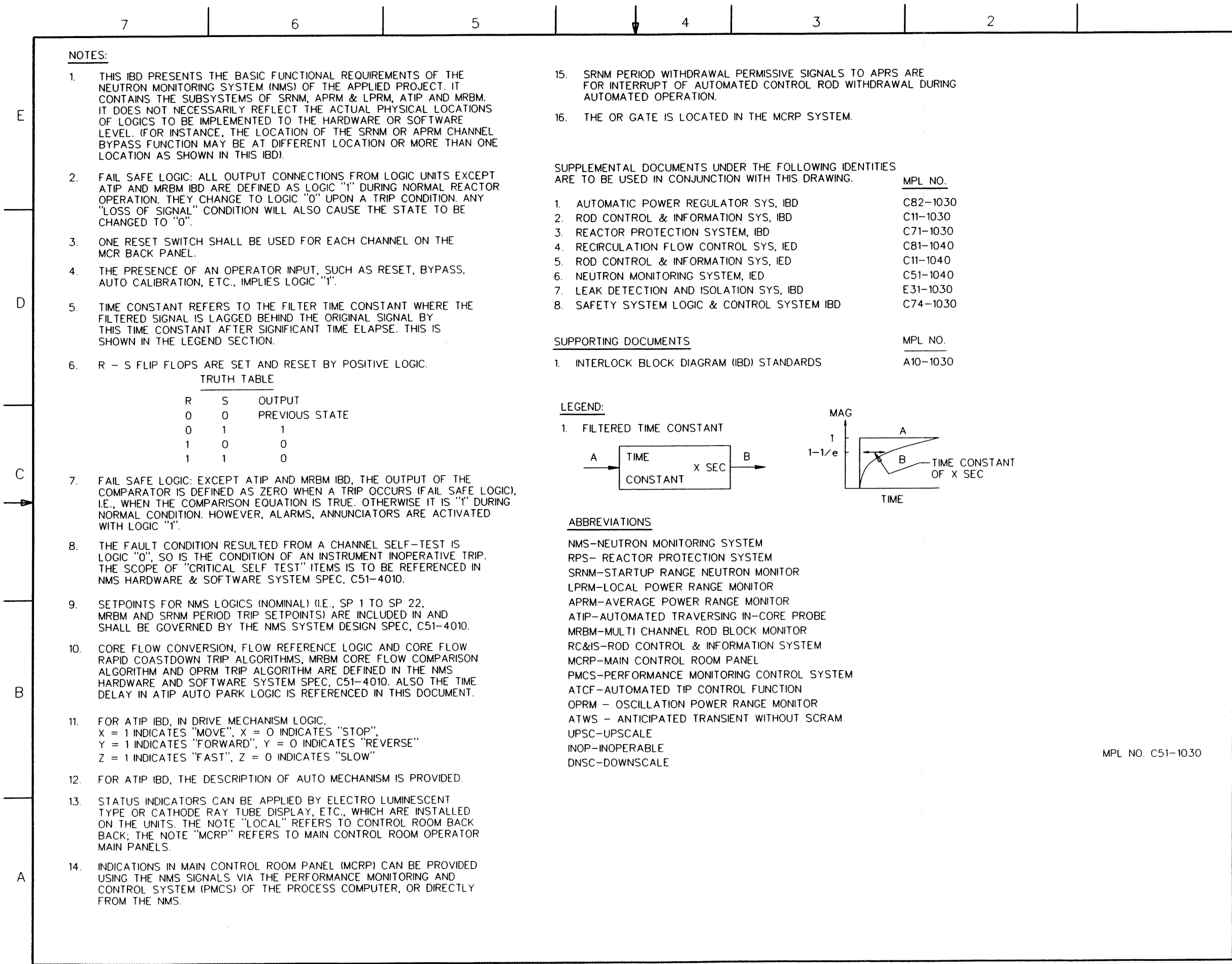


FIGURE 7.6-1 NEUTRON MONITORING SYSTEM I/FD SH 4 OF 4



E

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SH NO.	TITLE
1	REFERENCES, NOTES, ABBREVIATIONS, LEGEND
2	TABLE OF CONTENTS
3	SRNM INTERFACE TO RPS
4	SRNM INTERFACE TO RC&IS
5	SRNM INTERFACE TO RC&IS
6	SRNM CHANNEL A (TYPICAL OF 10)
7	SRNM CHANNEL A (TYPICAL OF 10)
8	SRNM CHANNEL A (TYPICAL OF 10)
9	APRM INTERFACE TO RPS & TO RC&IS
10	APRM CHANNEL A (LPRM) (TYPICAL OF 4)
11	APRM CHANNEL A (TYPICAL OF 4)
12	APRM CHANNEL A (TYPICAL OF 4)
13	APRM CHANNEL A (TYPICAL OF 4)
14	ATIP FUNCTIONAL CONFIGURATION
15	ATCF LOGIC OF ATIP (TYPICAL OF 3)
16	ATCF LOGIC OF ATIP (TYPICAL OF 3)
17	ATCF LOGIC OF ATIP (TYPICAL OF 3)
18	SRNM & APRM BYPASS SWITCH ASSIGNMENT
19	MRBM FUNCTIONAL CONFIGURATION
20	MRBM CONFIGURATION DESCRIPTION
21	MRBM CHANNEL A ROD BLOCK LOGIC (TYP FOR MRBM CH B)
22	MRBM CHANNEL A ROD BLOCK LOGIC (TYP FOR MRBM CH B)
23	MRBM CHANNEL A ROD BLOCK LOGIC (TYP FOR MRBM CH B)
24	MRBM ROD BLOCK SETPOINT SETUP LOGIC
25	SRNM "ATWS PERMISSIVE" INTERFACE TO SSLC
26	APRM "ATWS PERMISSIVE" INTERFACE TO SSLC
27	OPRM CHANNEL A (TYPICAL OF 4)
28	OPRM CONFIGURATION DESCRIPTION

D

C

B

A

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FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 2 of 28)

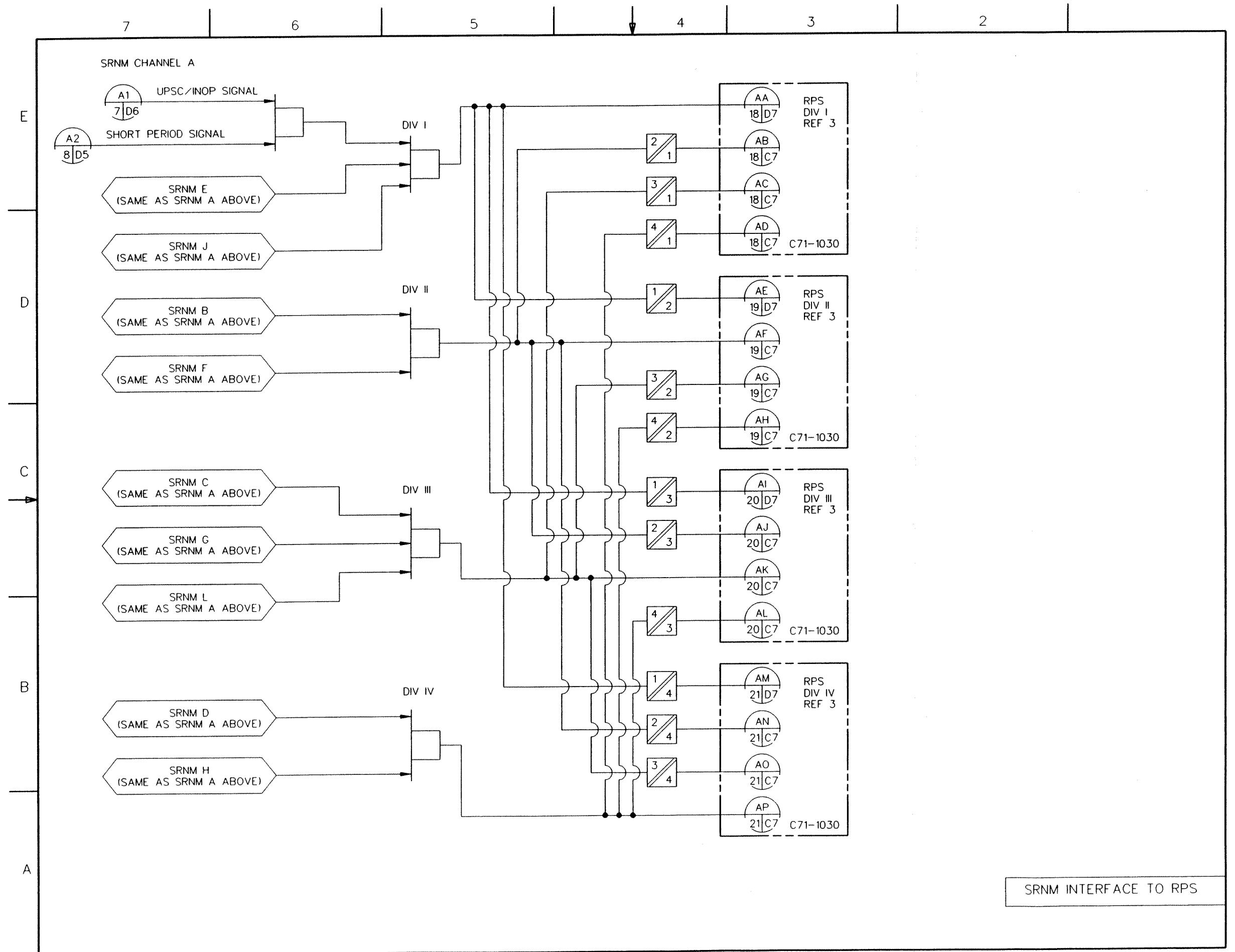


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 3 of 28)

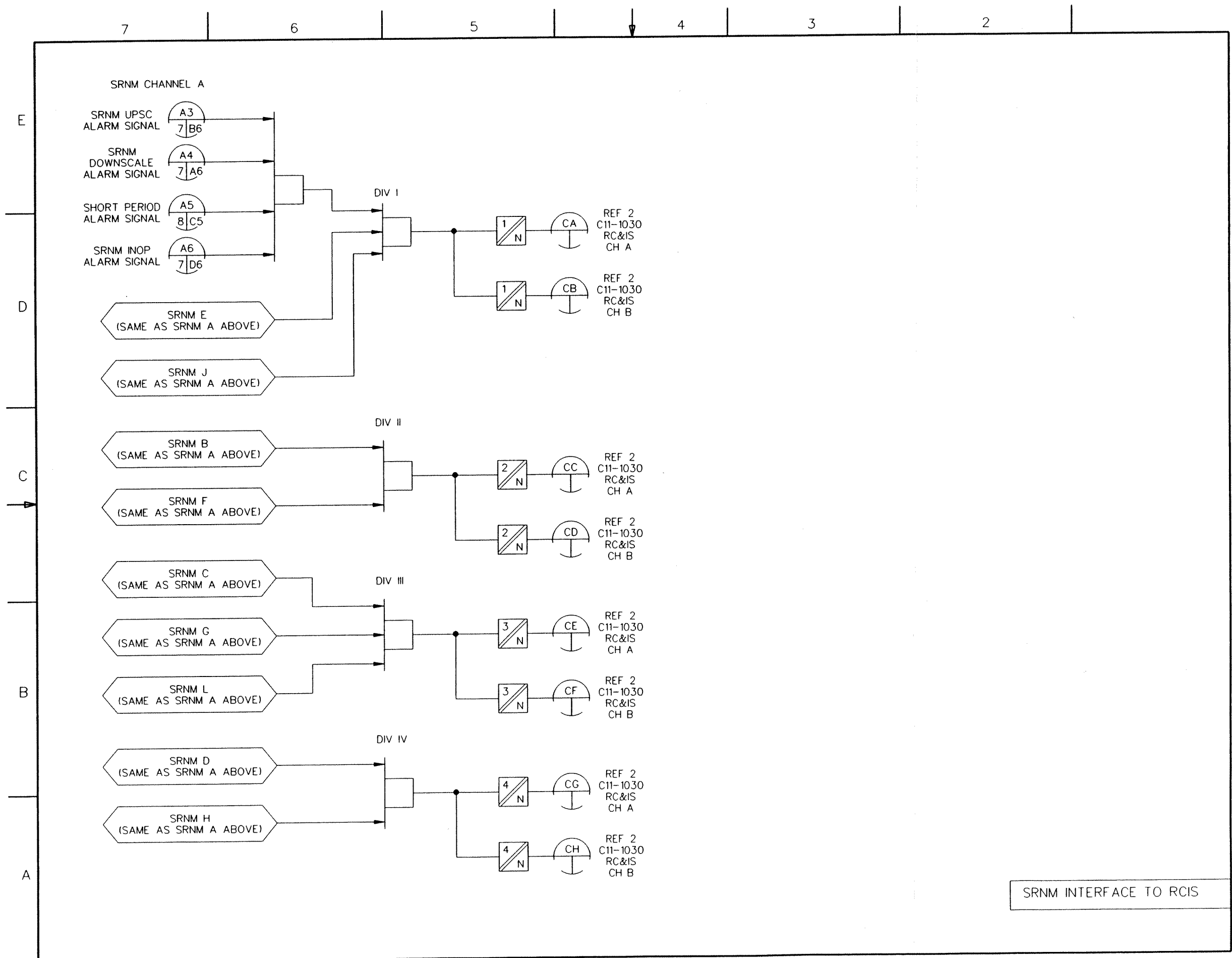


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 4 of 28)
 ABWR DCD/Tier 2 Rev. 0 21-310.4

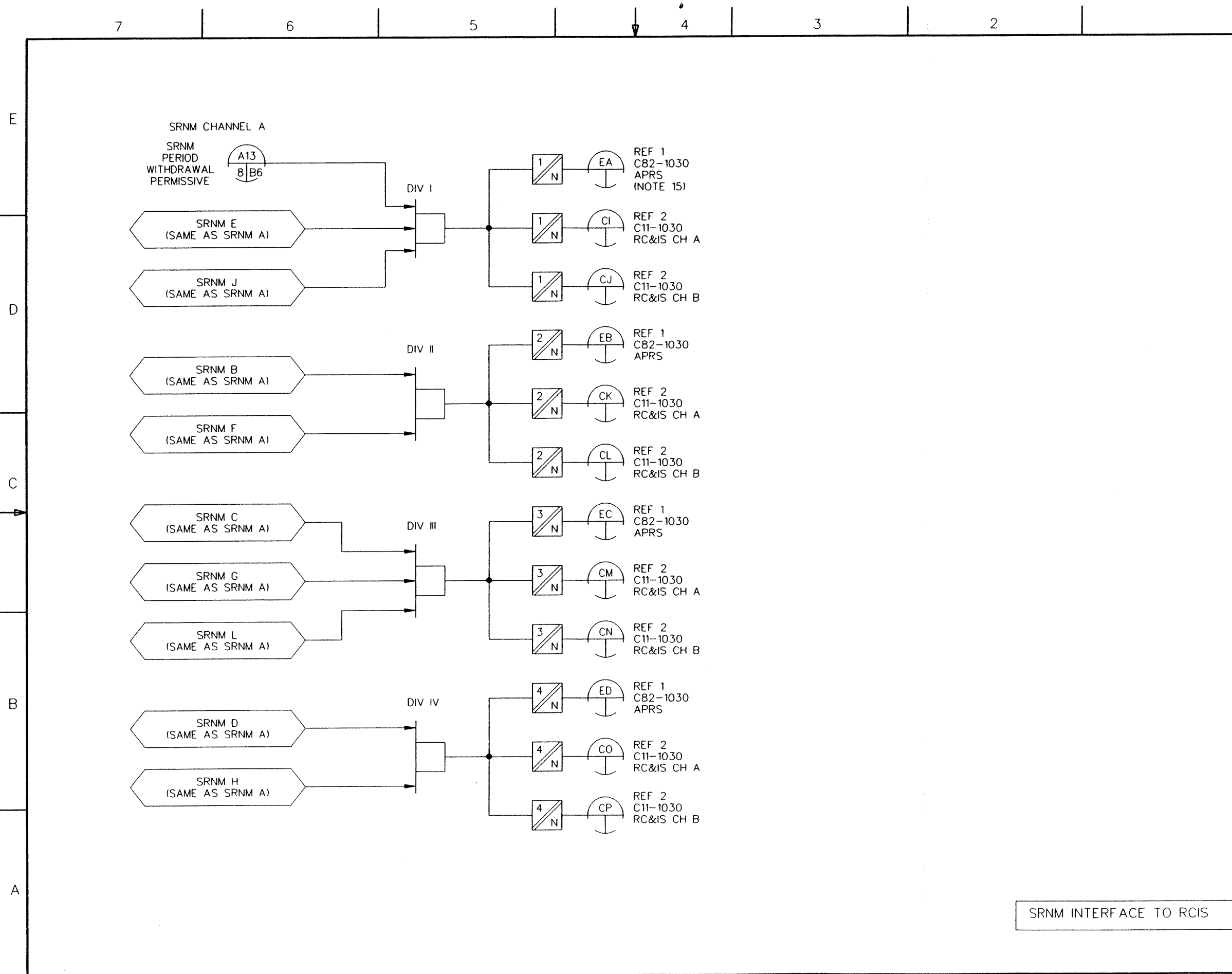
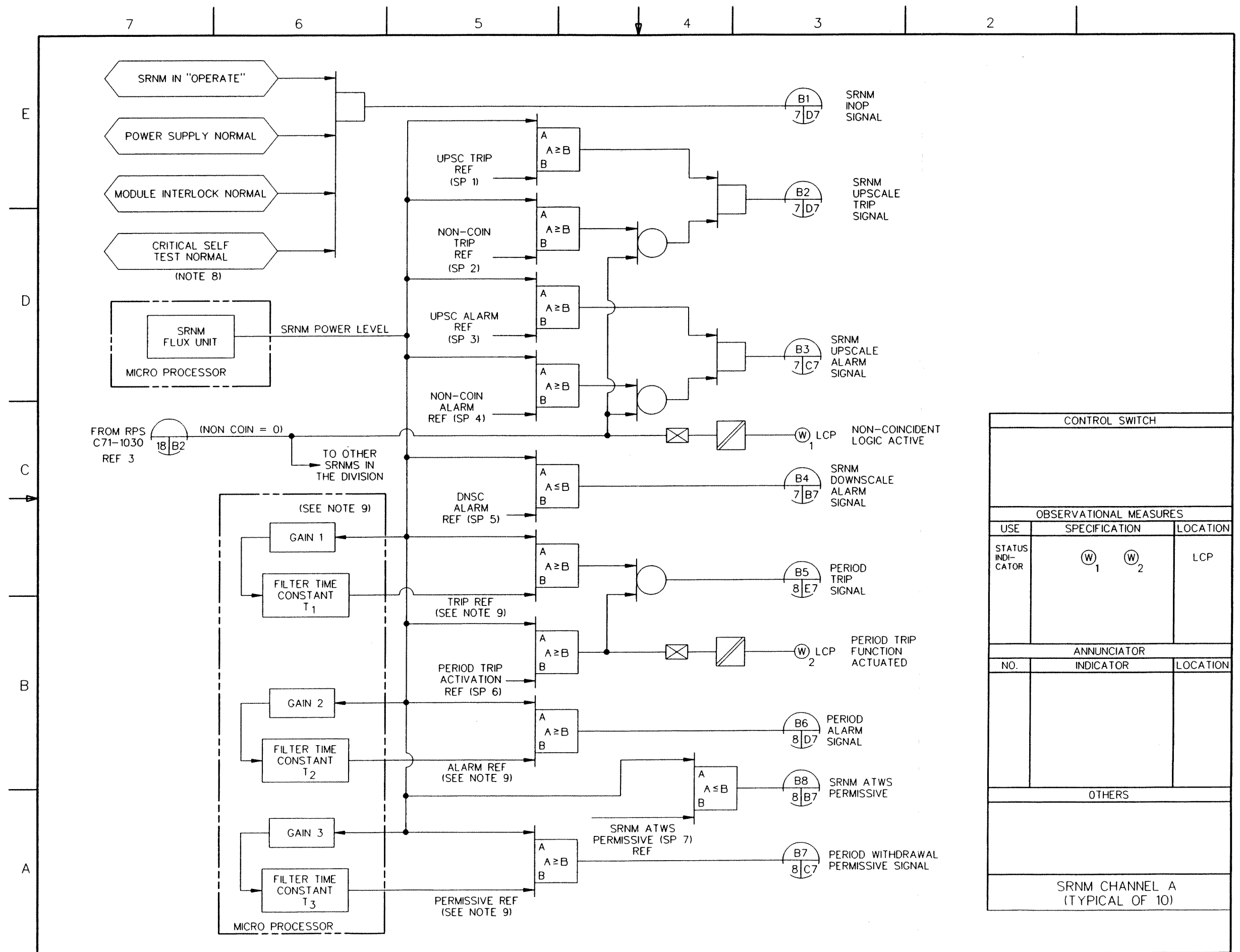


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 5 of 28)



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
STATUS INDICATOR	W1 W2	LCP
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
SRNM CHANNEL A (TYPICAL OF 10)		

FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 6 of 28)

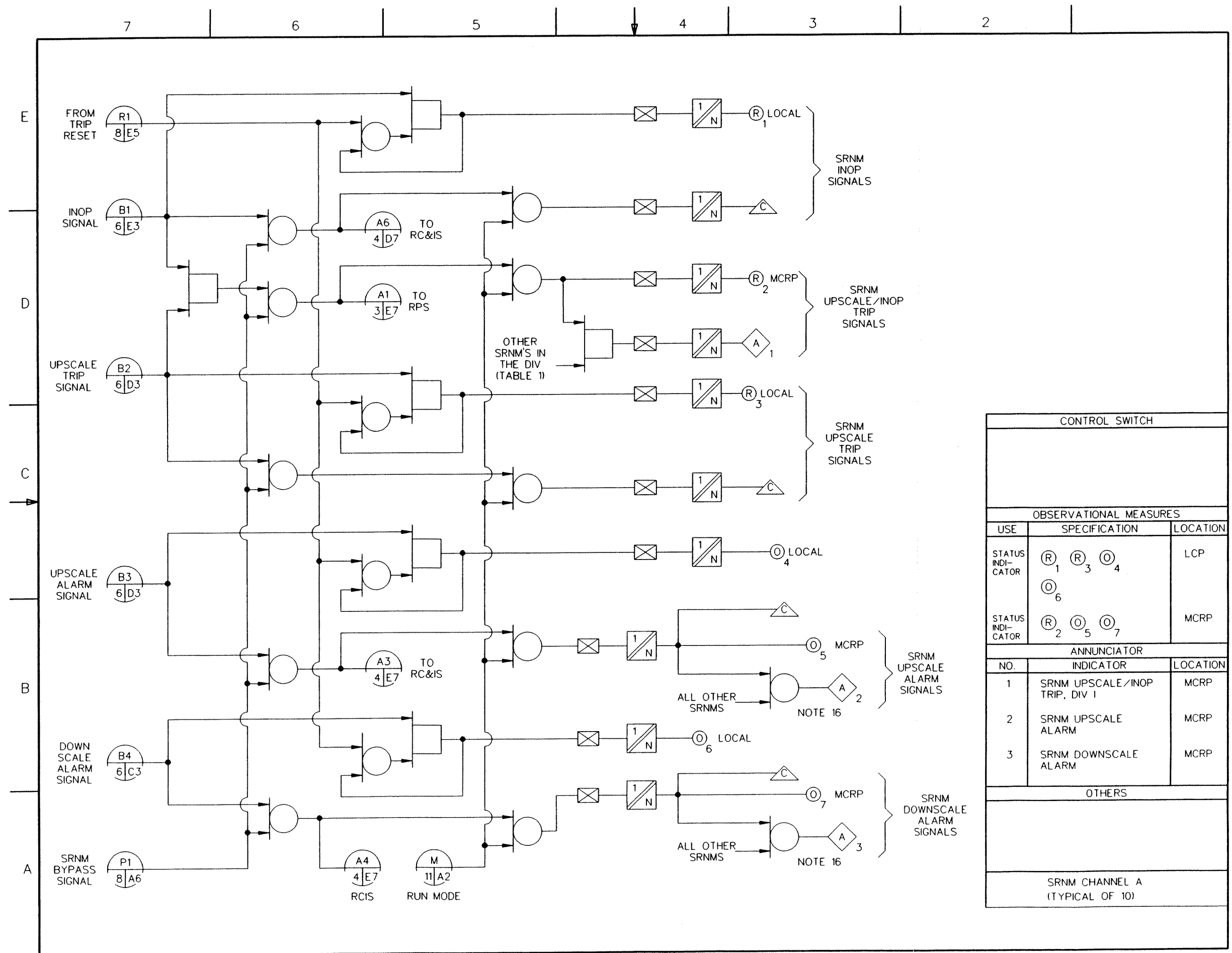
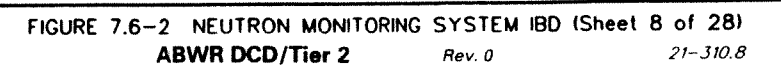


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 7 of 28)



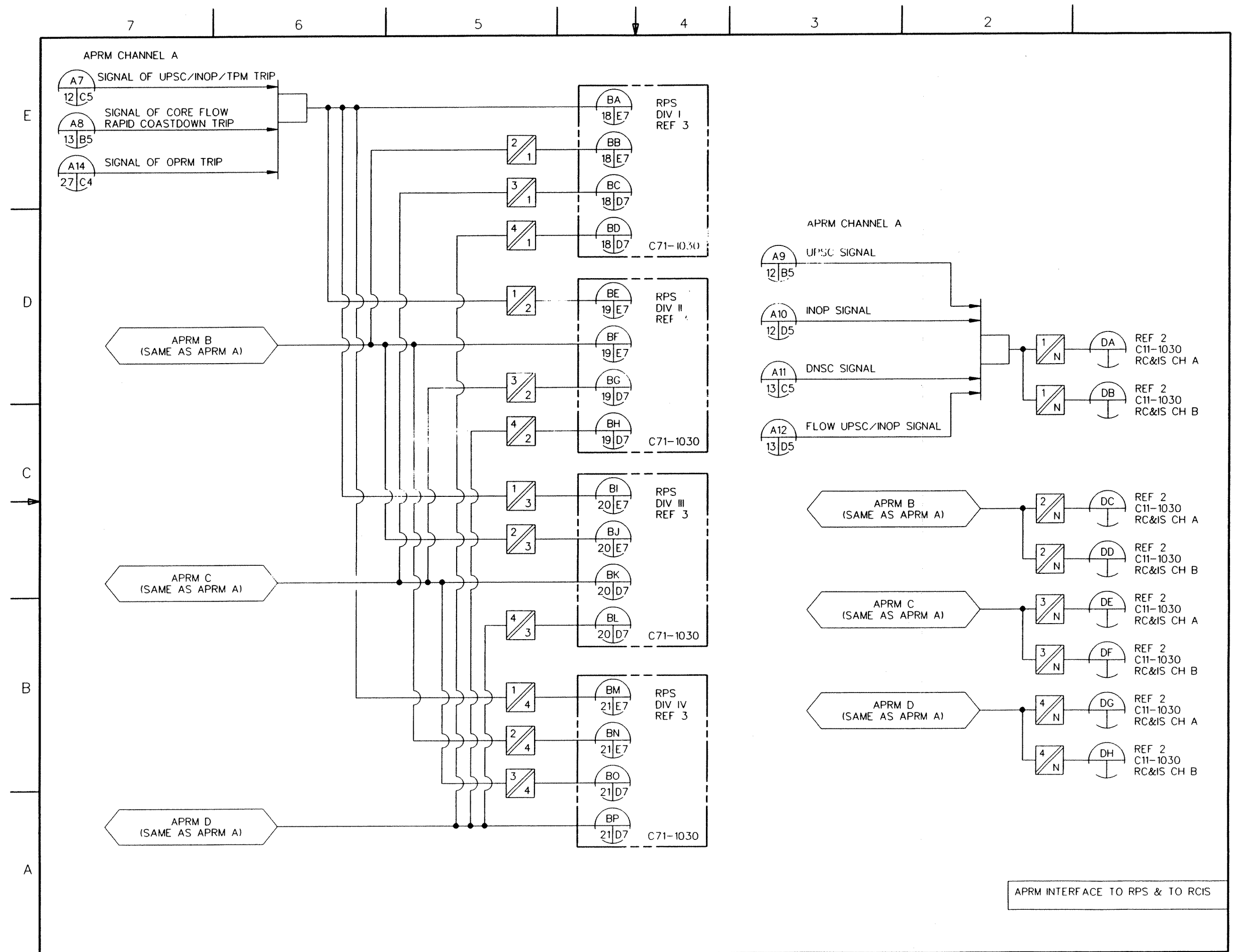


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 9 of 28)
 ABWR DCD/Tier 2 Rev. 0 21-310.9

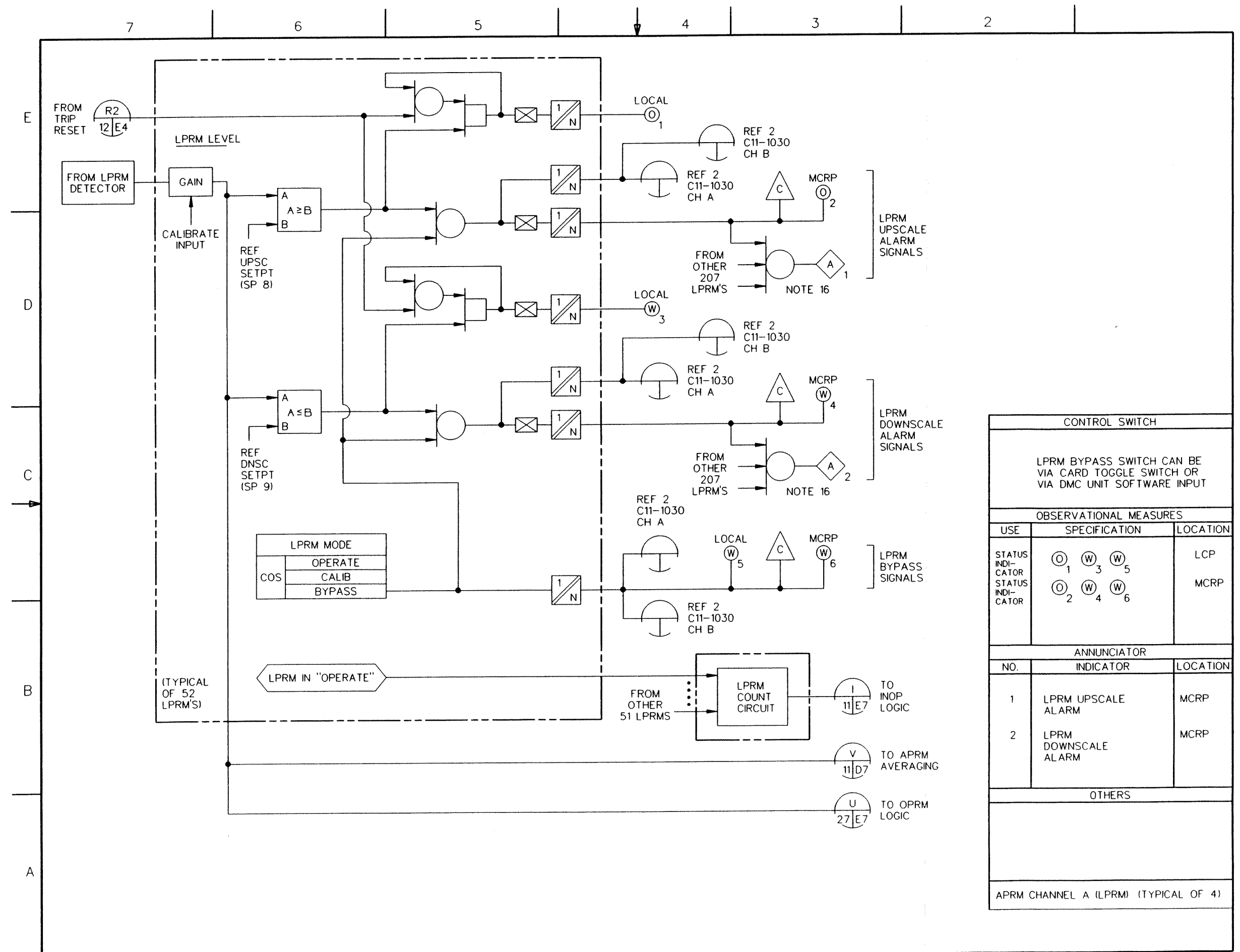
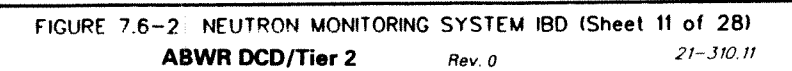


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 10 of 28)
ABWR DCD/Tier 2 Rev. 0 21-310.10



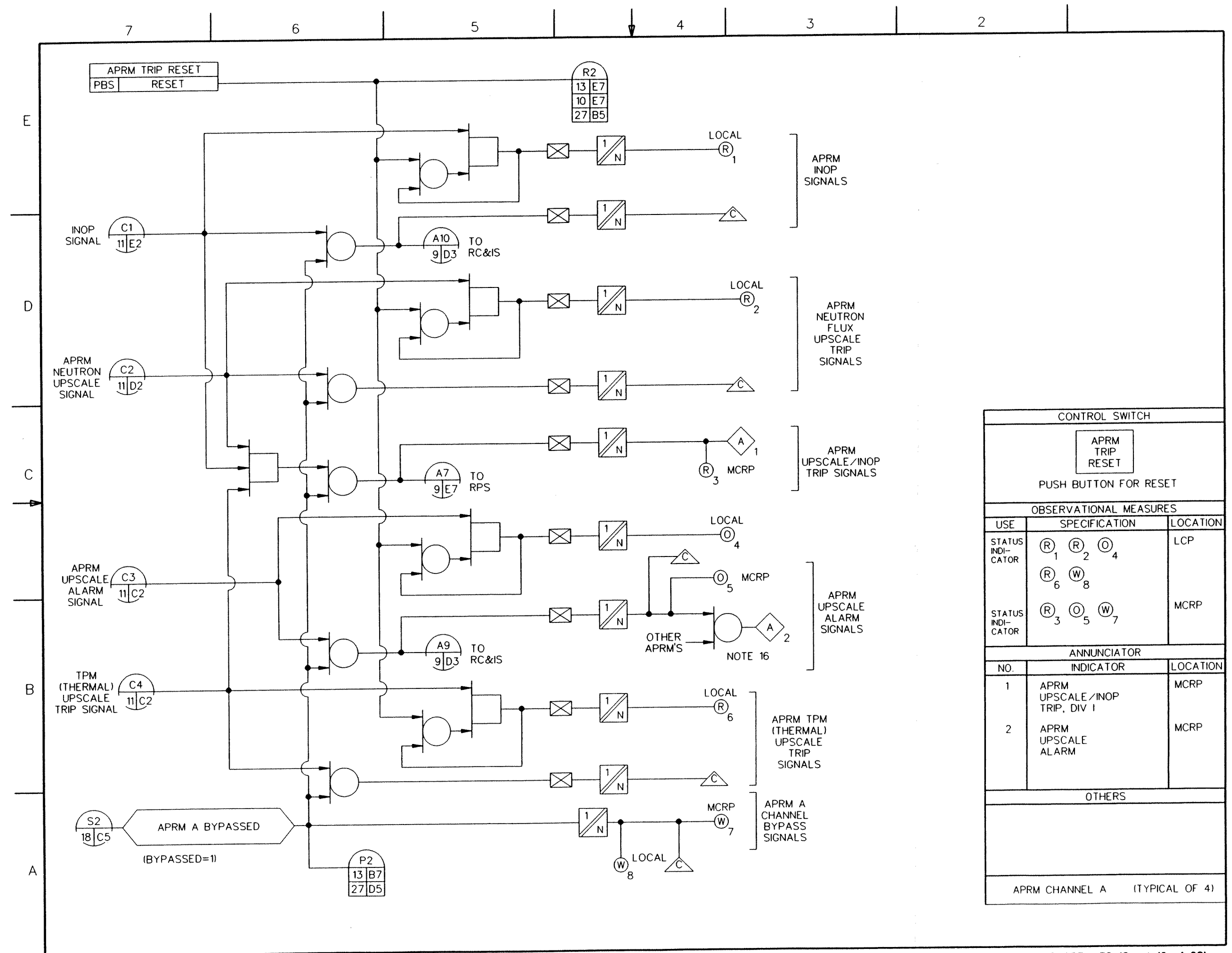


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 12 of 28)

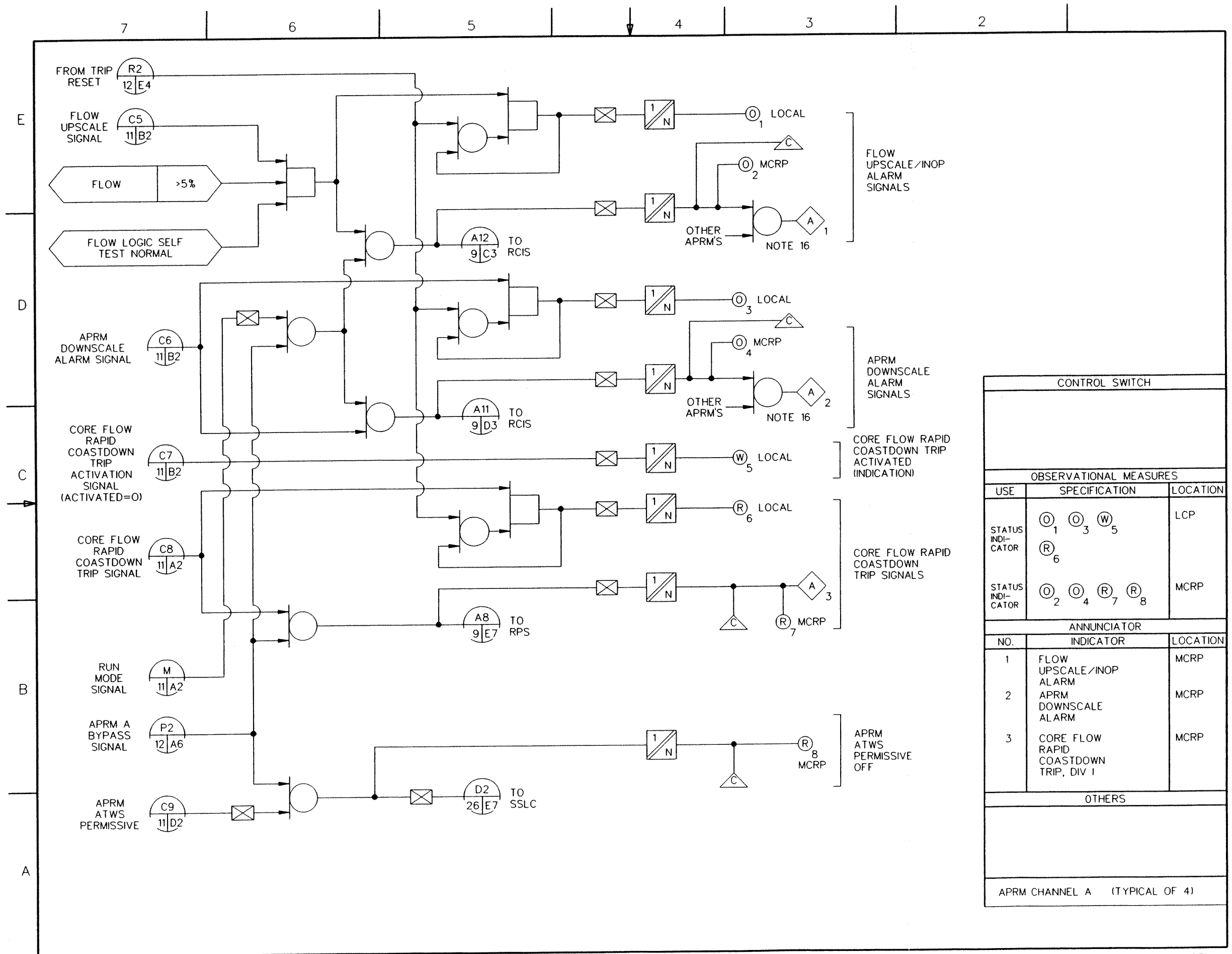


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 13 of 28)

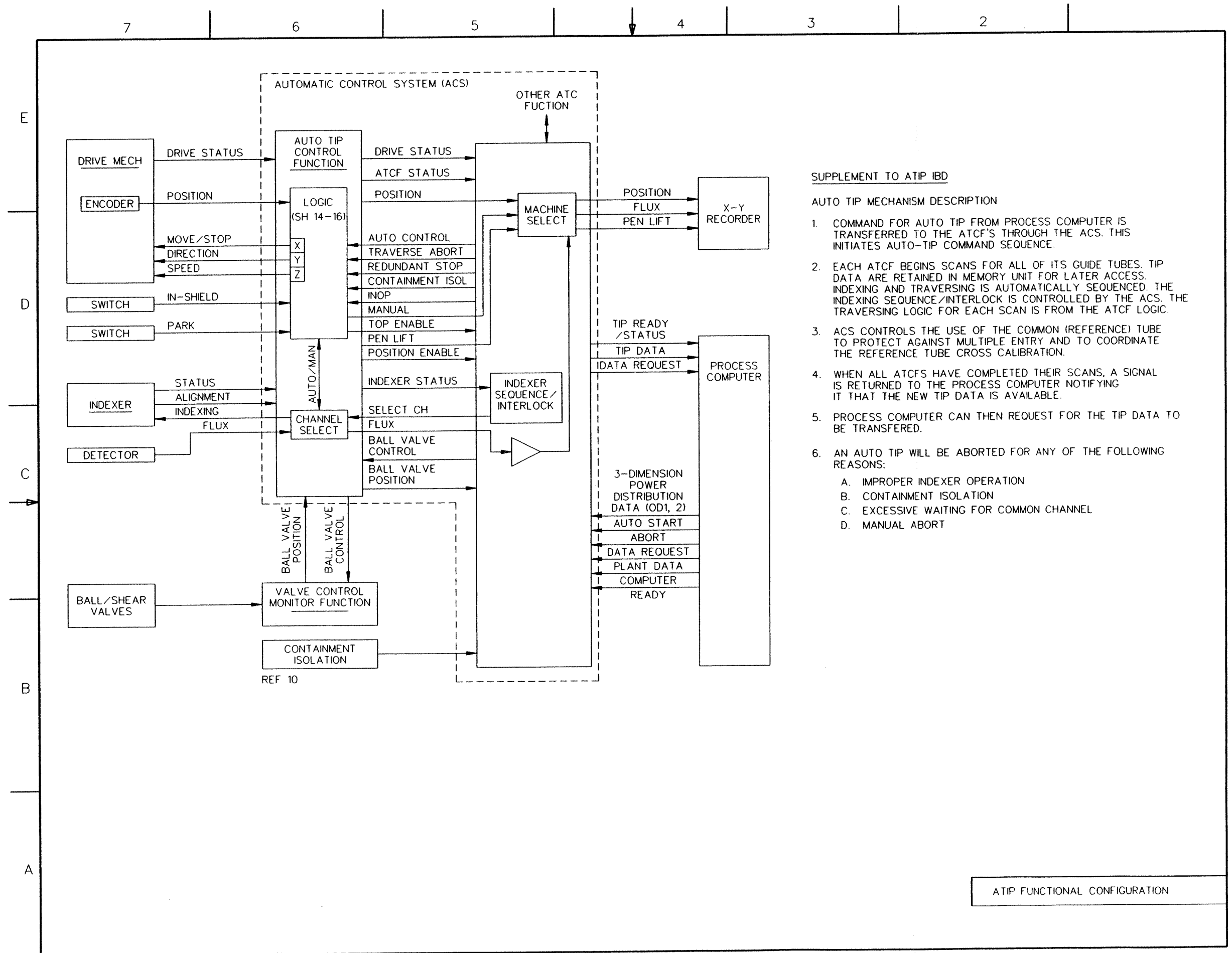


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 14 of 28)
ABWR DCD/Tier 2 Rev. 0 21-310.14

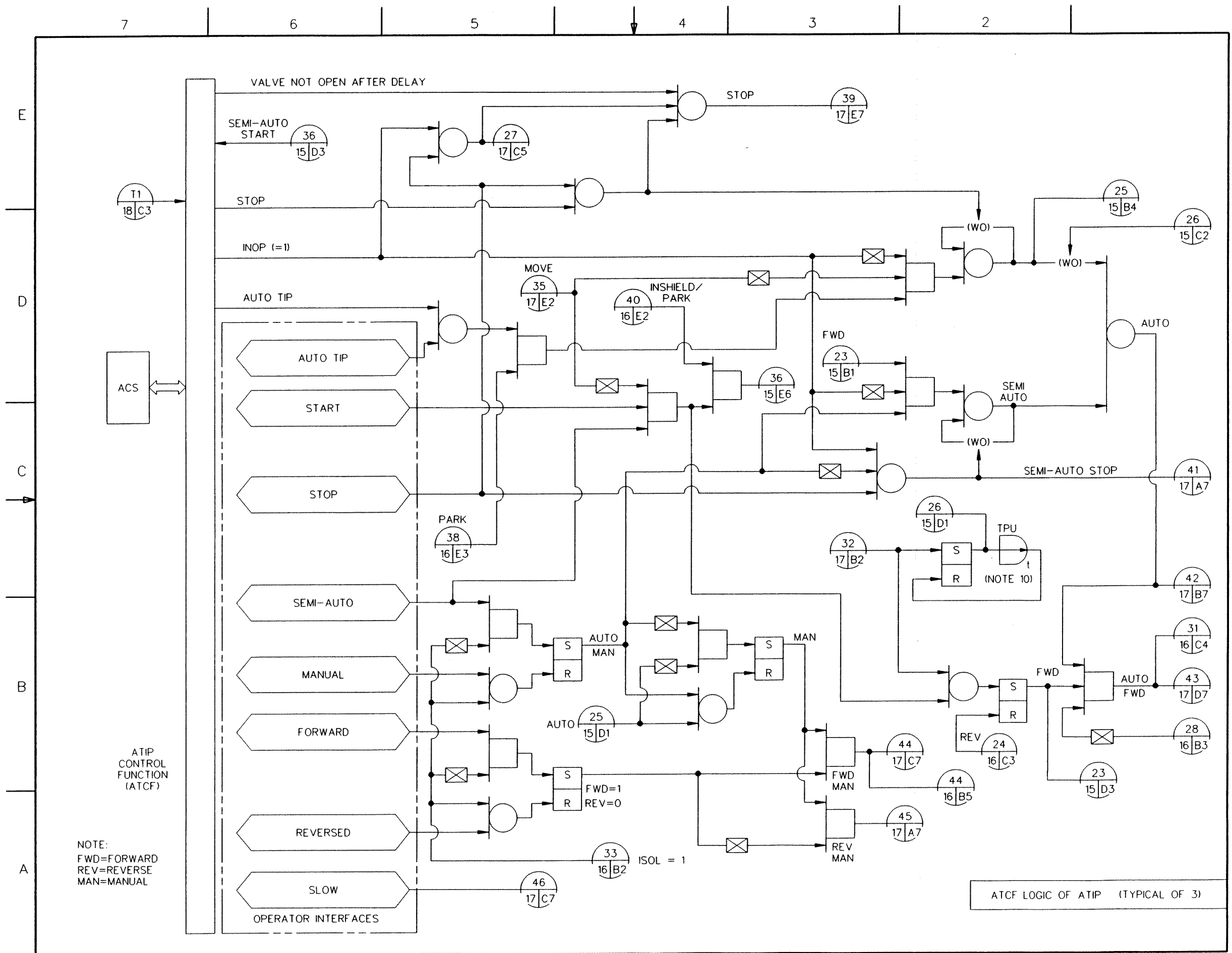


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 15 of 28)

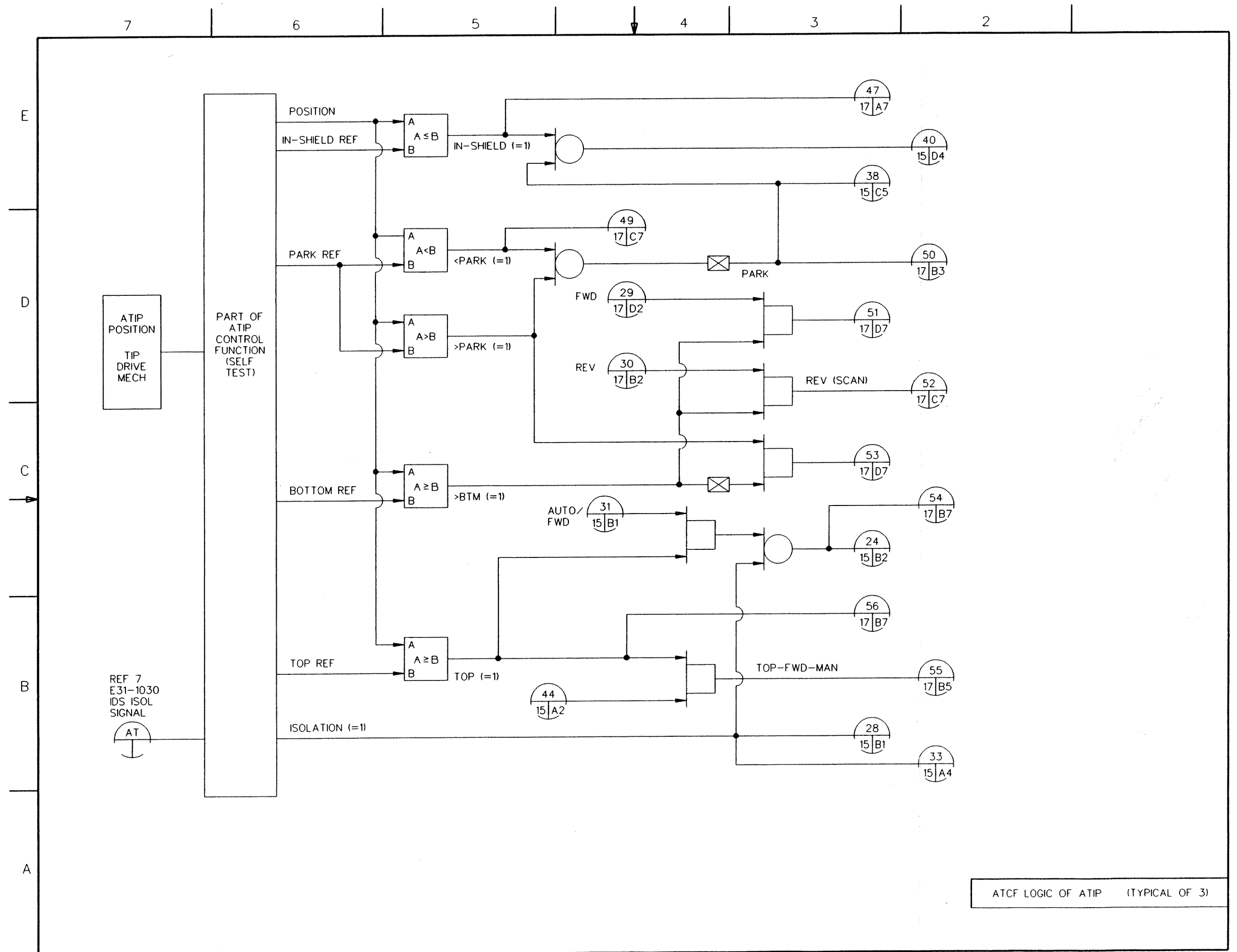


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 16 of 28)

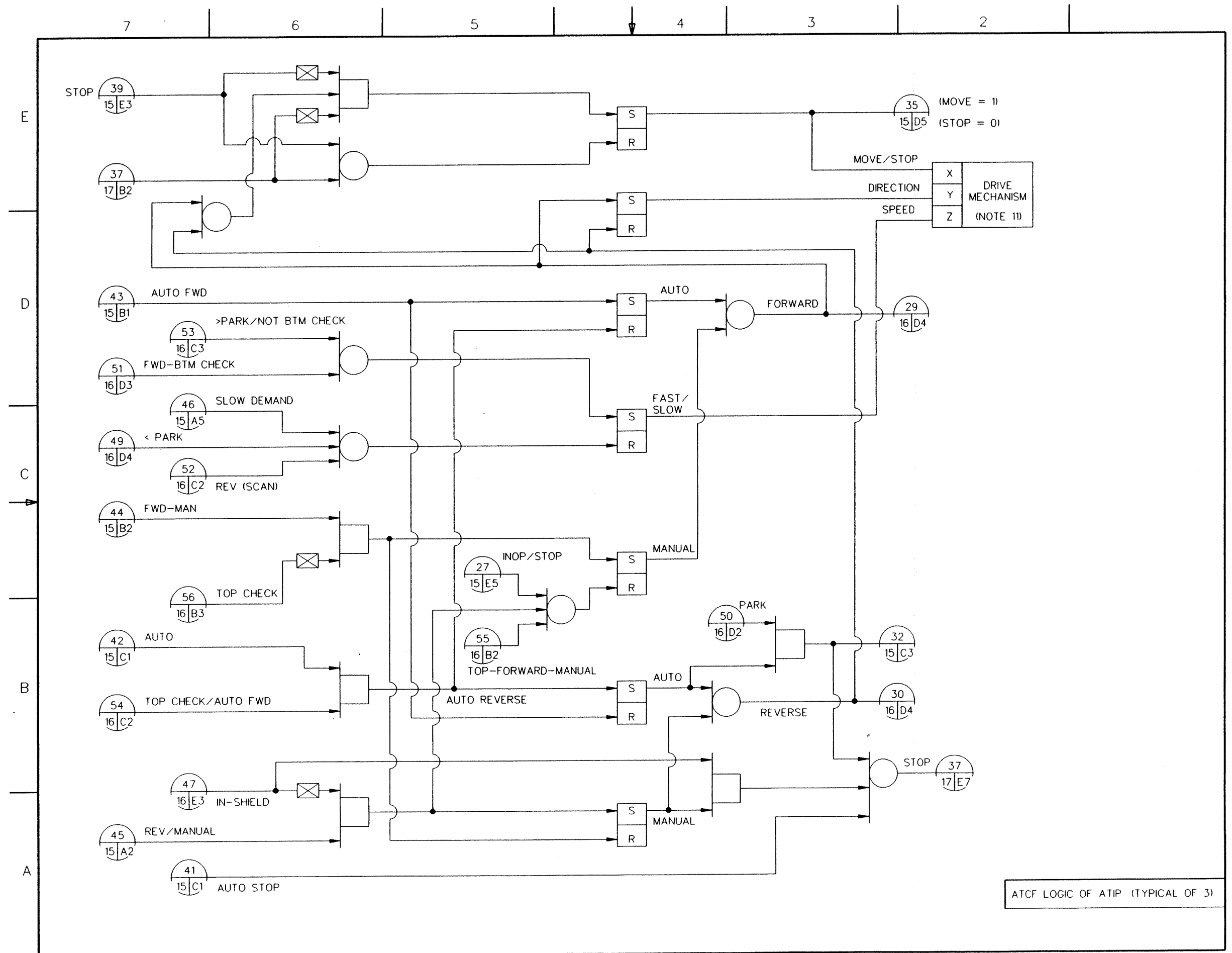
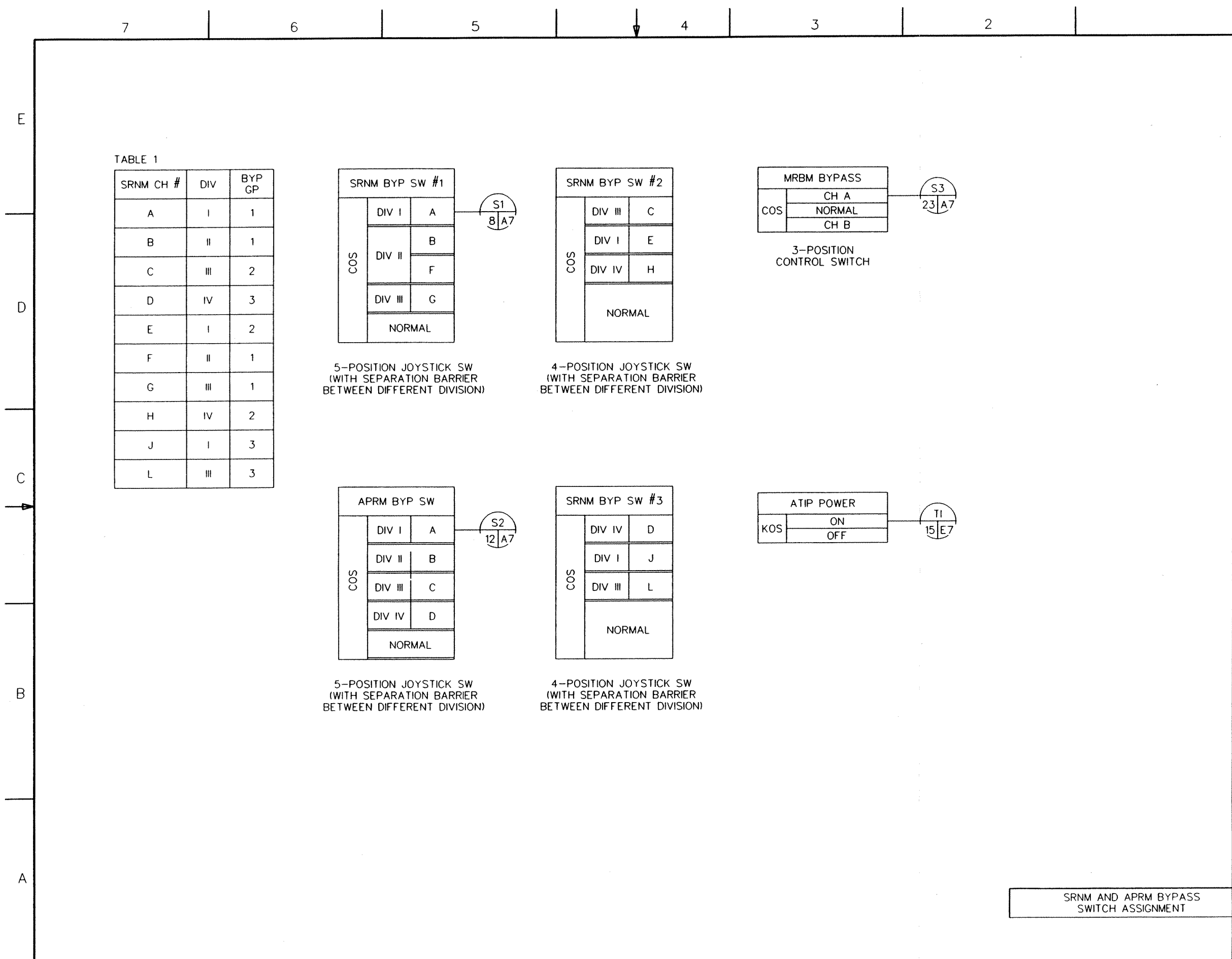


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 17 of 28)



SRNM AND APRM BYPASS
SWITCH ASSIGNMENT

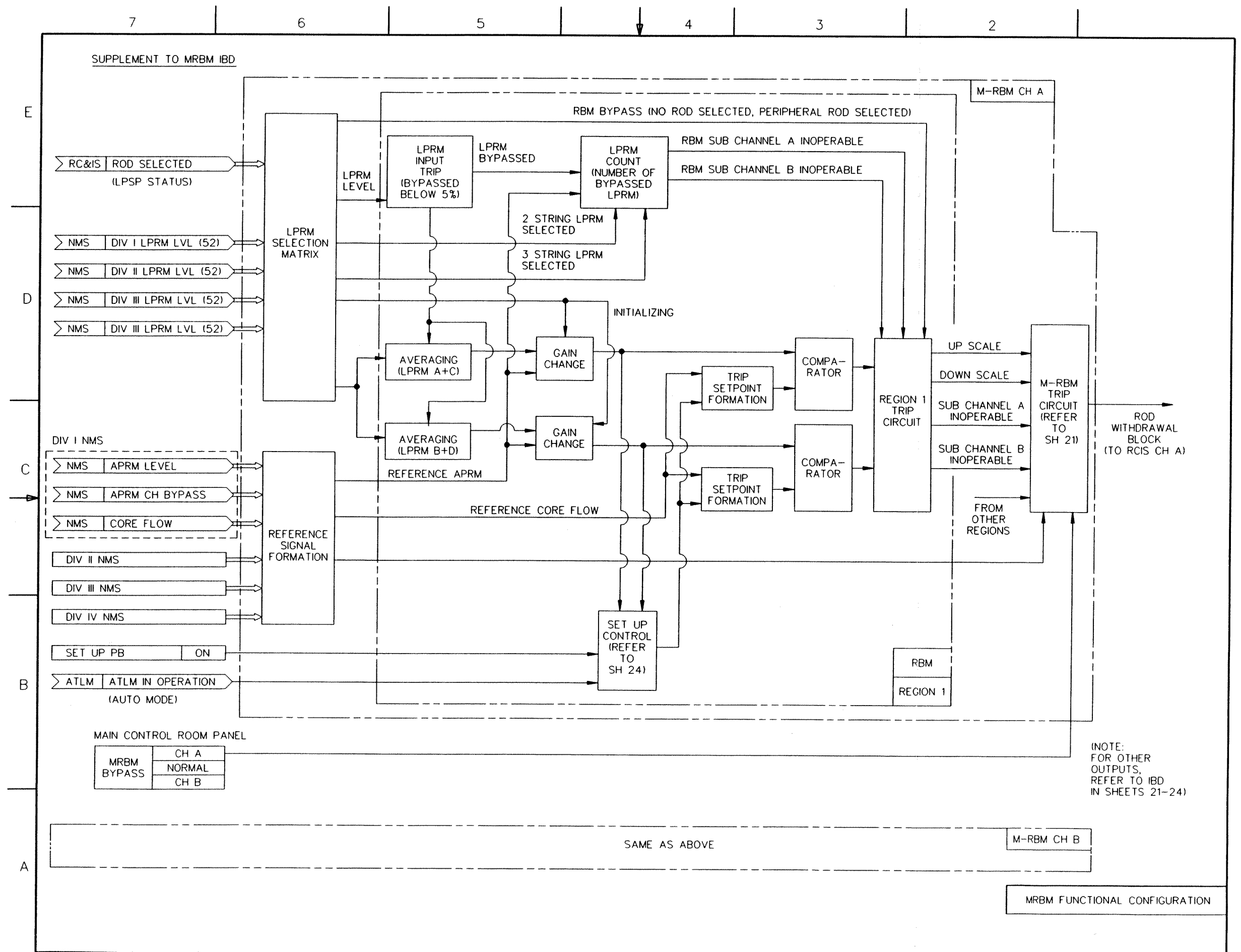


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 19 of 28)

E

D

C

→

B

A

SUPPLEMENT TO MRBM IBD

MRBM BASIC CONFIGURATION:

EACH MRBM CHANNEL HAS EIGHT RBM'S THAT COVER EIGHT CORE REGIONS, CORRESPONDING TO THE EIGHT CONTROL RODS BEING WITHDRAWN. MRBM GENERATES ROD BLOCK TRIP ACCORDING TO EACH RBM TRIP CONDITIONS. CONTROL ROD REGION ASSIGNMENT IS SHOWN IN THE FIGURE ON THIS SHEET. FOR DETAIL DESCRIPTION OF MRBM FUNCTION, REFER TO NMS HARDWARE AND SOFTWARE SYSTEM SPEC, C51-4010.

SUBSYSTEMS OF THE MRBM:

LPRM SELECTION MATRIX

WHEN ROD(S) ARE SELECTED, A MAXIMUM OF 16 LPRMS OF EACH REGION IS SELECTED WITH THE LPRM READINGS TRANSFERRED TO THE AVERAGING UNIT.

LPRM AVERAGING FUNCTION

ACTIVE LPRM SIGNALS OF EACH REGION ARE DIVIDED INTO TWO SUBCHANNELS "A & C" LEVELS AND "B & D" LEVELS. THE SIGNALS ARE SUMMED AND AVERAGE, RESPECTIVELY.

LPRM INPUT TRIP FUNCTION

EACH LPRM SIGNAL MUST BE GREATER THAN THE PRESET VALUE (5%) TO AVOID INPUT TRIP.

LPRM COUNT FUNCTION

IF THE NUMBER OF TRIPPED LPRMS EXCEEDS THE SETPOINT (SHEET 21) THIS SUBCHANNEL BECOMES INOPERABLE. CONDITIONS FOR MRBM CHANNEL INOPERABLE: 1) BOTH SUBCHANNEL A AND B IN ONE REGION ARE INOPERABLE; 2) ALL SUBCHANNEL A'S (OR B) IN ALL REGIONS ARE INOPERABLE; 3) SINGLE ROD CASE FOLLOWS CONDITION 1 ABOVE.

REFERENCE SIGNAL FORMATION FUNCTION

IT FORMS THE REFERENCE APRM SIGNAL AND THE CORE FLOW SIGNAL BY TAKING THE AVERAGE OF THE FOUR DIVISIONAL INPUTS.

GAIN CHANGE FUNCTION

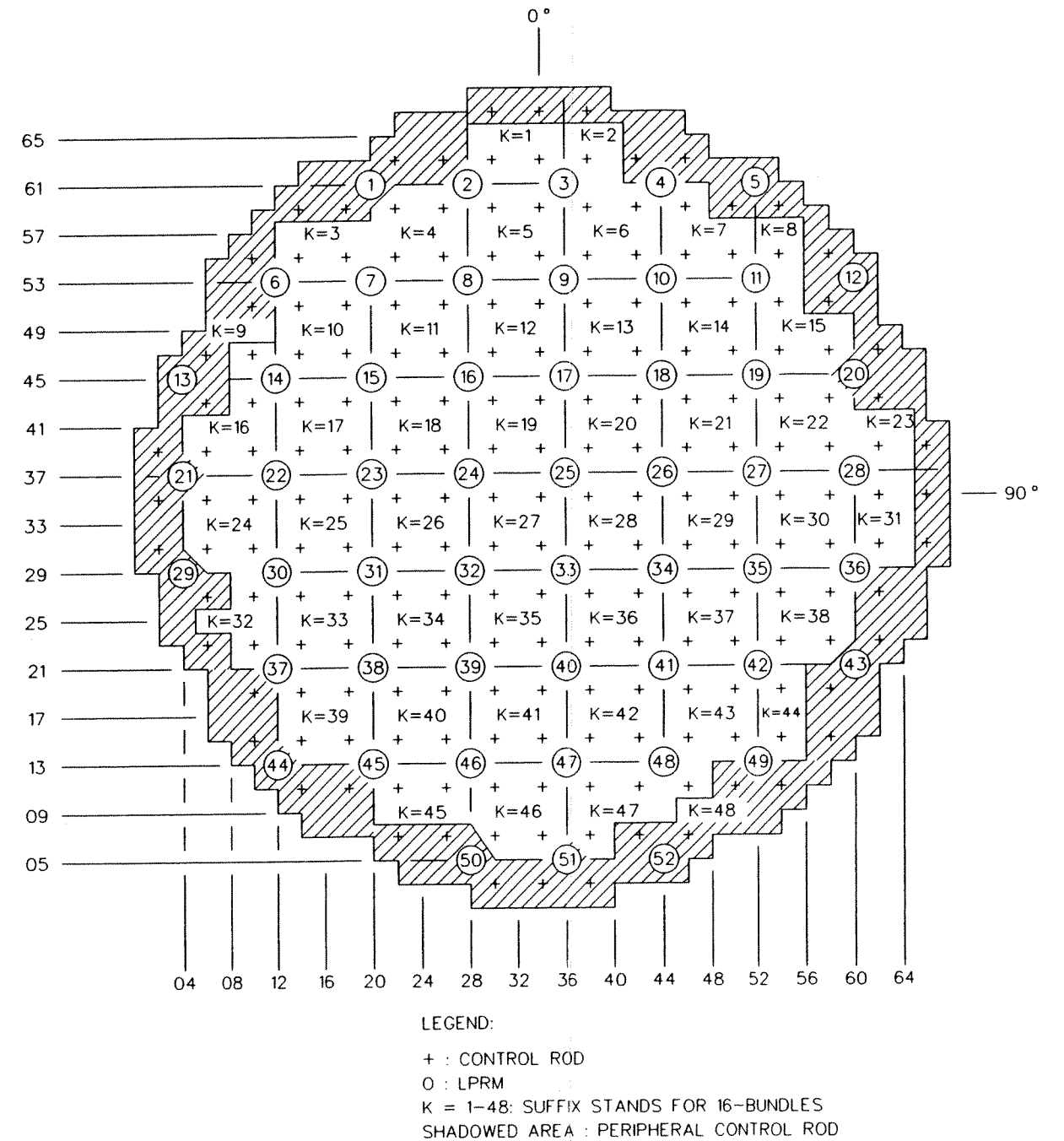
THE AVERAGED RBM SUBCHANNEL SIGNAL IS ADJUSTED TO BE EQUAL TO OR GREATER THAN THE REFERENCE APRM. SEE NMS HARDWARE AND SOFTWARE SYSTEM SPEC, C51-4010.

TRIP SETPOINT FORMATION AND TRIP FUNCTION

RBM TRIP SETPOINTS ARE CORE FLOW BIASED AND HAVE MULTIPLE ROD BLOCK LINES.

CORE FLOW COMPARISON FUNCTION

IT PERFORMS CORE FLOW COMPARISON FUNCTION BETWEEN EACH INDIVIDUAL PAIR OF CORE FLOW VALUES. IT ISSUES ROD BLOCK SIGNAL IF ABNORMAL CORE FLOW CONDITION EXISTS.



MRBM CONFIGURATION DESCRIPTION

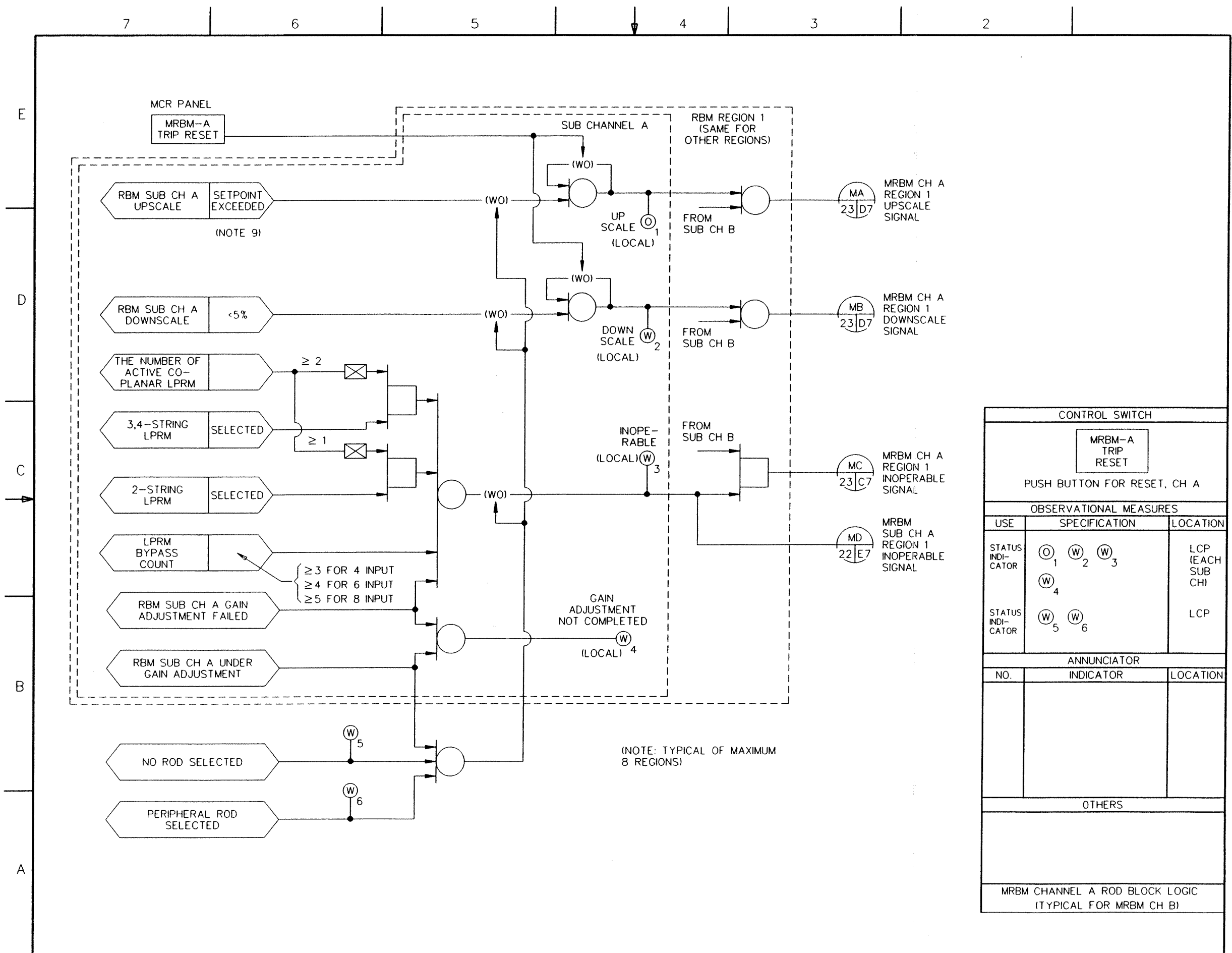
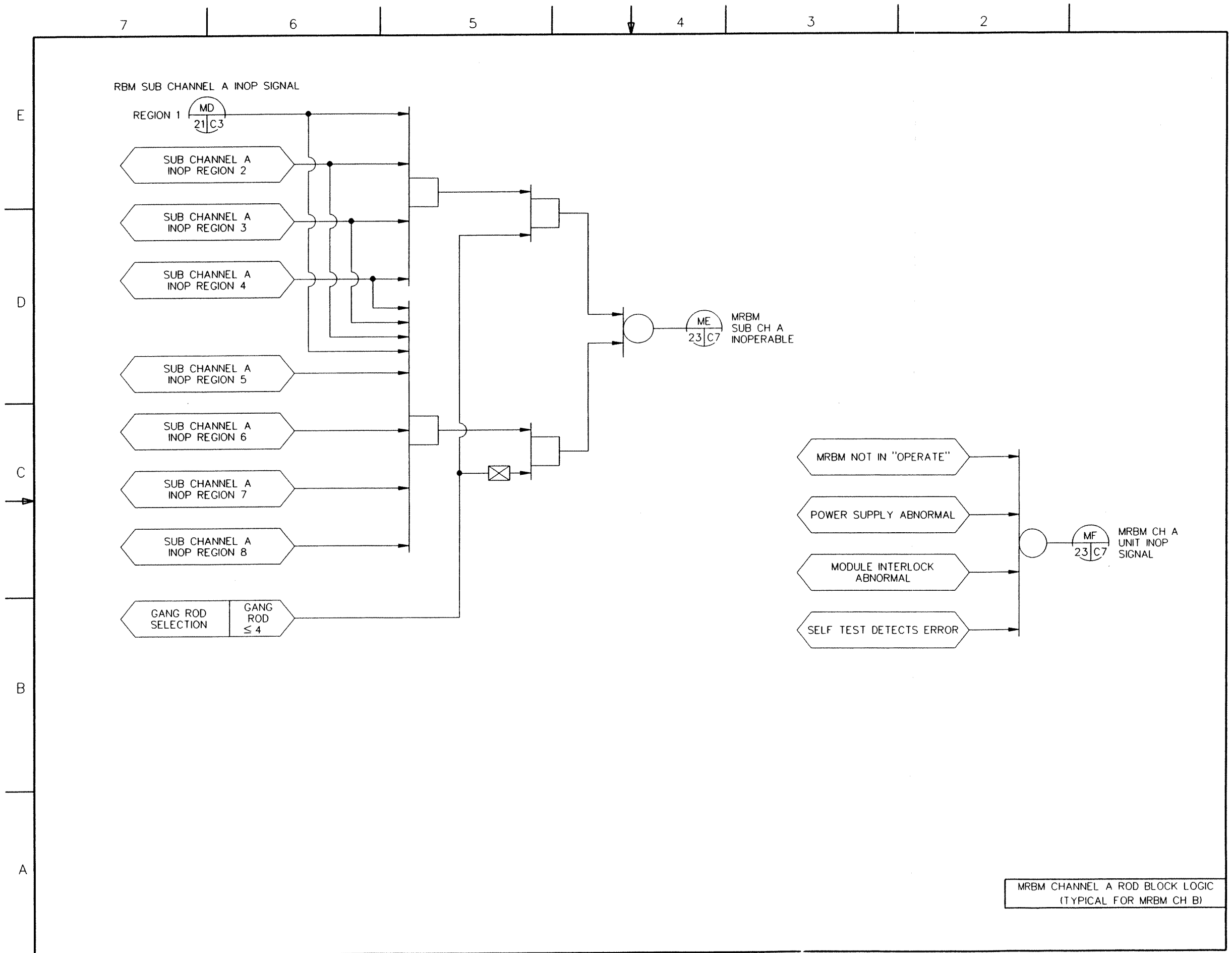


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 21 of 28)



MRBM CHANNEL A ROD BLOCK LOGIC
(TYPICAL FOR MRBM CH B)

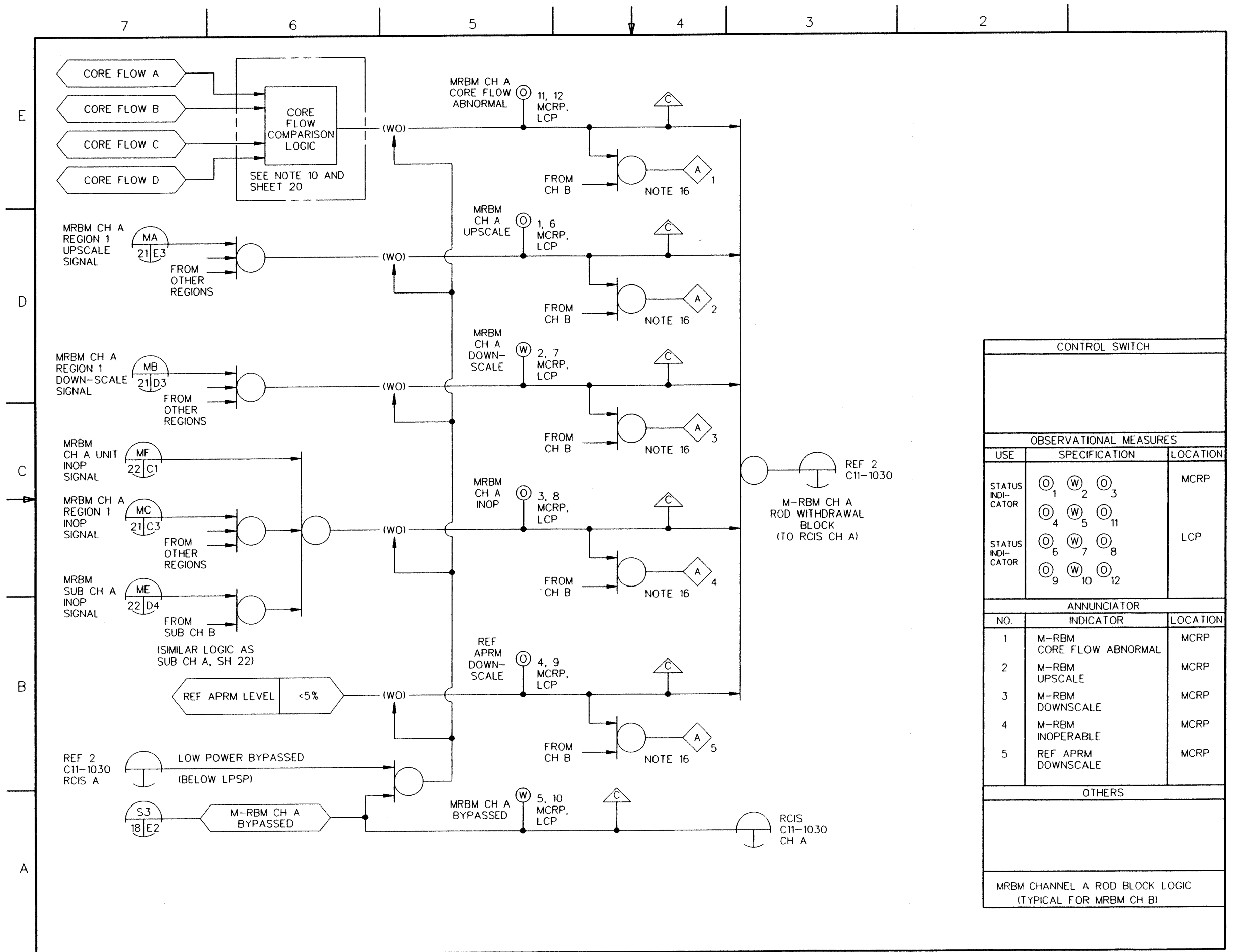


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 23 of 28)
 ABWR DCD/Tier 2 Rev. 0 21-310.23

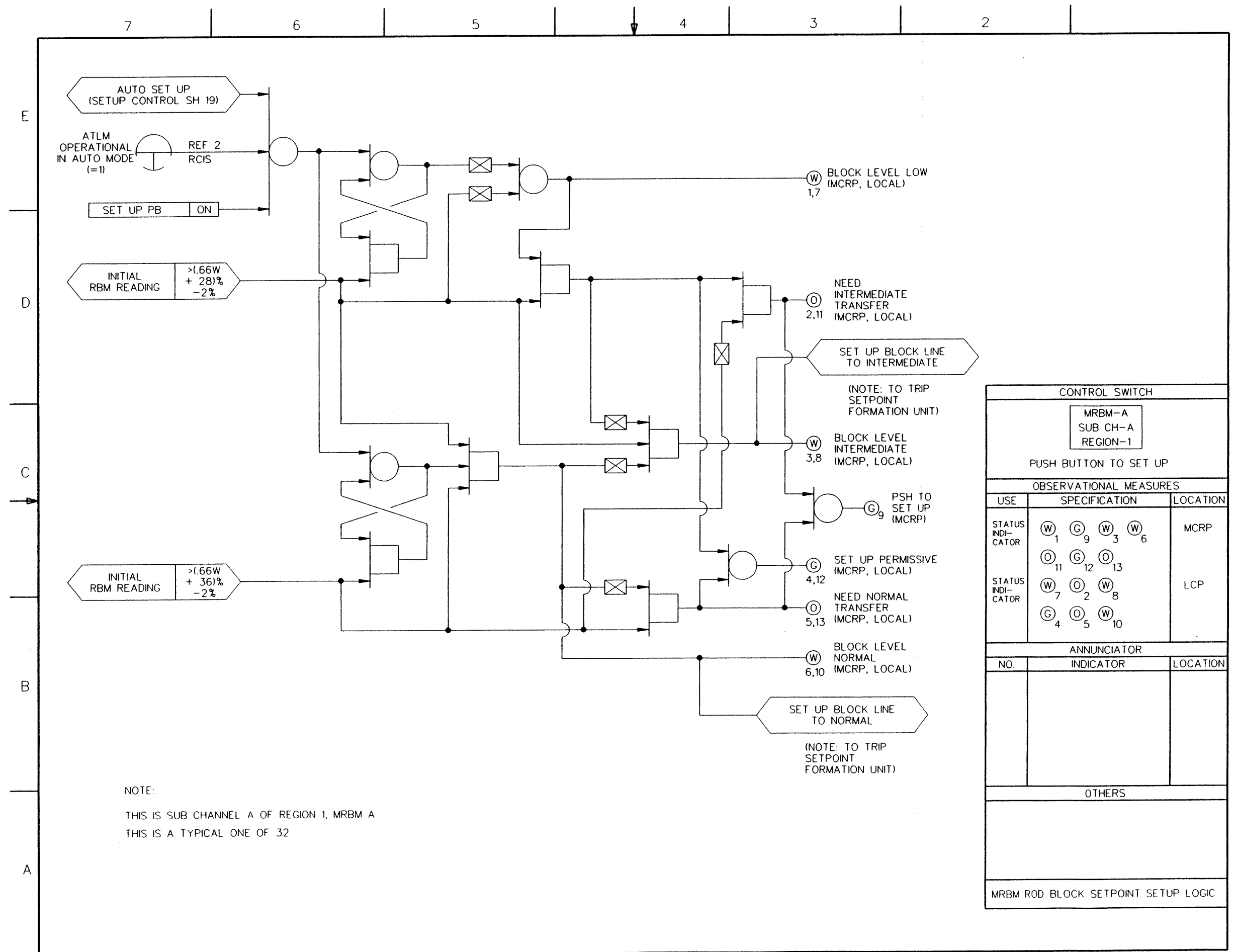


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 24 of 28)
ABWR DCD/Tier 2 Rev. 0 21-310.24

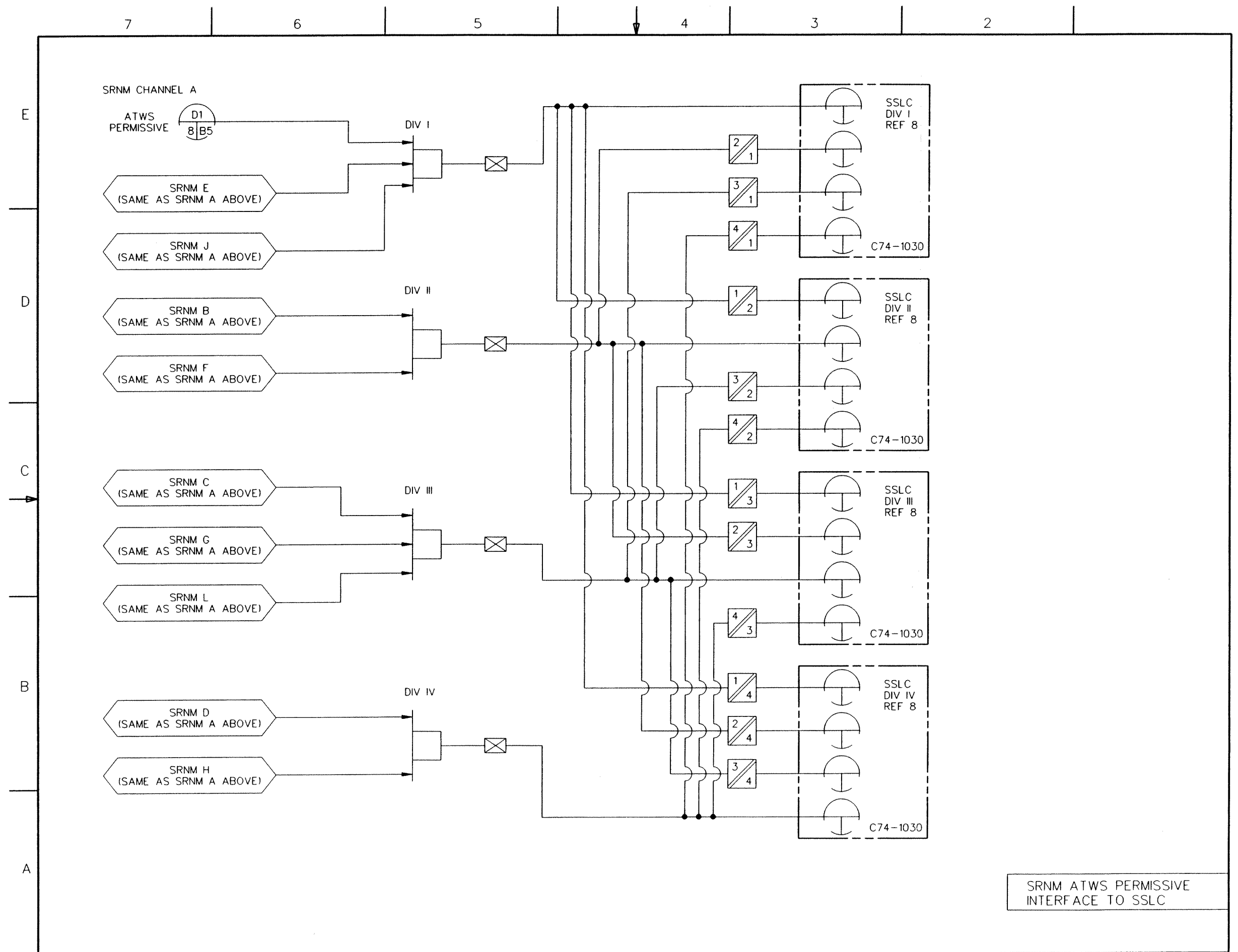


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 25 of 28)
 ABWR DCD/Tier 2 Rev. 0 21-310.25

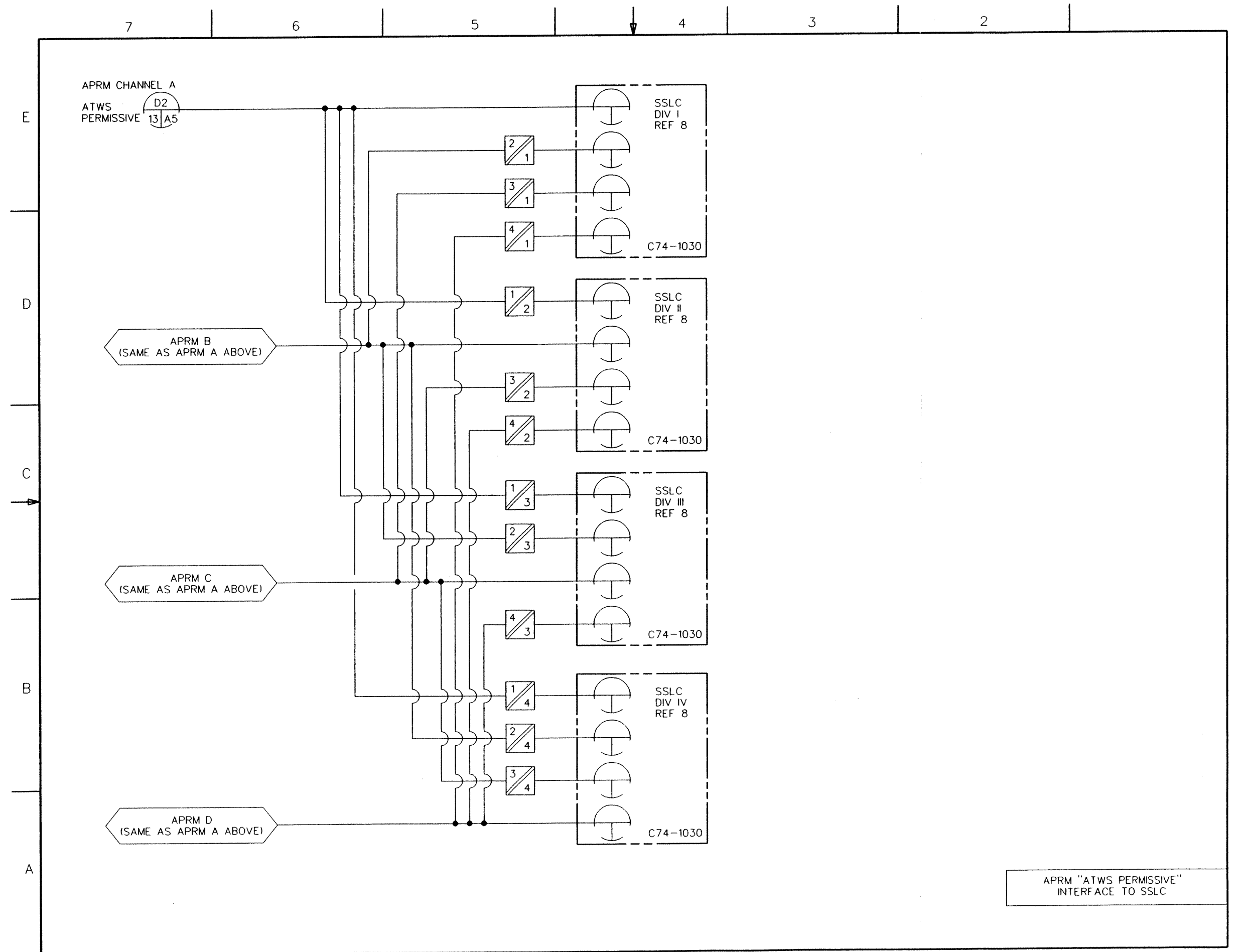
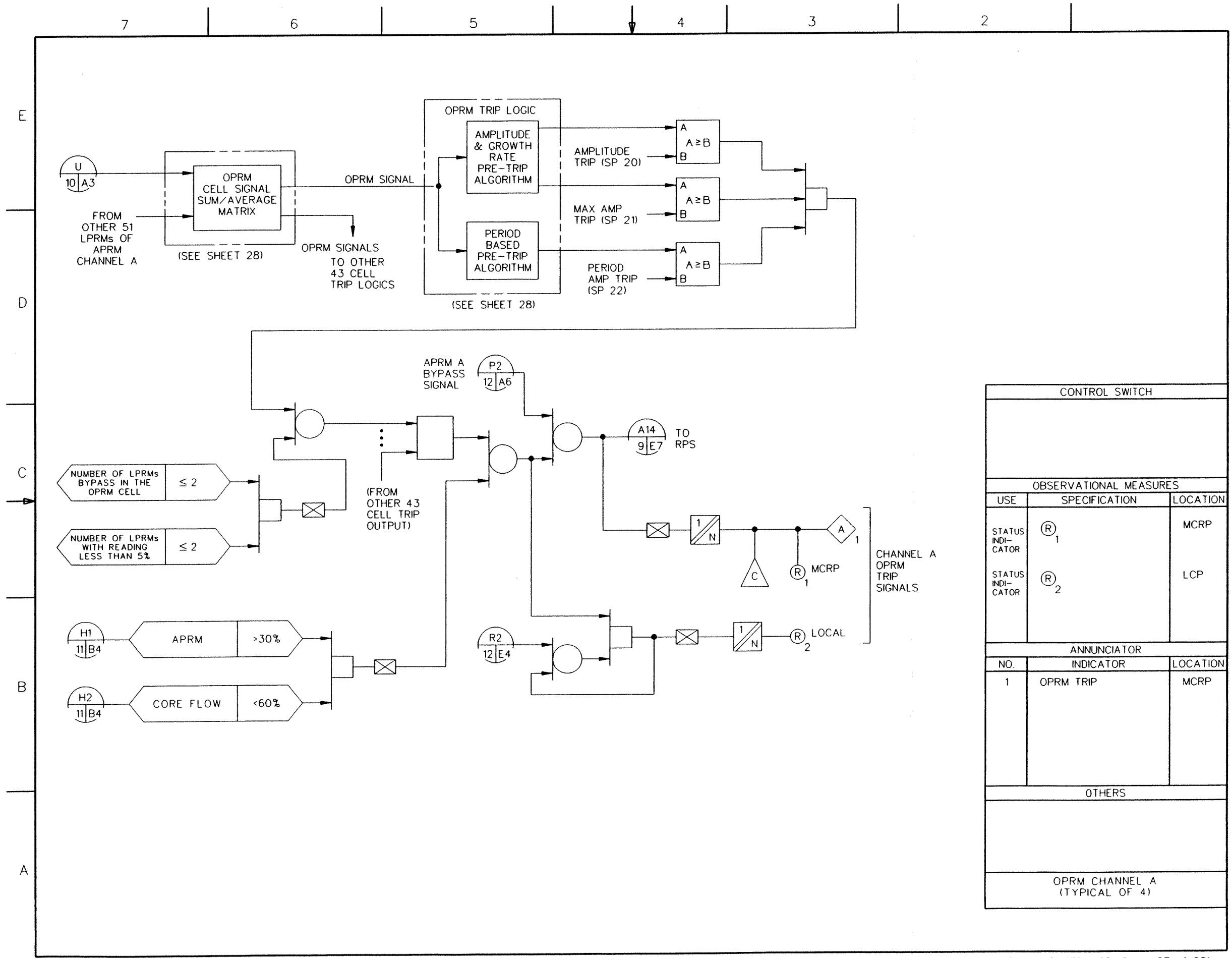


FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 26 of 28)



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
STATUS INDICATOR	(R) ₁	MCRP
STATUS INDICATOR	(R) ₂	LCP
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	OPRM TRIP	MCRP
OTHERS		
OPRM CHANNEL A (TYPICAL OF 4)		

FIGURE 7.6-2 NEUTRON MONITORING SYSTEM IBD (Sheet 27 of 28)
 ABWR DCD/Tier 2 Rev. 0 21-310.27

SUPPLEMENT TO OPRM IBD

A. OPRM BASIC CONFIGURATION:

THE ASSIGNMENT OF LPRM SIGNALS TO DIFFERENT OPRM CHANNELS IS SHOWN IN THE RIGHT FIGURE. THE OPRM CHANNEL A RECEIVES THE SAME 52 LPRM SIGNALS AS FROM THE SAME APRM CHANNEL A. THERE ARE 44 OPRM CELLS. EACH CELL TAKES FOUR LPRM SIGNALS AS INPUT, EXCEPT NEAR THE PERIPHERY WHERE THREE LPRMS ARE USED. THE FOUR LPRMS ARE FROM THE LPRM STRINGS AT THE FOUR CORNERS OF A 4X4 FUEL BUNDLE SQUARE REGION. EACH OPRM CELL MONITORS THE NEUTRON FLUX IN THE ASSIGNED CORE REGION AND ISSUES TRIP SIGNAL AS AN OPRM TRIP IF THE TRIP SETPOINT IS EXCEEDED.

B. OPRM DESIGN LOGICS & ALGORITHM

OPRM CELL SIGNAL CALCULATION

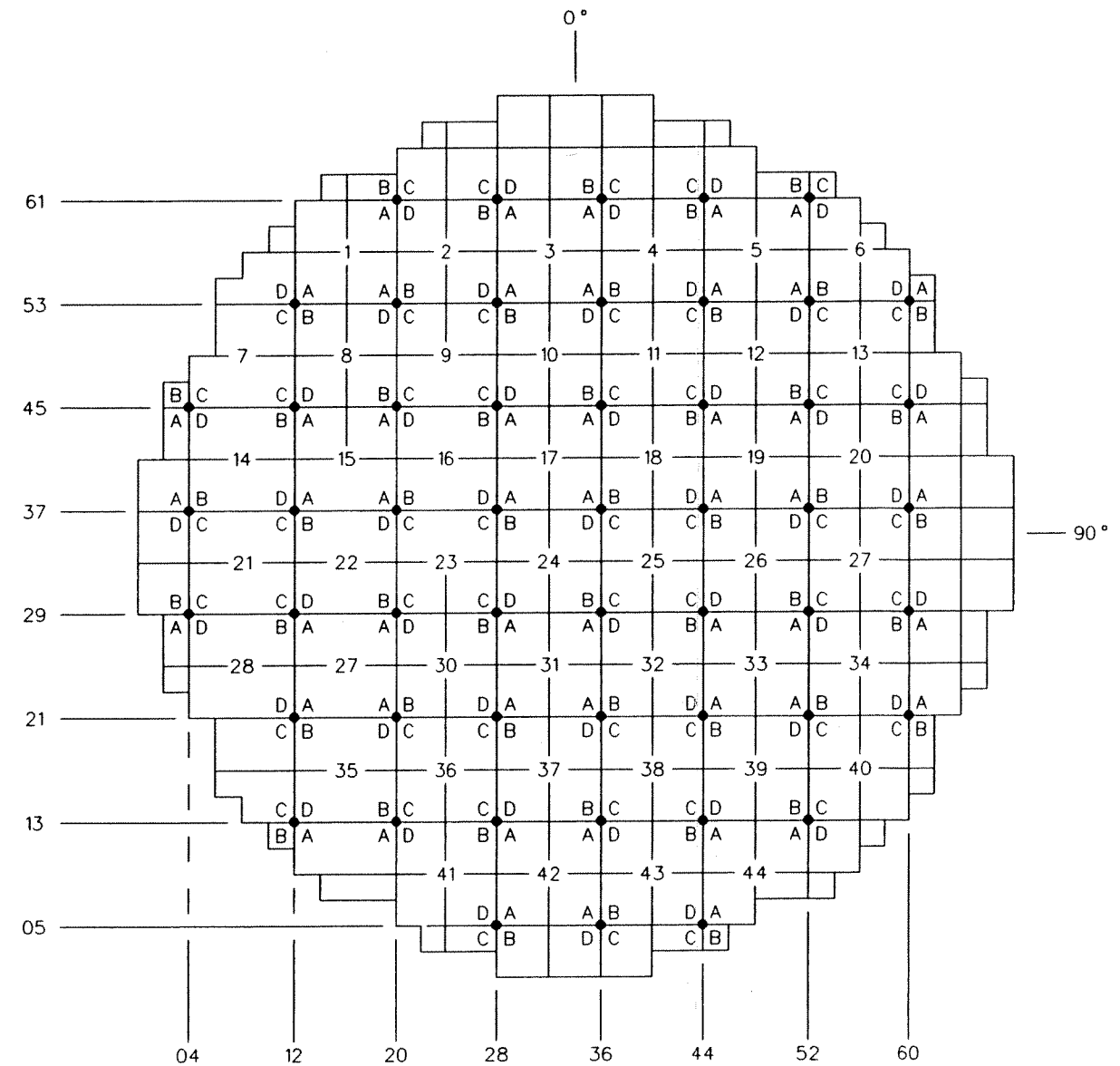
FOR EACH CELL, THE LPRM SIGNALS ARE FIRST FILTERED TO ELIMINATE HIGH FREQUENCY NOISES AND THEN NORMALIZED WITH RESPECT TO A TIME-AVERAGED SIGNAL OF THIS LPRM. THE FOUR (OR THREE) PROCESSED LPRM SIGNALS ARE THEN SUMMED AND AVERAGE TO PROVIDE A RELATIVE OPRM SIGNAL FOR THIS CELL.

OPRM CELL TRIP ALGORITHM

THERE ARE TWO INDEPENDENT OPRM TRIP ALGORITHMS. ONE IS THE AMPLITUDE AND GROWTH RATE BASED ALGORITHM, WHERE THE ALGORITHM IS TO FIRST DETECT A FLUX OSCILLATION WITHIN A DEFINED PERIOD RANGE FOR ONE CYCLE AND THEN DETECT THE OSCILLATING FLUX MAGNITUDE FOR TRIP. THE SECOND ONE IS THE PERIOD BASED ALGORITHM WHERE IT MUST DETECT A DEFINED NUMBER OF OSCILLATIONS BEFORE IT WILL ISSUE TRIP BASED ON A DEFINED OSCILLATING MAGNITUDE. FOR DETAILED DESCRIPTION OF THE ALGORITHM, REFER TO NMS HARDWARE AND SOFTWARE SYSTEM SPEC, C51-4010.

OPRM TRIP BYPASS FUNCTION

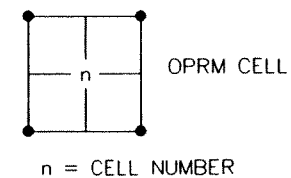
FOR EACH CELL, IF THE NUMBER OF BYPASS LPRMS IS GREATER THAN TWO OR IF THE NUMBER OF LPRMS WITH A READING OF 5% OR LESS IS GREATER THAN TWO THEN THIS OPRM CELL IS AUTOMATICALLY BYPASSED FOR TRIP OUTPUT. THE OPRM CHANNEL TRIP IS BYPASSED IF THE APRM IS LESS THAN 30% OR IF THE CORE FLOW IS GREATER THAN 60%. ALSO, THE APRM CHANNEL BYPASS WILL AUTOMATICALLY BYPASS THE TRIP OUTPUT OF THE OPRM OF THIS APRM CHANNEL.



A	B
D	C

 LPRMS PROVIDING INPUT TO OPRM CHANNELS A,B,C,D

UPPER LEFT LETTER = INPUT FOR OPRM CHANNEL A
 UPPER RIGHT LETTER = INPUT FOR OPRM CHANNEL B
 LOWER RIGHT LETTER = INPUT FOR OPRM CHANNEL C
 LOWER LEFT LETTER = INPUT FOR OPRM CHANNEL D
 (LETTERS IN THE MAP REFER TO LPRM DETECTORS A,B,C,D PER ASSEMBLY)



OPRM CONFIGURATION DESCRIPTION

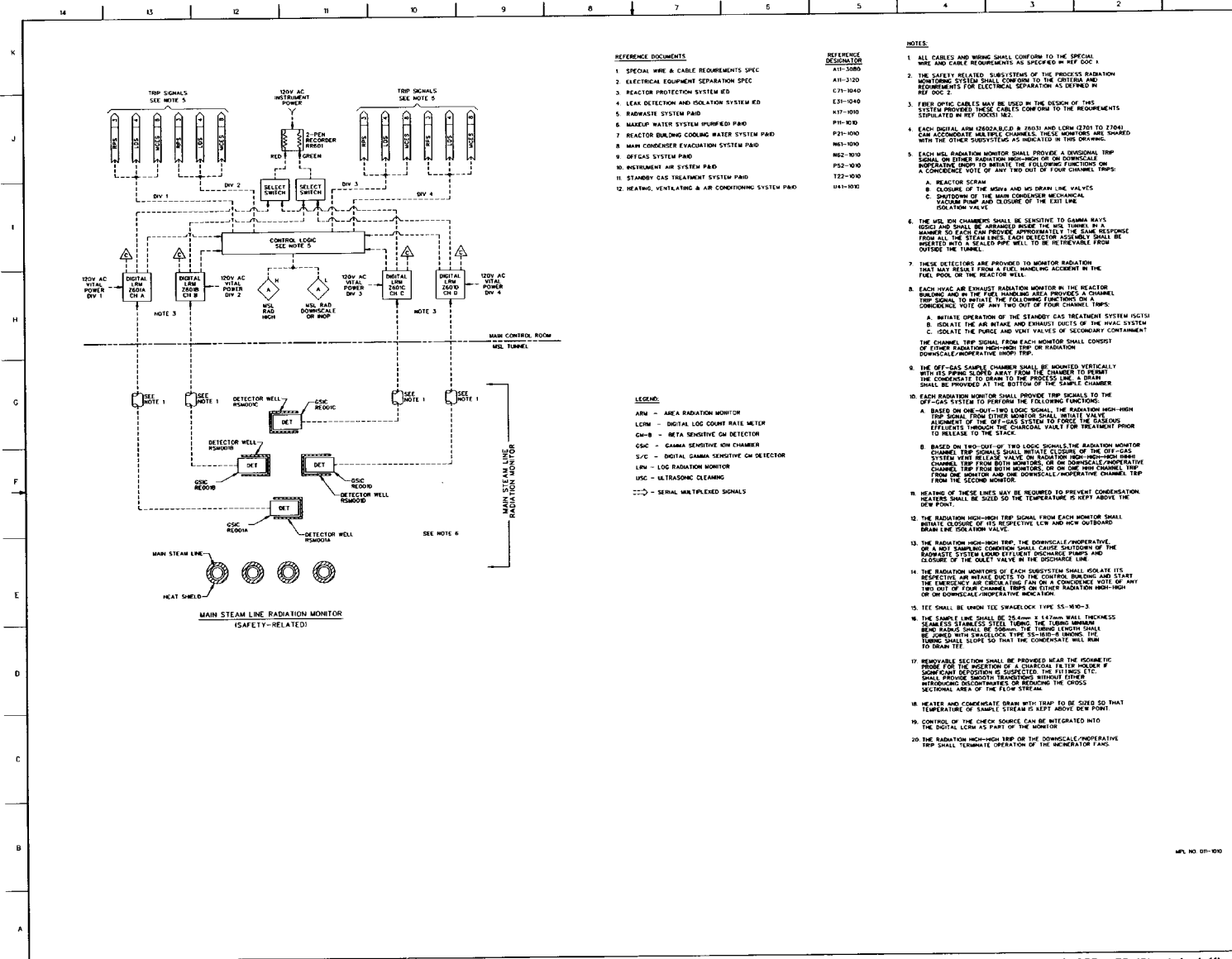


FIGURE 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 1 of 11)
ABWR DCD/Tier 2 Rev. 0 21-311

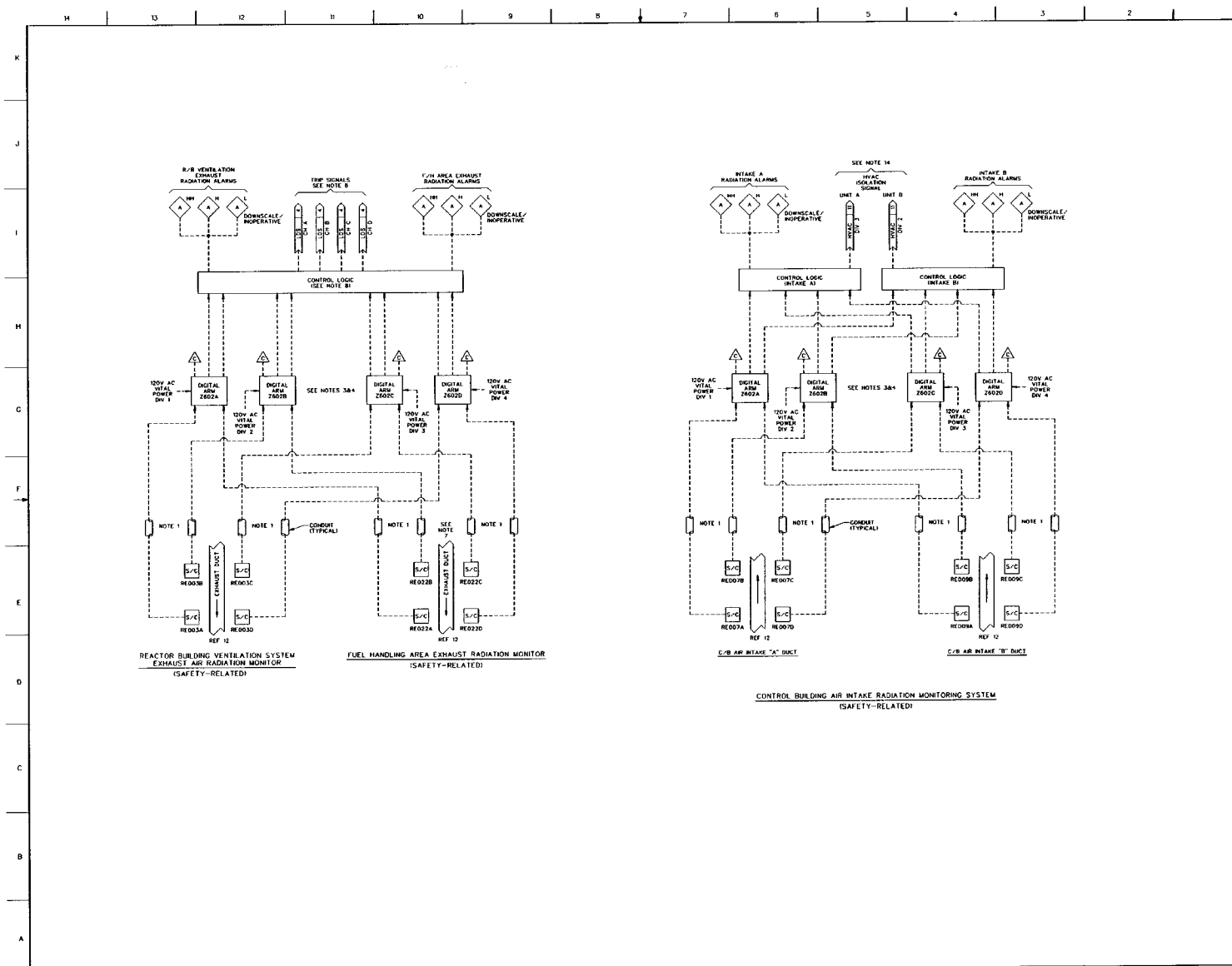


FIGURE 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 2 of 11)
 ABWR DCD/Tier 2 Rev. 0 21-312

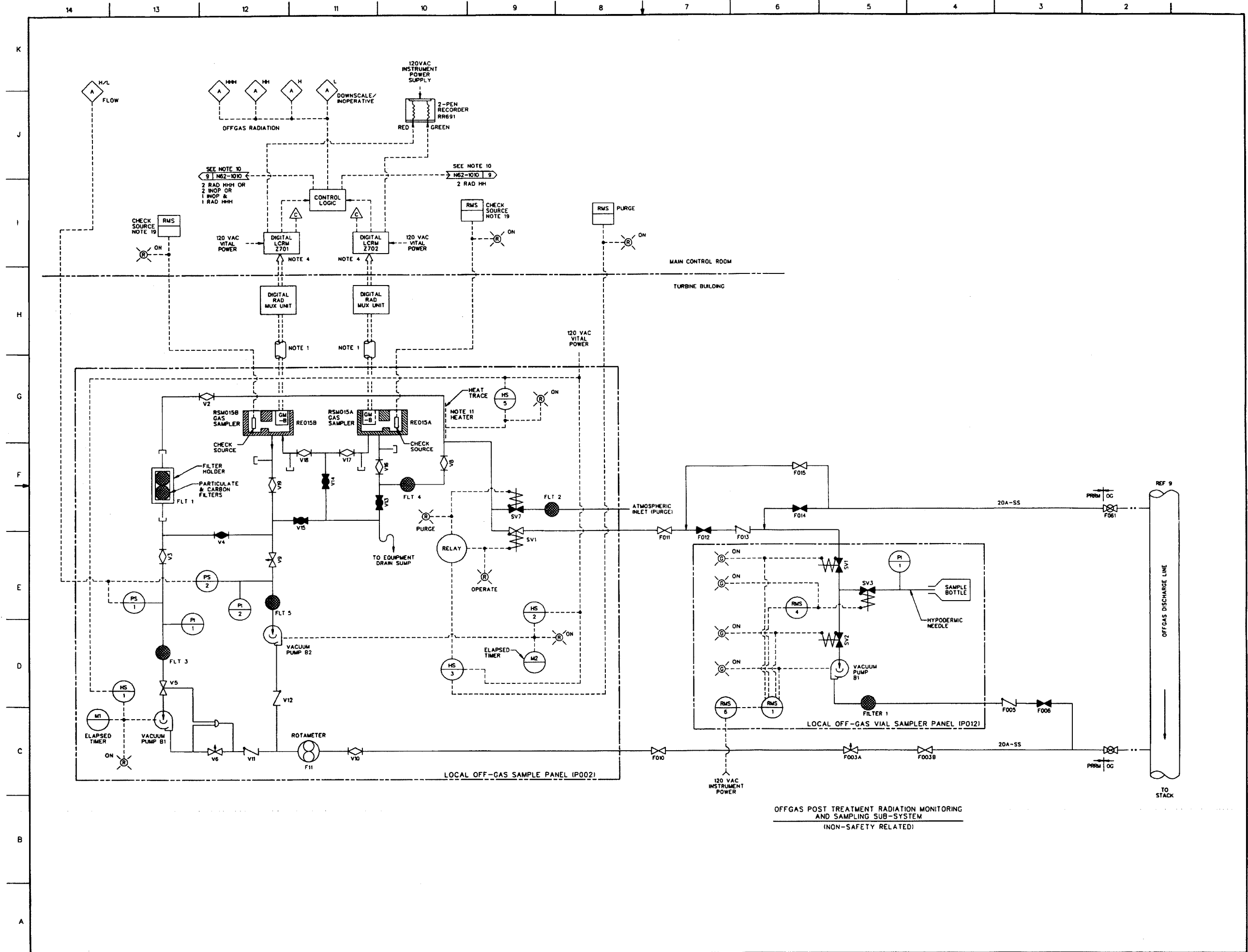


Figure 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 4 of 11)

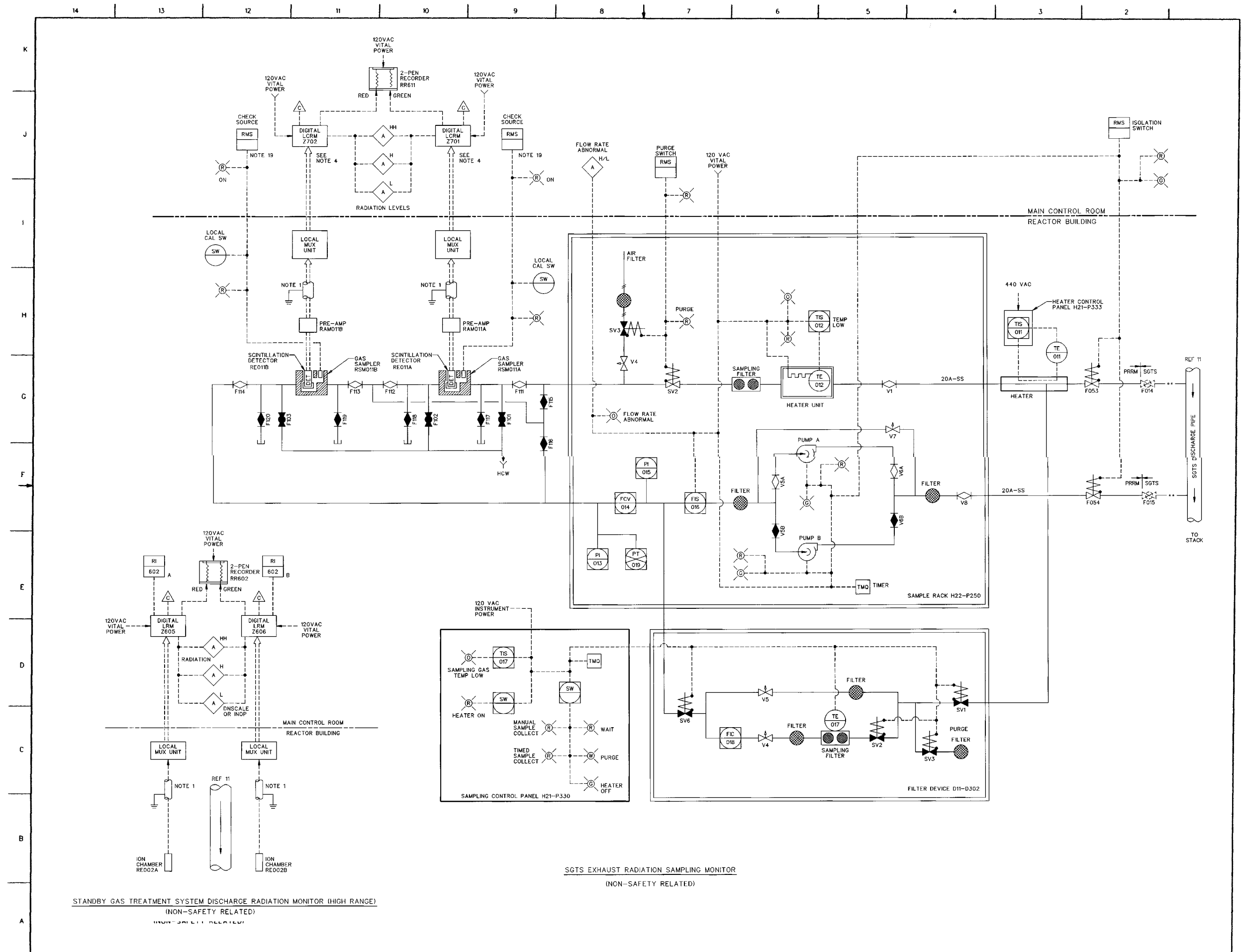


FIGURE 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 5 of 11)
 ABWR DCD/Tier 2 Rev. 0 21-315

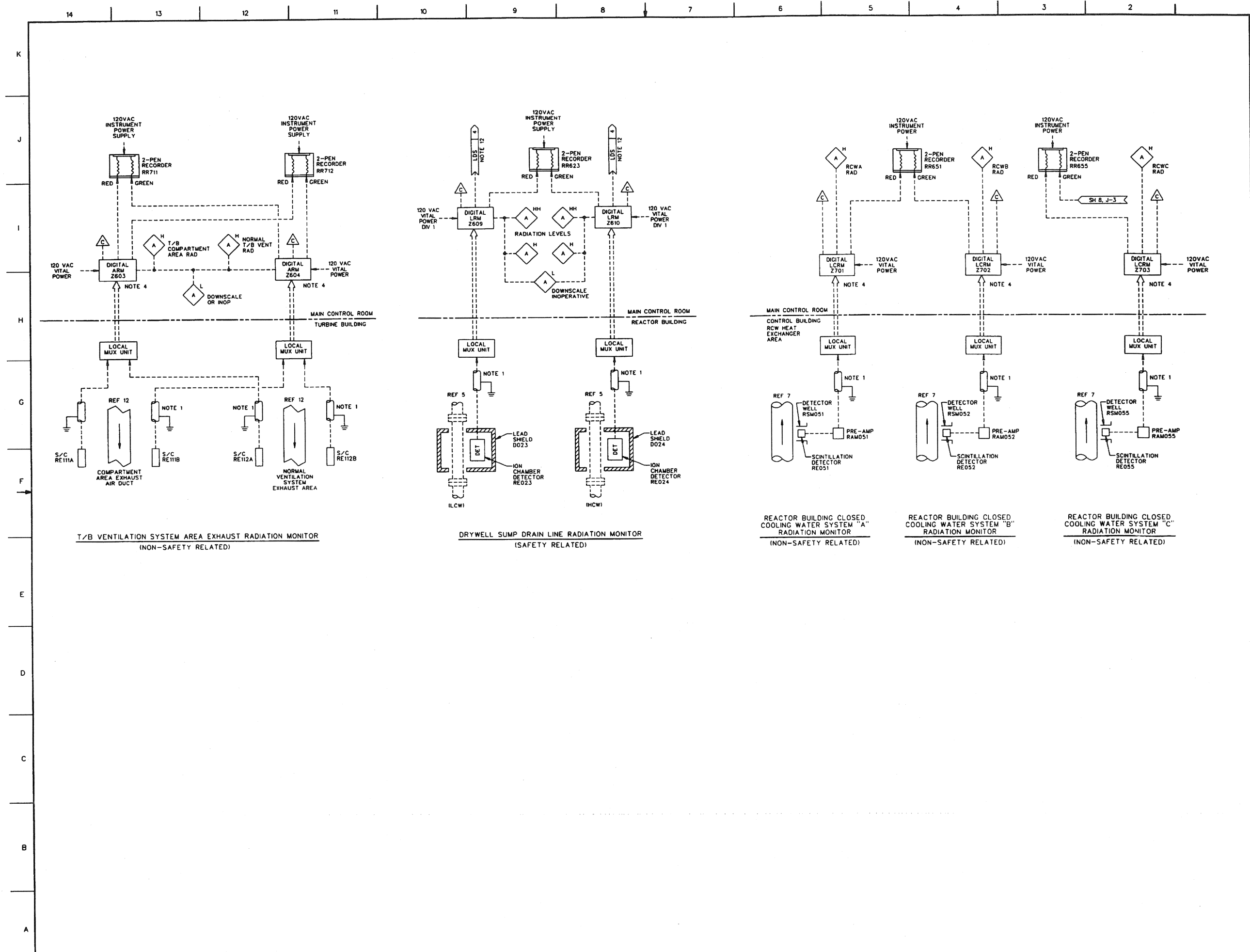


Figure 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 6 of 11)
 ABWR DCD/Tier 2 Rev. 0 21-316

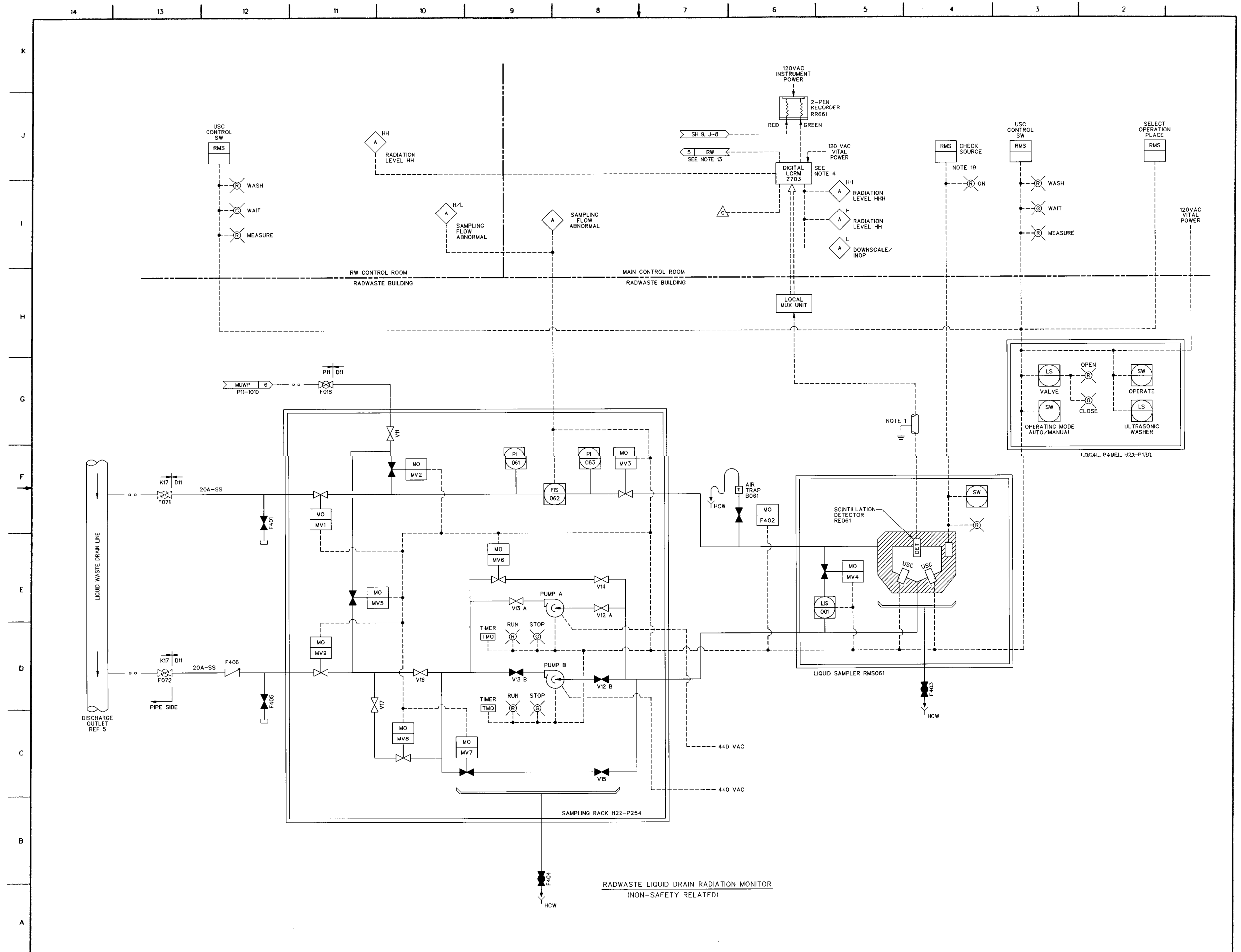


FIGURE 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 7 of 11)
 ABWR DCD/Tier 2 Rev. 0 21-317

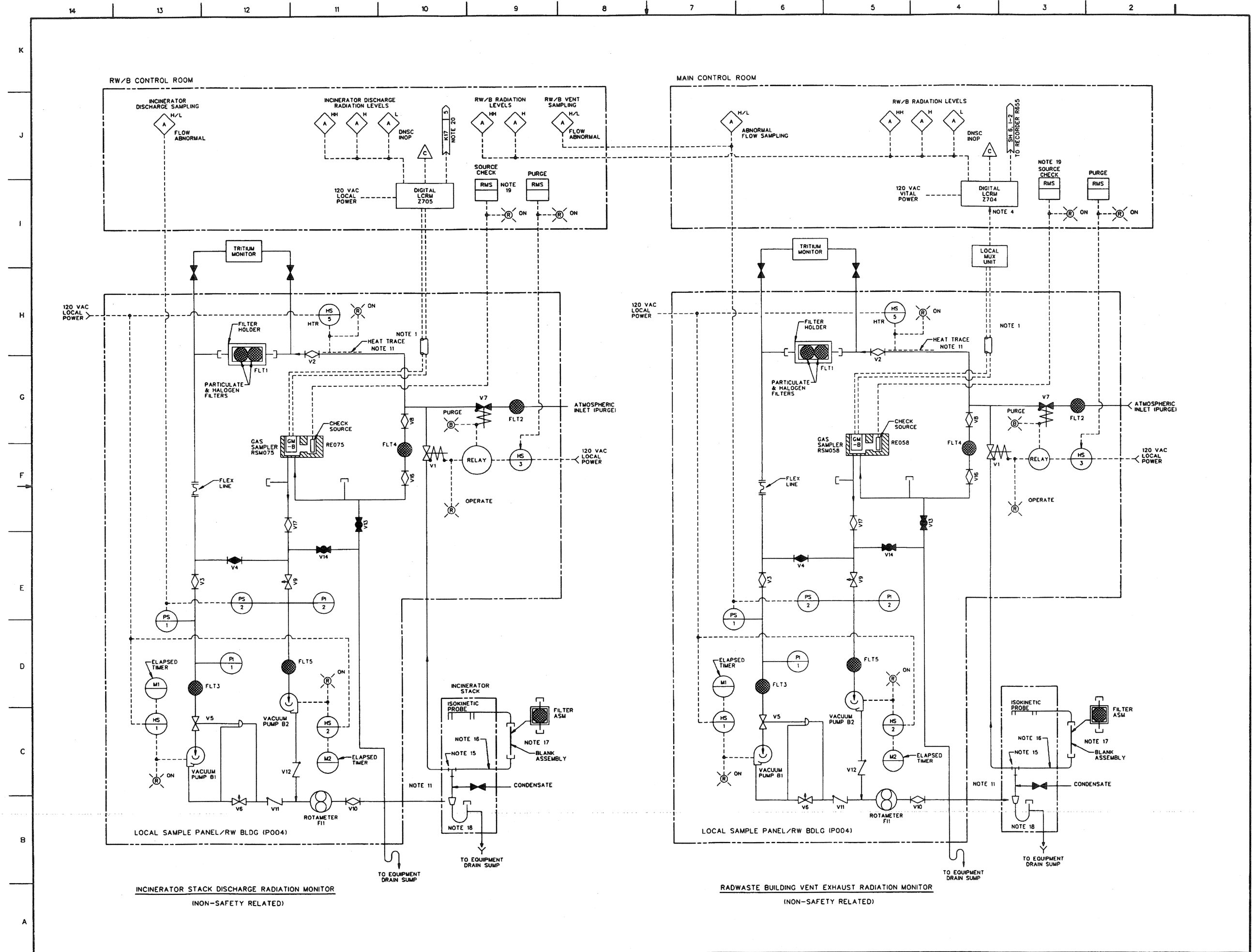


Figure 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 8 of 11)
ABWR DCD/Tier 2 Rev. 0 21-318

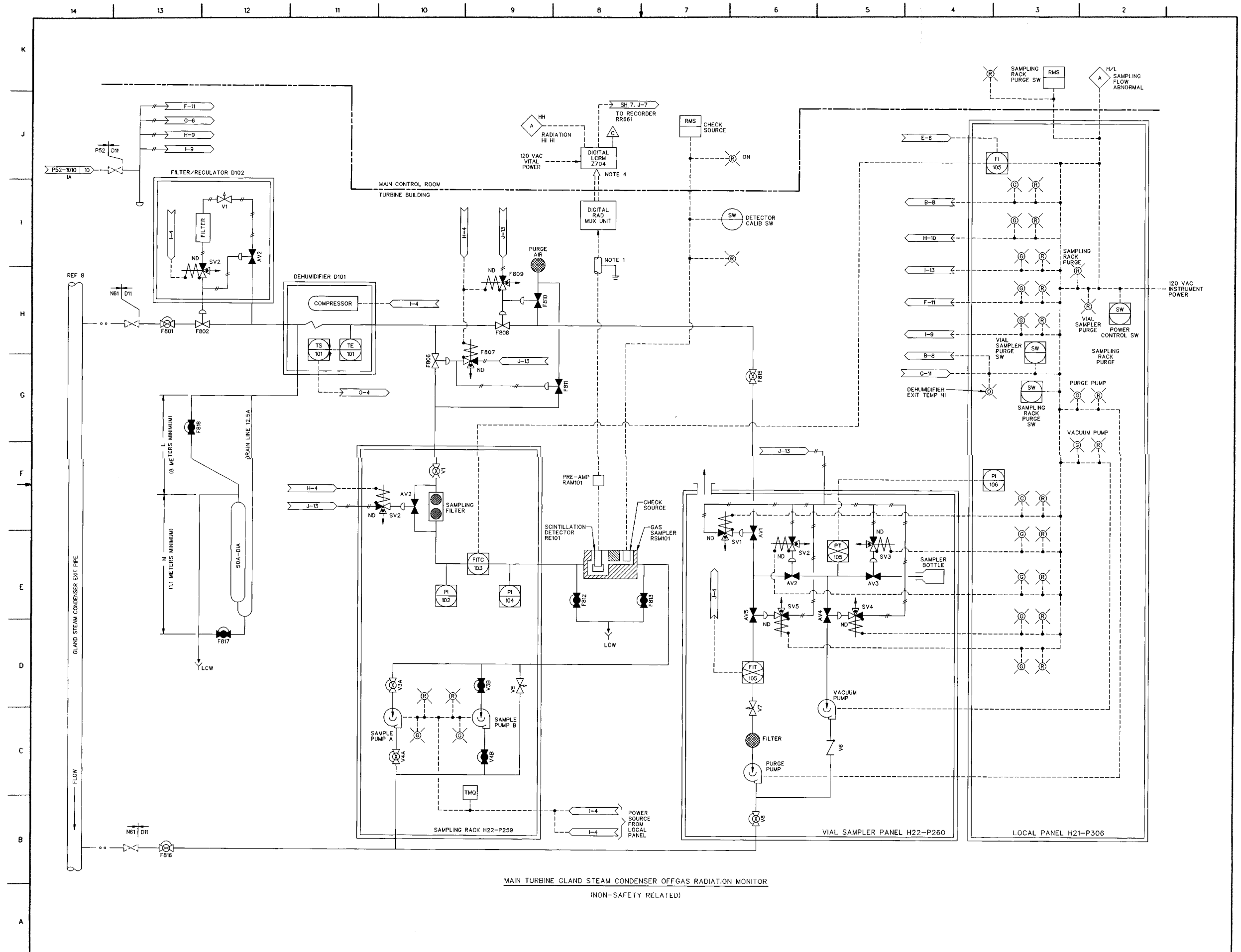


FIGURE 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 9 of 11)
ABWR DCD/Tier 2 Rev. 0 21-319

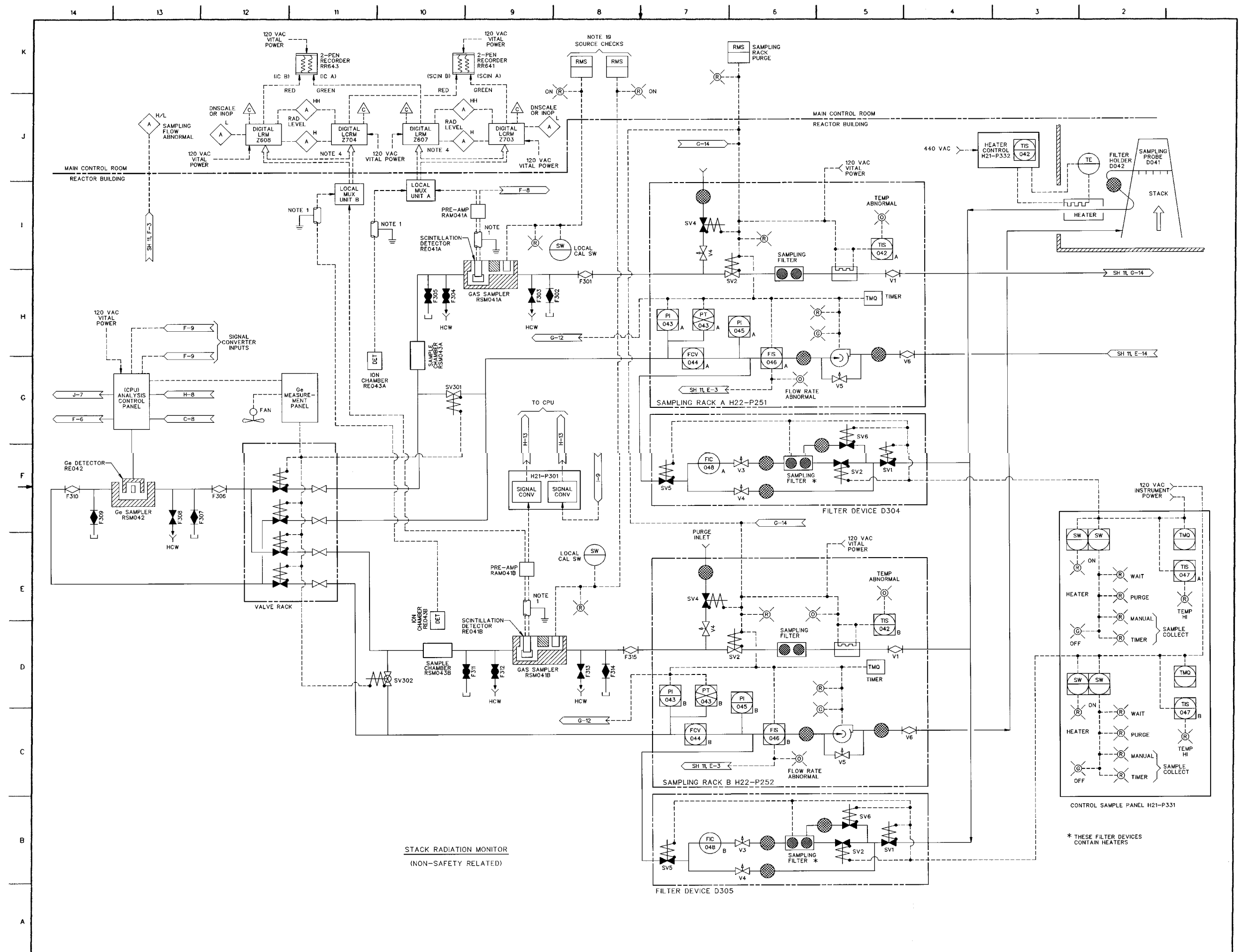


FIGURE 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 10 of 11)
ABWR DCD/Tier 2 Rev. 0 21-320

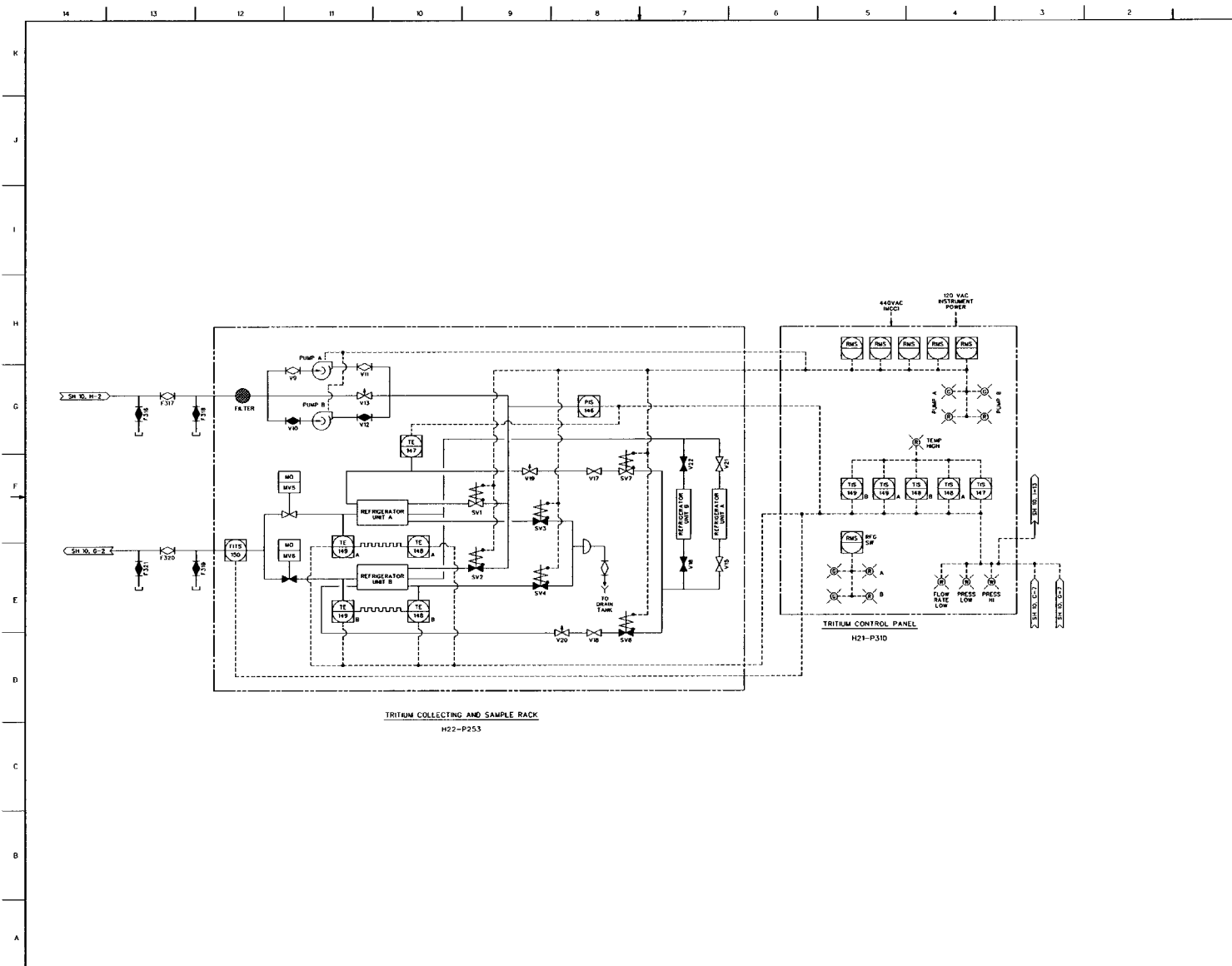
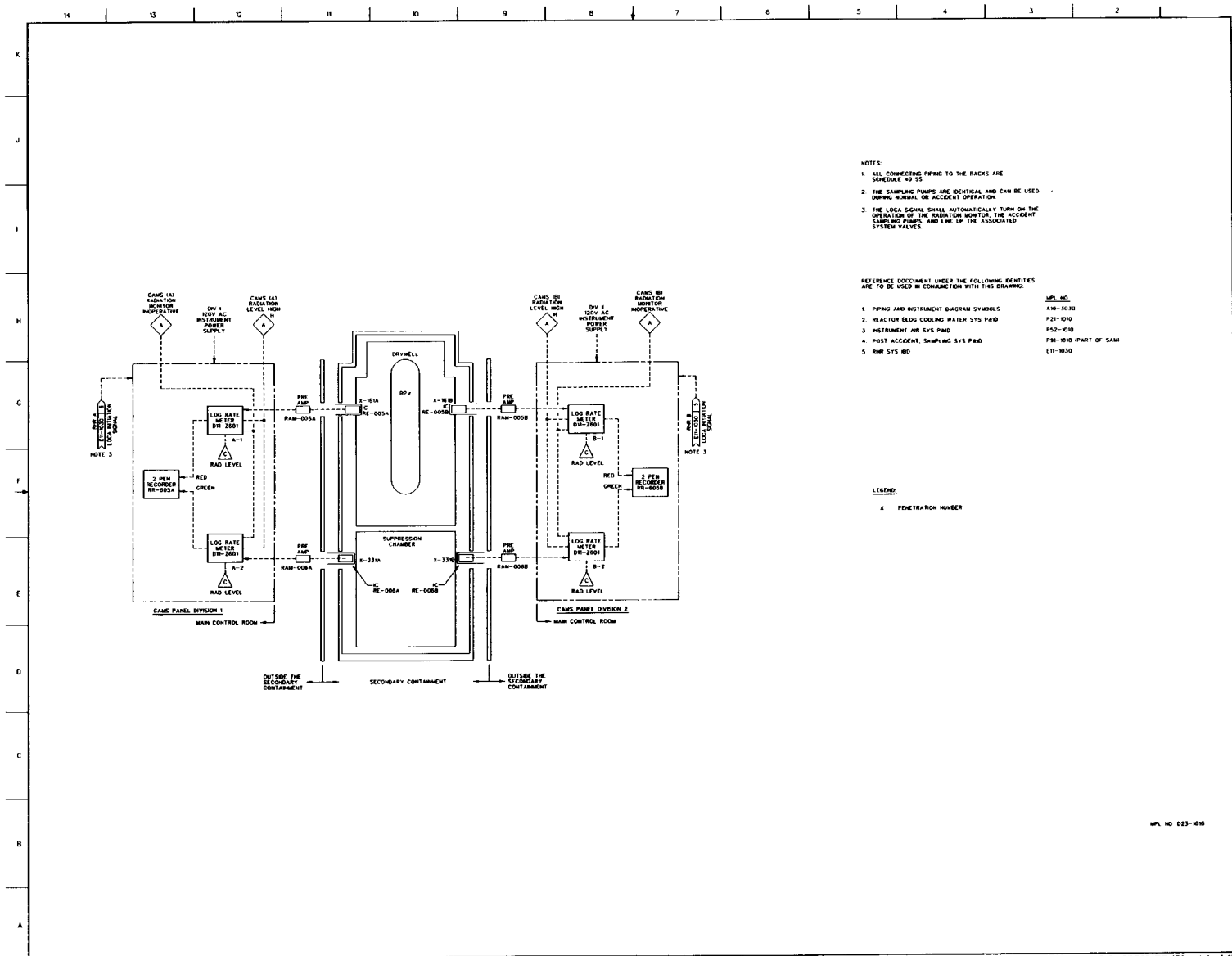


FIGURE 7.6-5 PROCESS RADIATION MONITORING SYSTEM IED (Sheet 11 of 11)
ABWR DCD/Tier 2 Rev. 0 21-321



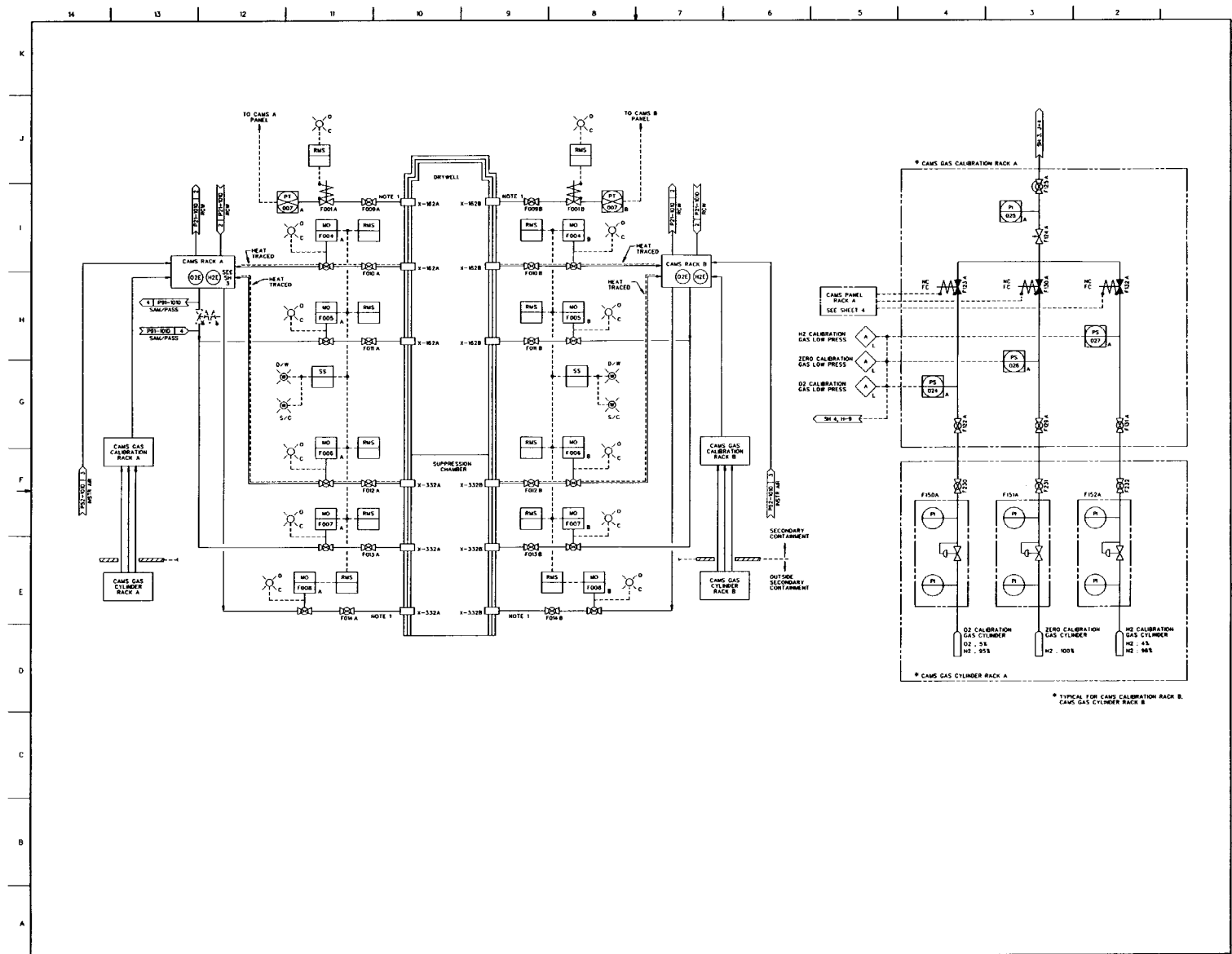


Figure 7.6-7 CONTAINMENT ATMOSPHERIC MONITORING SYSTEM IED (Sheet 2 of 4)
ABWR DCD/Tier 2 Rev. 0 21-323



	7	6	5	4	3	2
E						
D						
C						
B						
A						

TABLE OF CONTENTS

SH NO.	TITLE
1	TITLE SHEET, NOTES, CONTENTS
2	CAMS A, (B) ISOLATION VALVES CONTROL LOGIC
3	CAMS A (B) INITIATION & CONTROL LOGIC
4	CAMS A (B) SAMPLE PUMPS CONTROL LOGIC
5	CAMS A (B) GAS CALIBRATION & CONTROL
6	CAMS A (B) RACK DRAIN CONTROL
7	CAMS A (B) POWER-ON CONTROL
8	CAMS A (B) ANNUNCIATORS
9	CAMS A (B) ANNUNCIATORS (CONT'D)
10	CAMS A (B) ANNUNCIATORS (CONT'D)

NOTES:

1. THE SAMPLING PUMPS ARE IDENTICAL AND CAN BE USED DURING NORMAL OR ACCIDENT OPERATION.

REFERENCE DOCUMENTS

	MPL NO.
1. CONTAINMENT ATMOSPHERIC MONITORING SYS IED	D23-1010
2. RESIDUAL HEAT REMOVAL SYS IBD	E11-1030

MPL NO. D23-1030

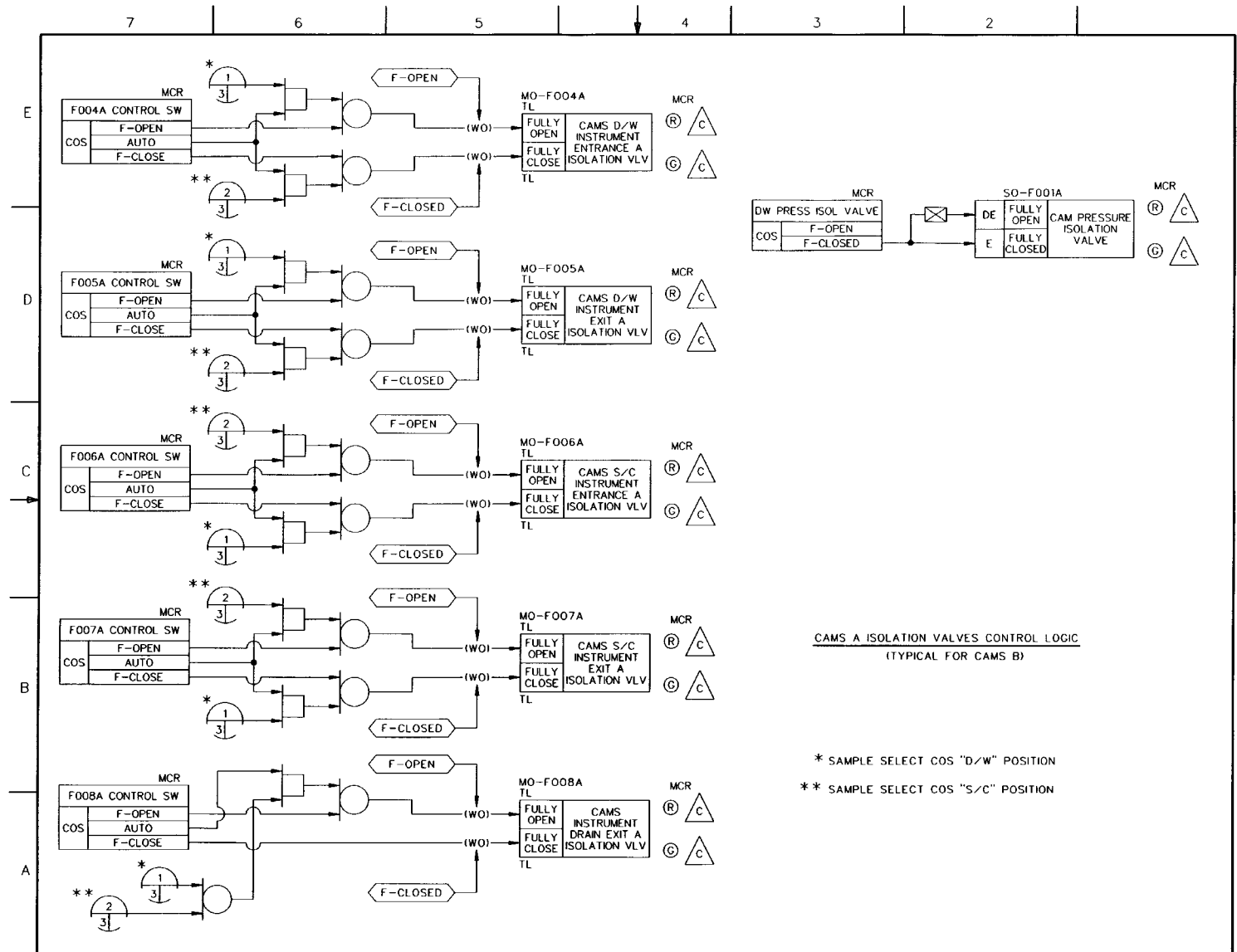


Figure 7.6-8 CONTAINMENT ATMOSPHERIC MONITORING SYSTEM IBD (Sheet 2 of 10)
ABWR DCD/Tier 2 Rev. 0 21-327

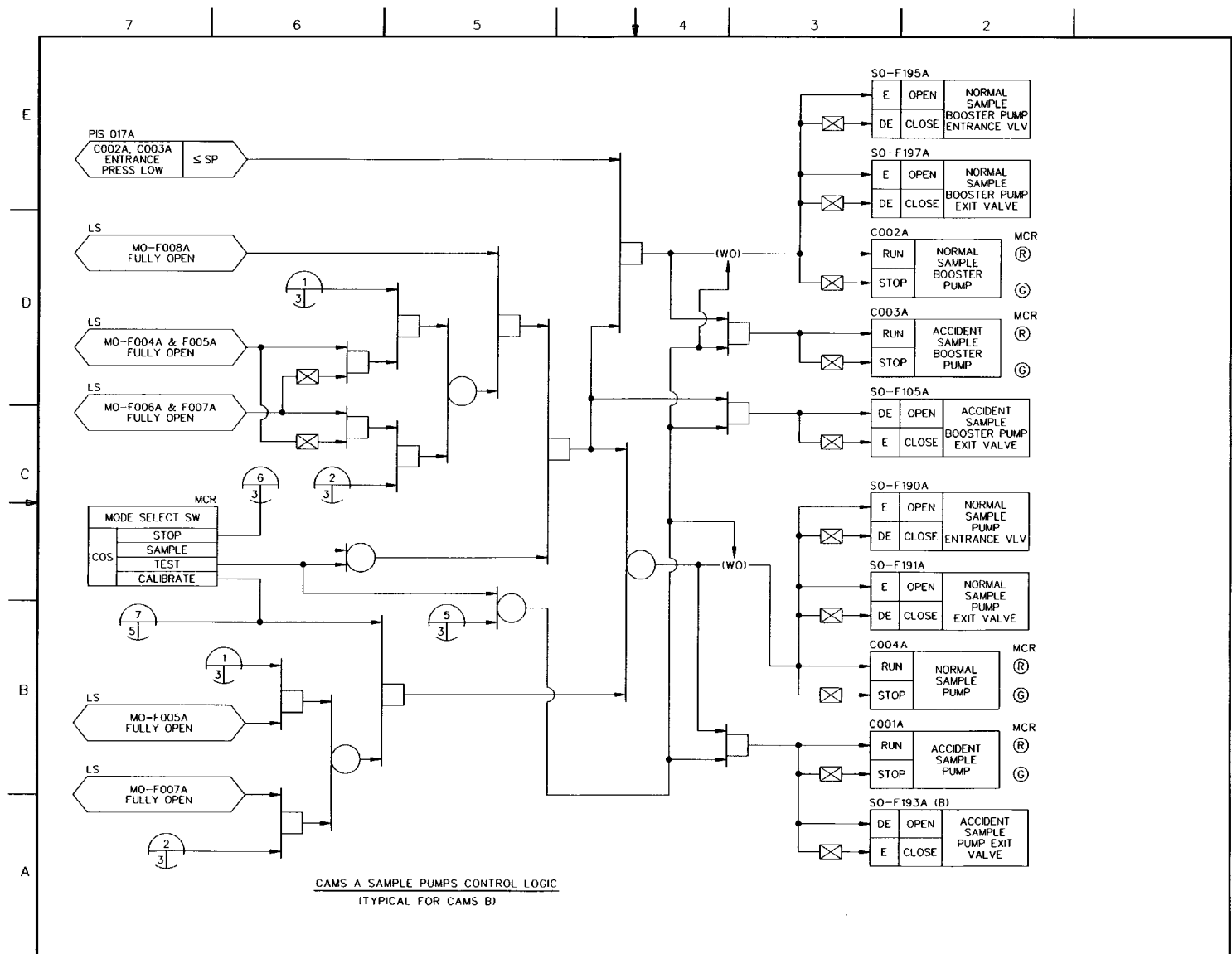


Figure 7.6-8 CONTAINMENT ATMOSPHERIC MONITORING SYSTEM IBD (Sheet 4 of 10)
 ABWR DCD/Tier 2 Rev. 0 21.329

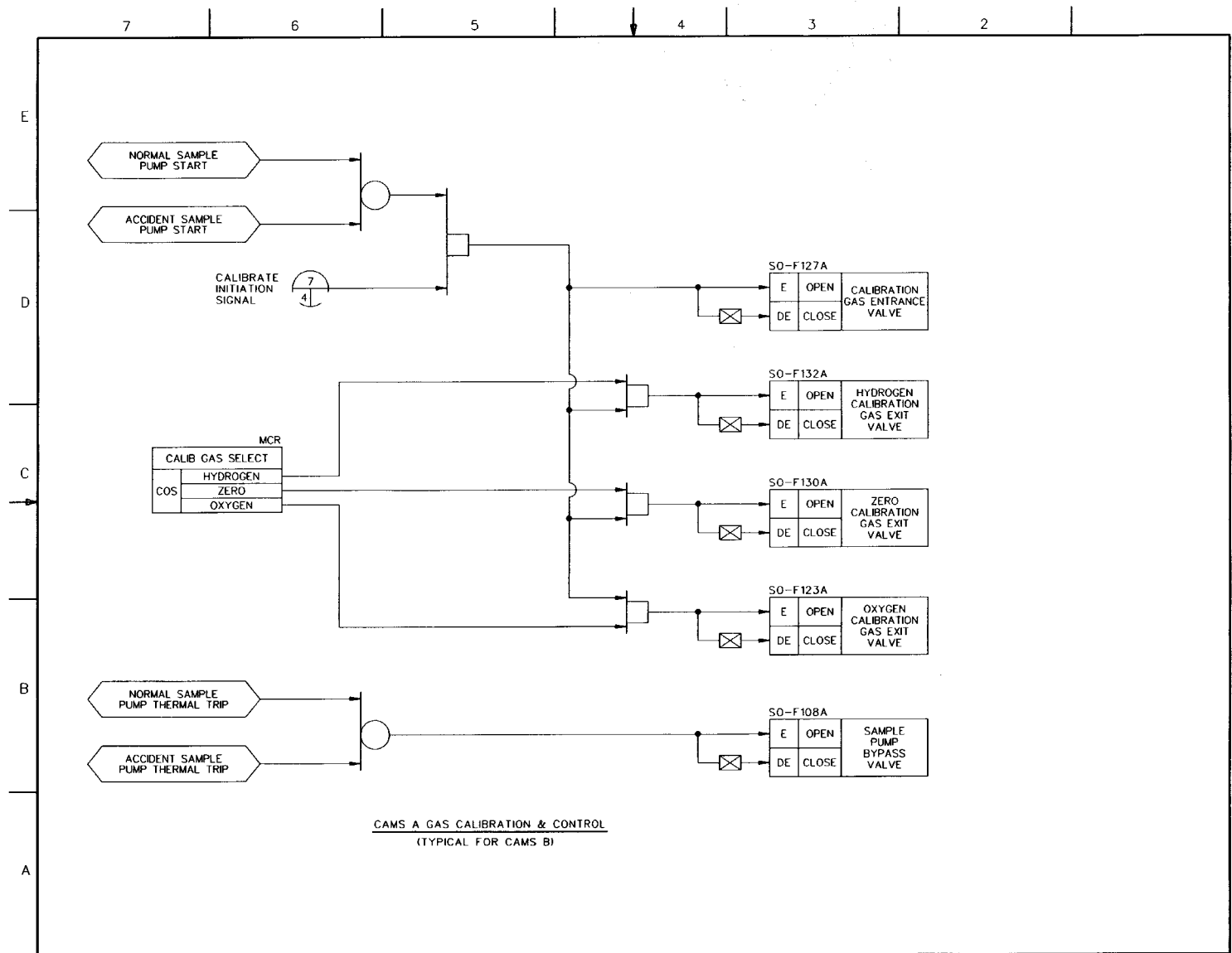


Figure 7.6-8 CONTAINMENT ATMOSPHERIC MONITORING SYSTEM IBD (Sheet 5 of 10)
ABWR DCD/Tier 2 Rev. 0 27-330

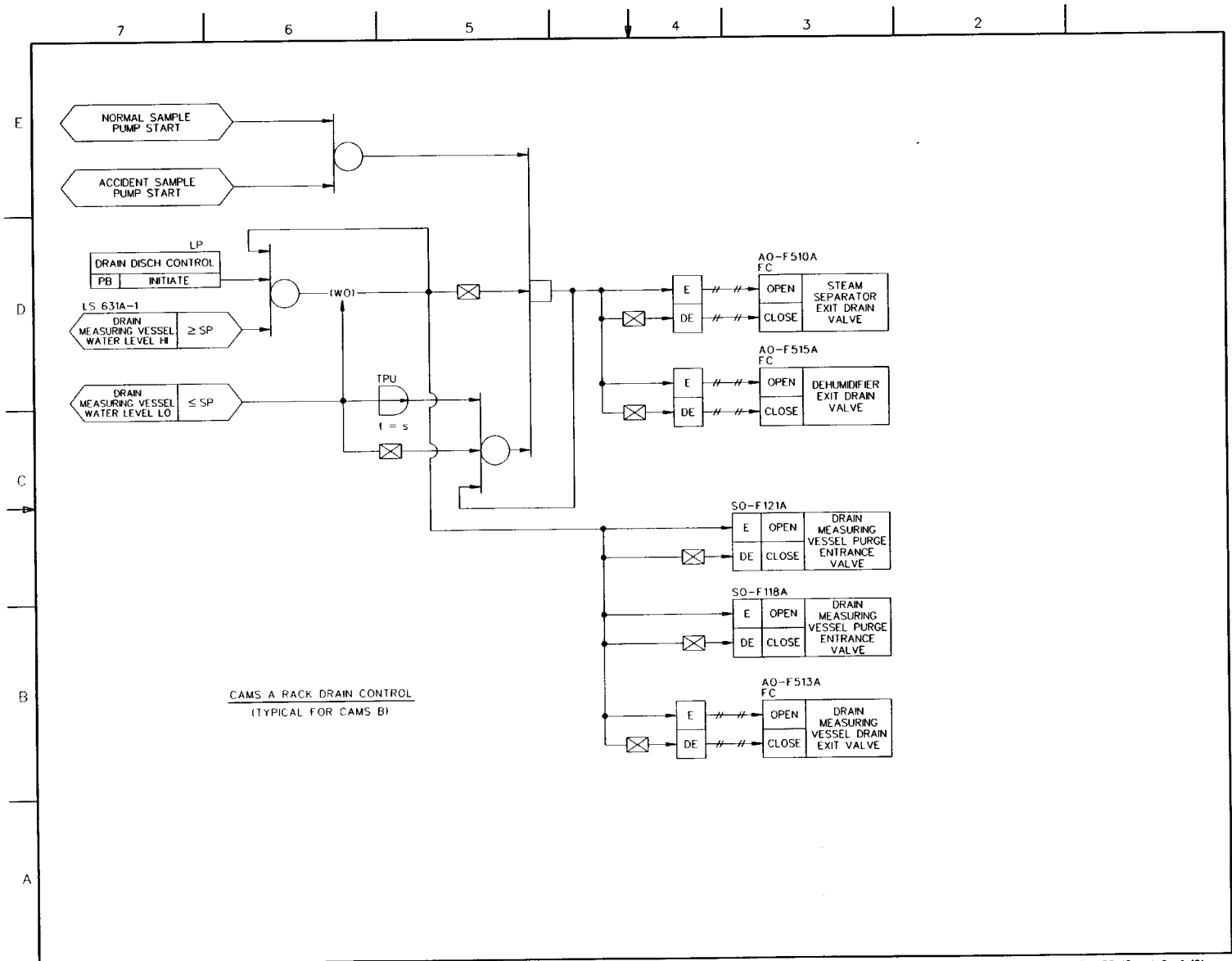


FIGURE 7.6-8 CONTAINMENT ATMOSPHERIC MONITORING SYSTEM IBD (Sheet 6 of 10)
ABWR DCD/Tier 2 Rev. 0 21-331

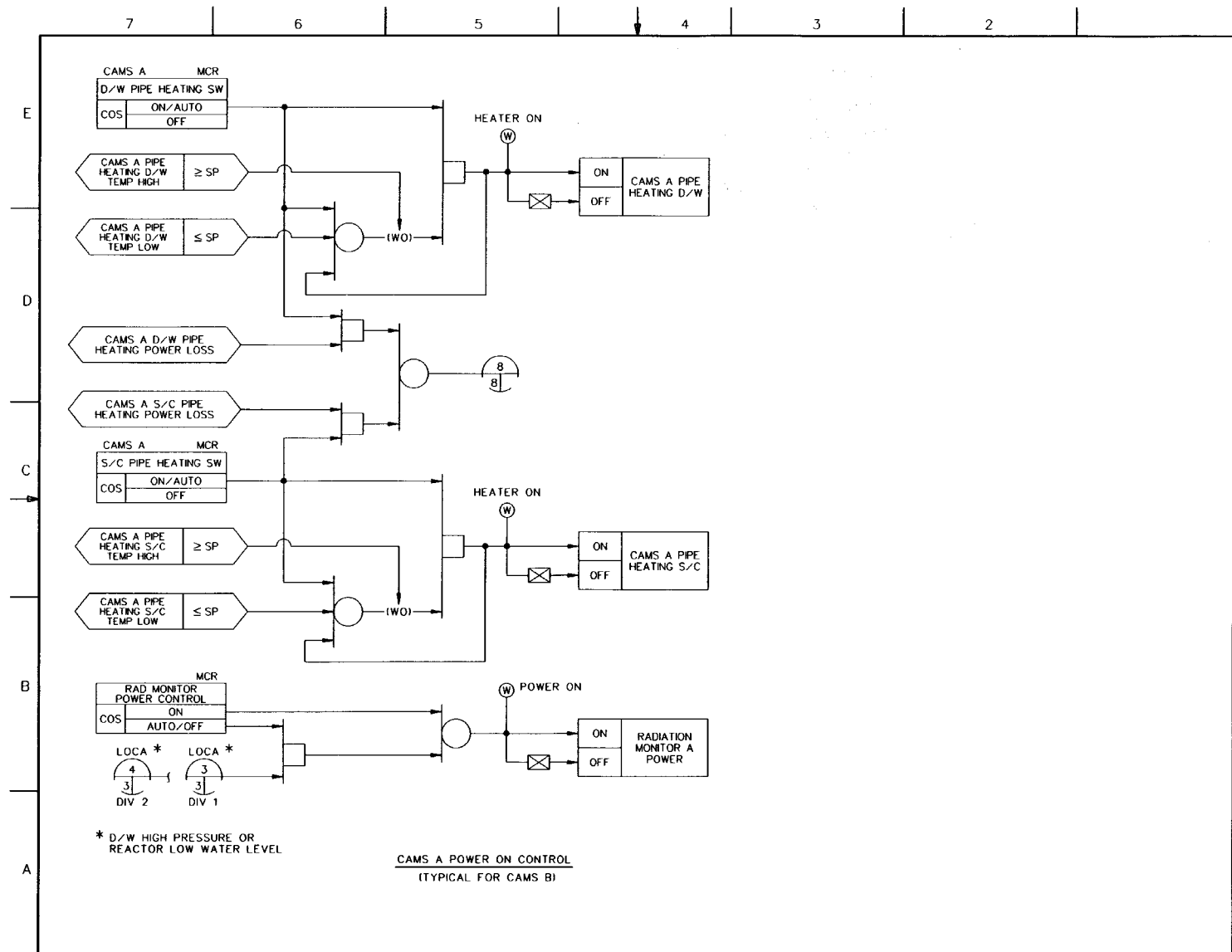


Figure 7.6-8 CONTAINMENT ATMOSPHERIC MONITORING SYSTEM IBD (Sheet 7 of 10)
 ABWR DCD/Tier 2 Rev. 0 21-332

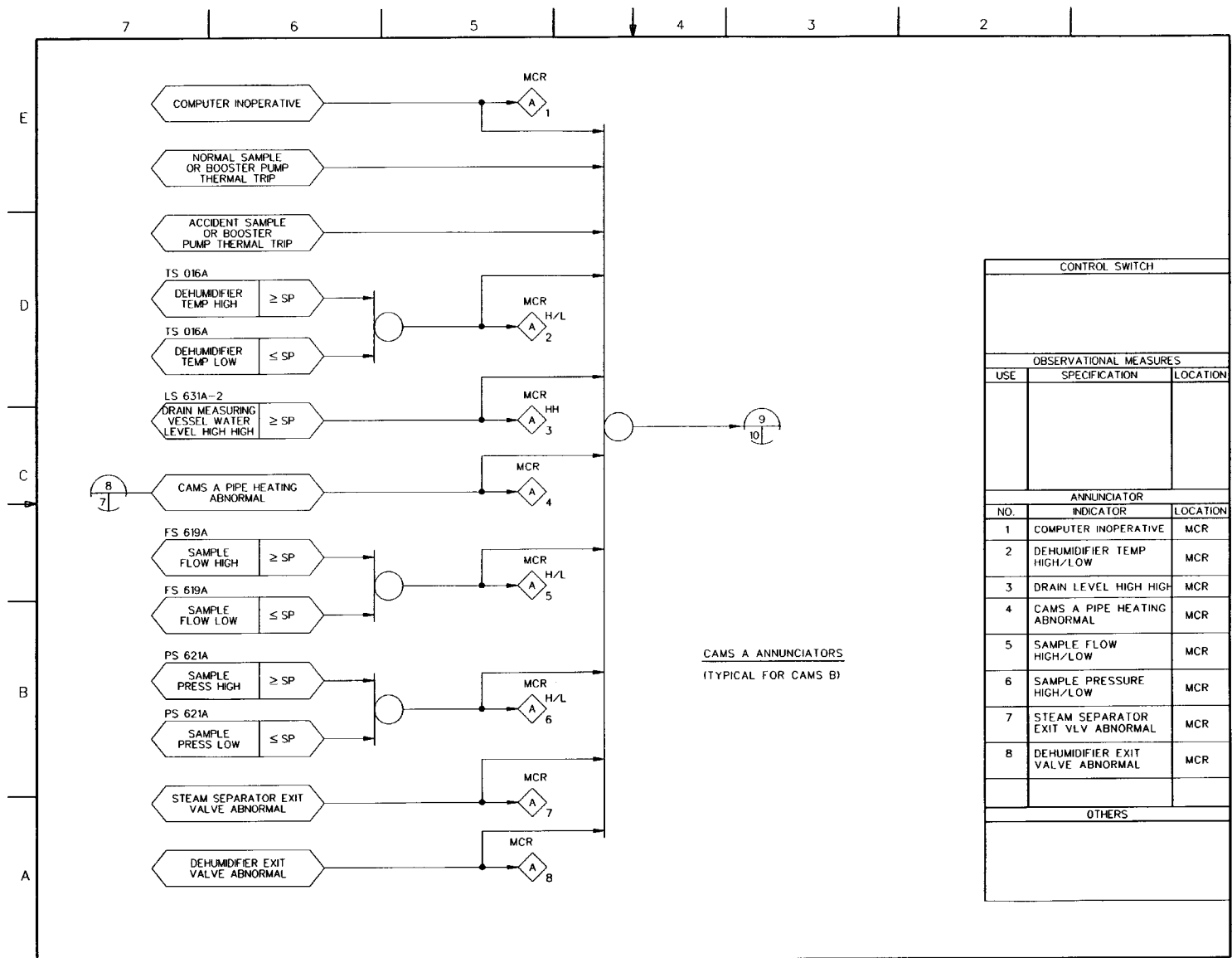
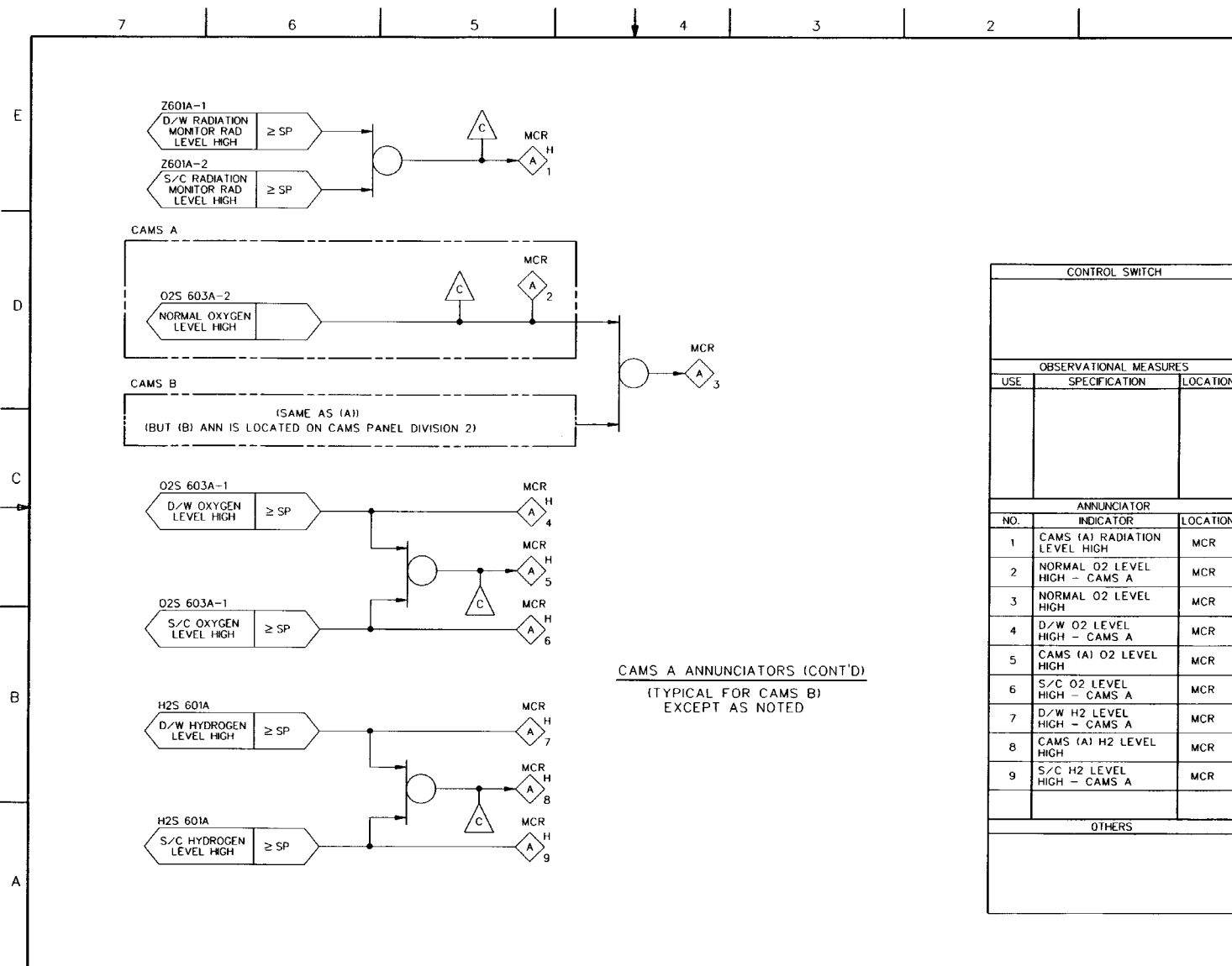
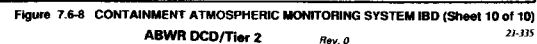
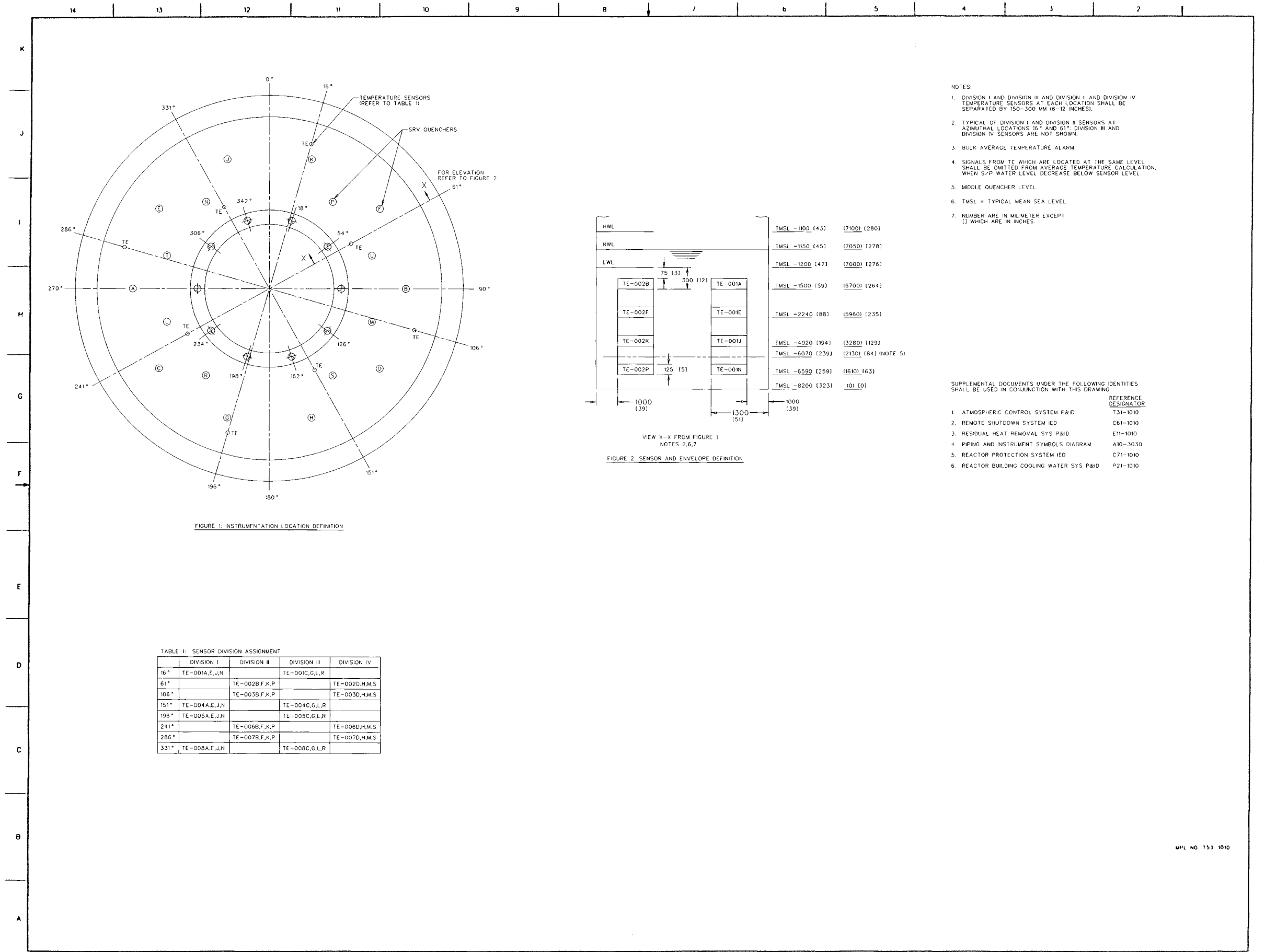


Figure 7.6-8 CONTAINMENT ATMOSPHERIC MONITORING SYSTEM IBD (Sheet 8 of 10)
 ABWR DCD/Tier 2 Rev. 0 21-333







- NOTES:
1. DIVISION I AND DIVISION III AND DIVISION II AND DIVISION IV TEMPERATURE SENSORS AT EACH LOCATION SHALL BE SEPARATED BY 150-300 MM (6-12 INCHES).
 2. TYPICAL OF DIVISION I AND DIVISION II SENSORS AT AZIMUTHAL LOCATIONS 16° AND 61°. DIVISION III AND DIVISION IV SENSORS ARE NOT SHOWN.
 3. BULK AVERAGE TEMPERATURE ALARM.
 4. SIGNALS FROM TE WHICH ARE LOCATED AT THE SAME LEVEL SHALL BE OMITTED FROM AVERAGE TEMPERATURE CALCULATION, WHEN S/P WATER LEVEL DECREASE BELOW SENSOR LEVEL.
 5. MIDDLE QUENCHER LEVEL.
 6. TMSL = TYPICAL MEAN SEA LEVEL.
 7. NUMBER ARE IN MILLIMETER EXCEPT [] WHICH ARE IN INCHES.

- SUPPLEMENTAL DOCUMENTS UNDER THE FOLLOWING IDENTITIES SHALL BE USED IN CONJUNCTION WITH THIS DRAWING.
- | | |
|--|-------------------------------|
| 1. ATMOSPHERIC CONTROL SYSTEM P&ID | REFERENCE DESIGNATOR T31-1010 |
| 2. REMOTE SHUTDOWN SYSTEM IED | C61-1010 |
| 3. RESIDUAL HEAT REMOVAL SYS P&ID | E11-1010 |
| 4. PIPING AND INSTRUMENT SYMBOLS DIAGRAM | A10-3030 |
| 5. REACTOR PROTECTION SYSTEM IED | C71-1010 |
| 6. REACTOR BUILDING COOLING WATER SYS P&ID | P21-1010 |

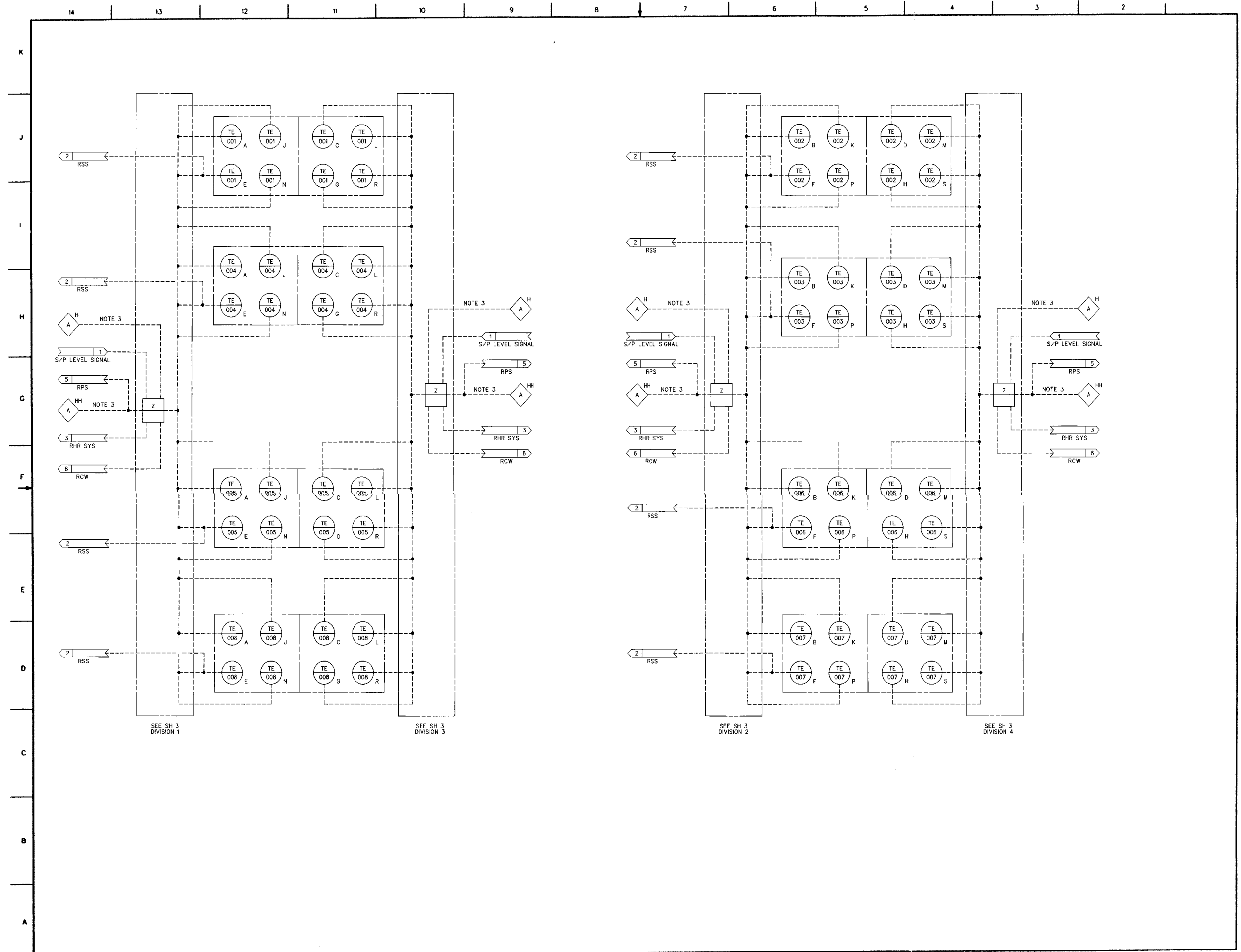


Figure 7.6-11 SUPPRESSION POOL TEMPERATURE MONITORING SYSTEM IED (Sheet 2 of 3)

ABWR DCD/Tier 2

Rev. 0

21-337

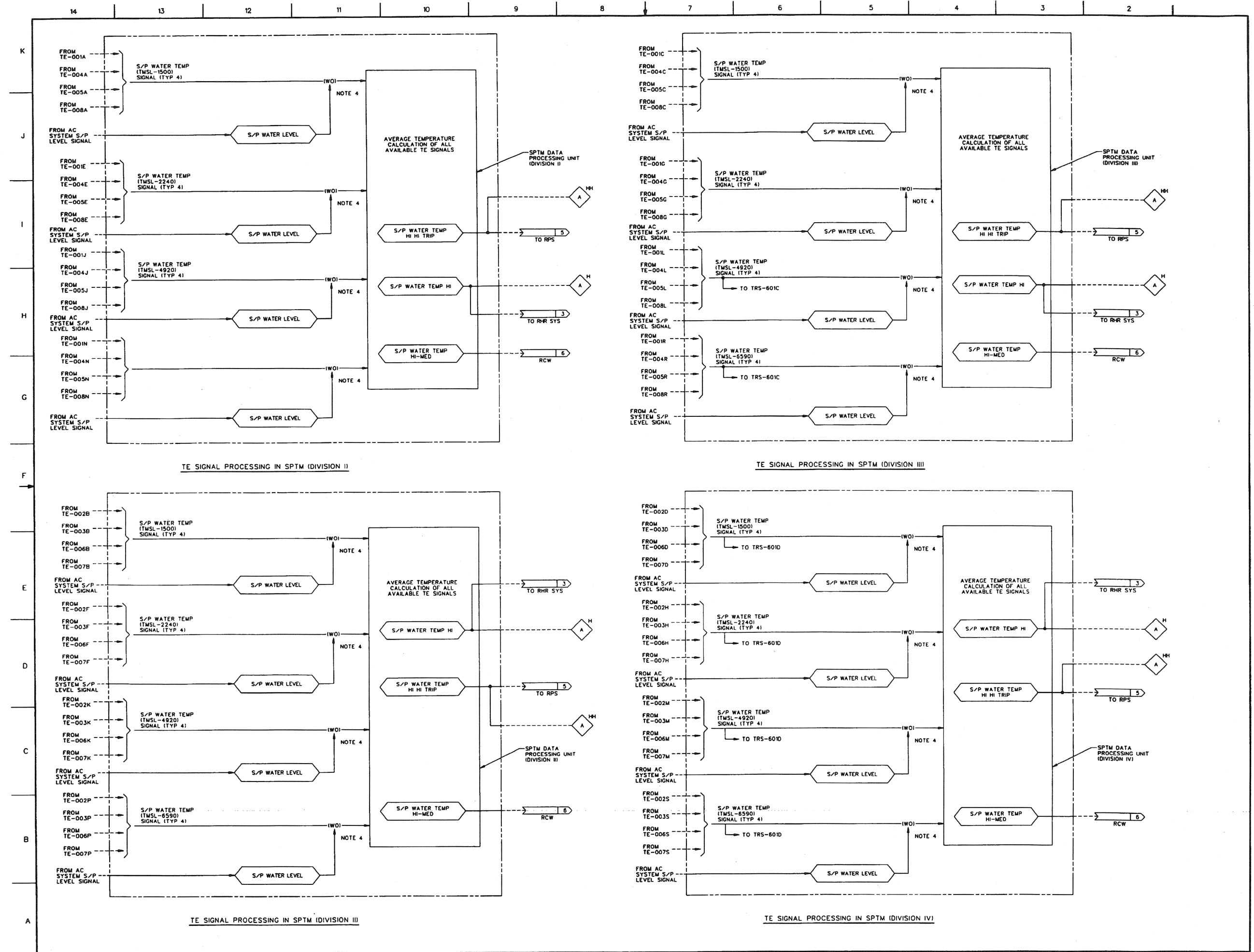
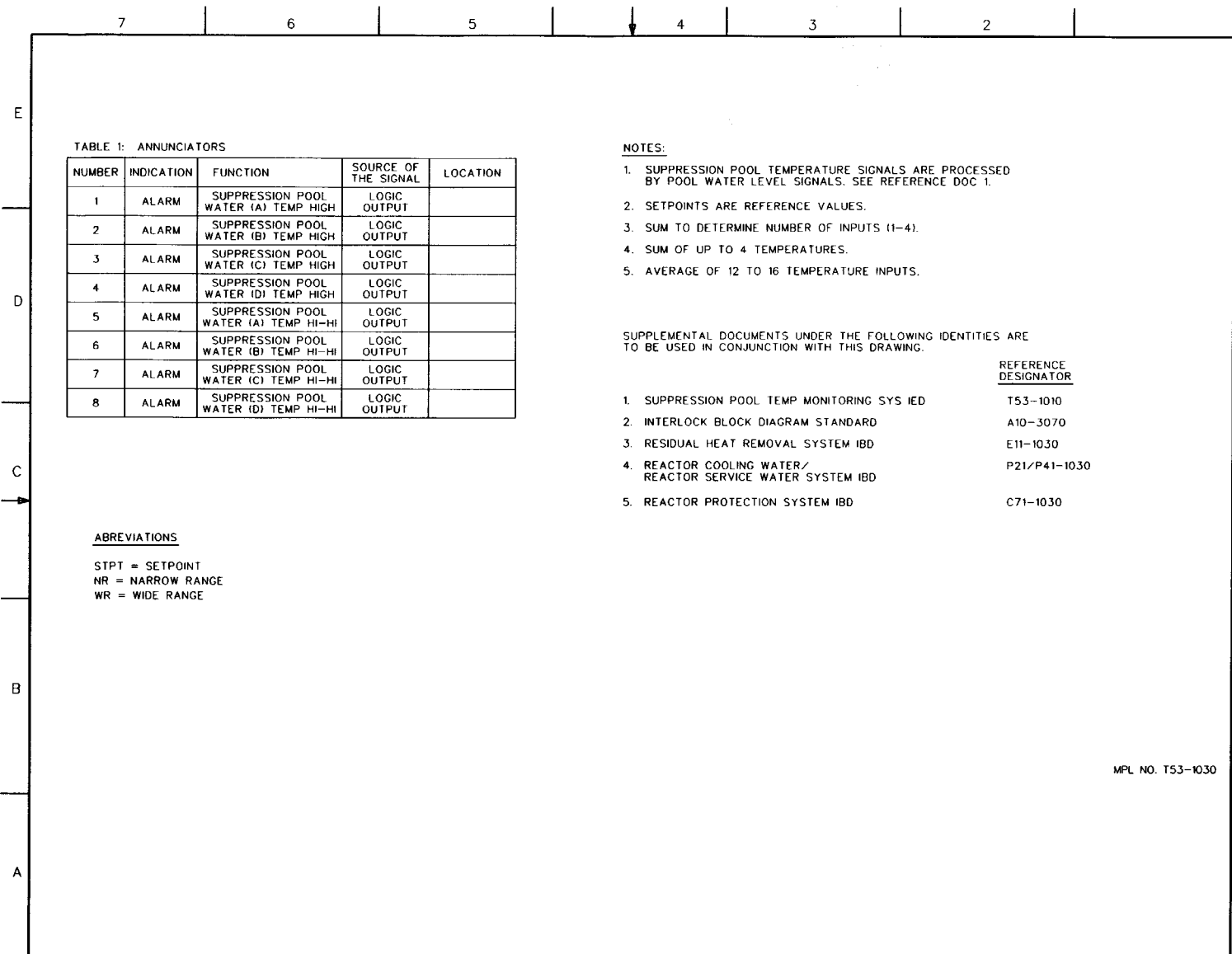
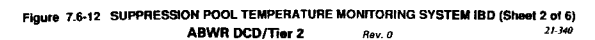


Figure 7.6-11 SUPPRESSION POOL TEMPERATURE MONITORING SYSTEM IED (Sheet 3 of 3)
ABWR DCD/Tier 2 Rev. 0





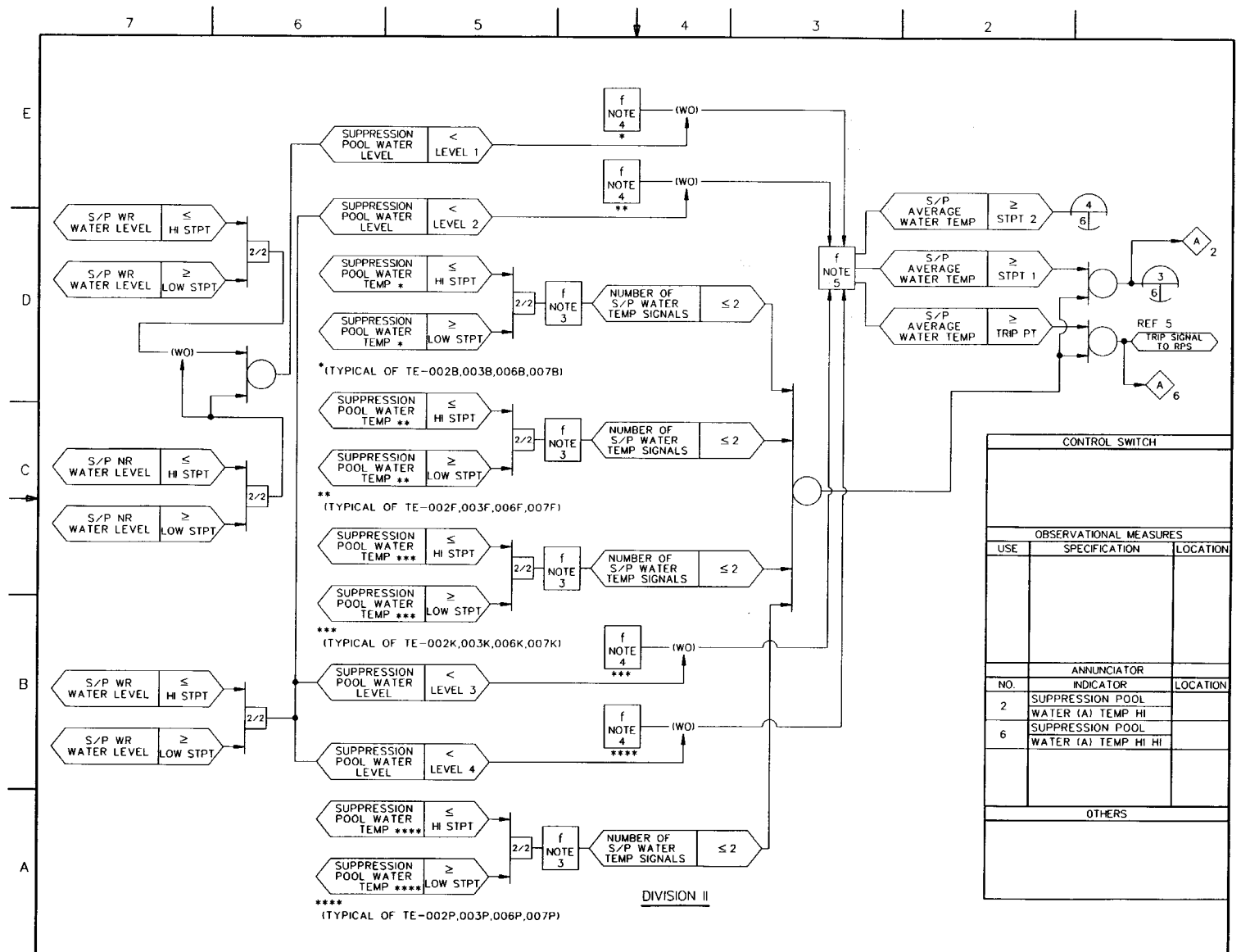


Figure 7.6-12 SUPPRESSION POOL TEMPERATURE MONITORING SYSTEM IBD (Sheet 3 of 6)

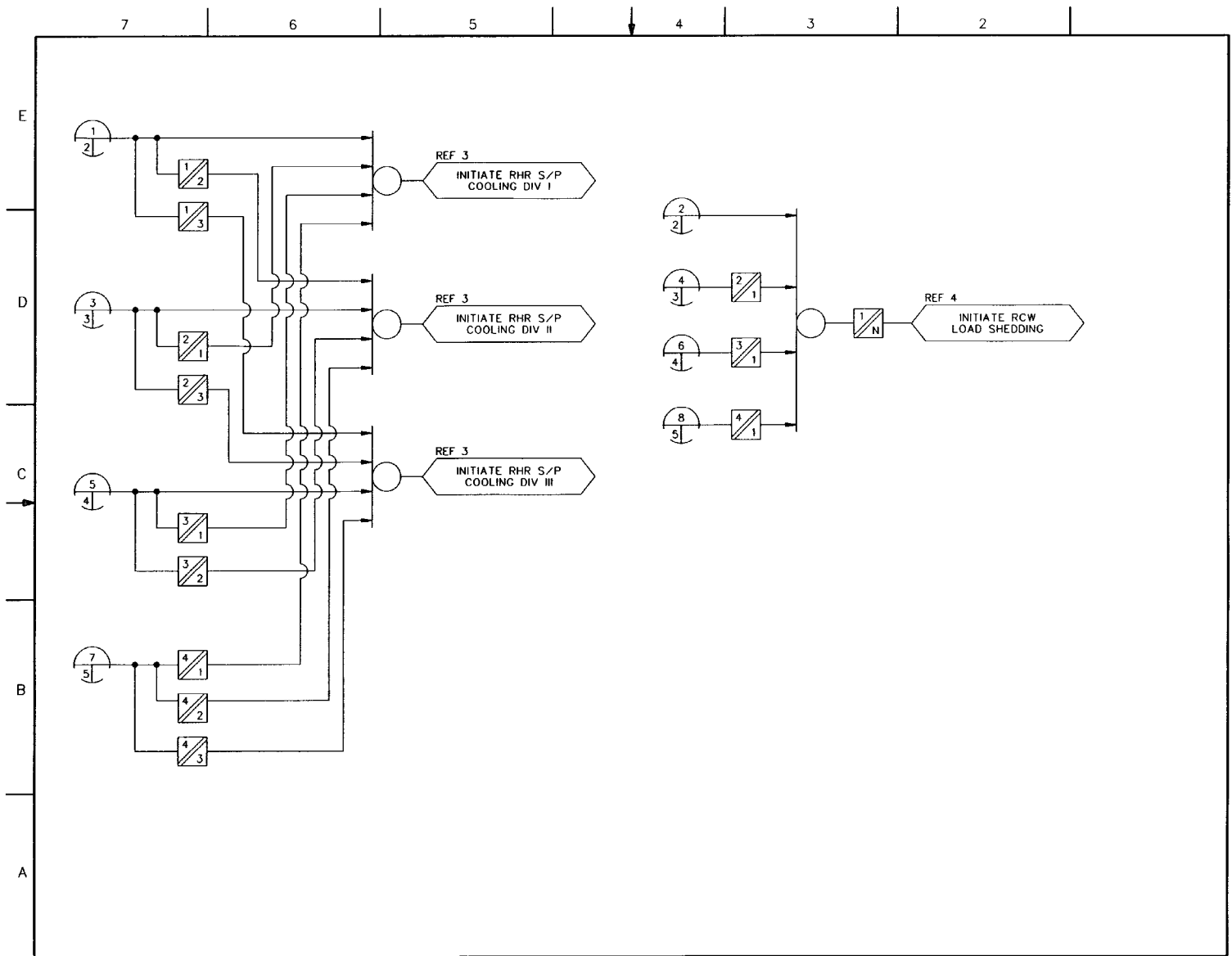
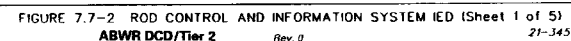


Figure 7.6-12 SUPPRESSION POOL TEMPERATURE MONITORING SYSTEM IBD (Sheet 6 of 6)
 ABWR DCD/Tier 2 Rev. 0 21-344



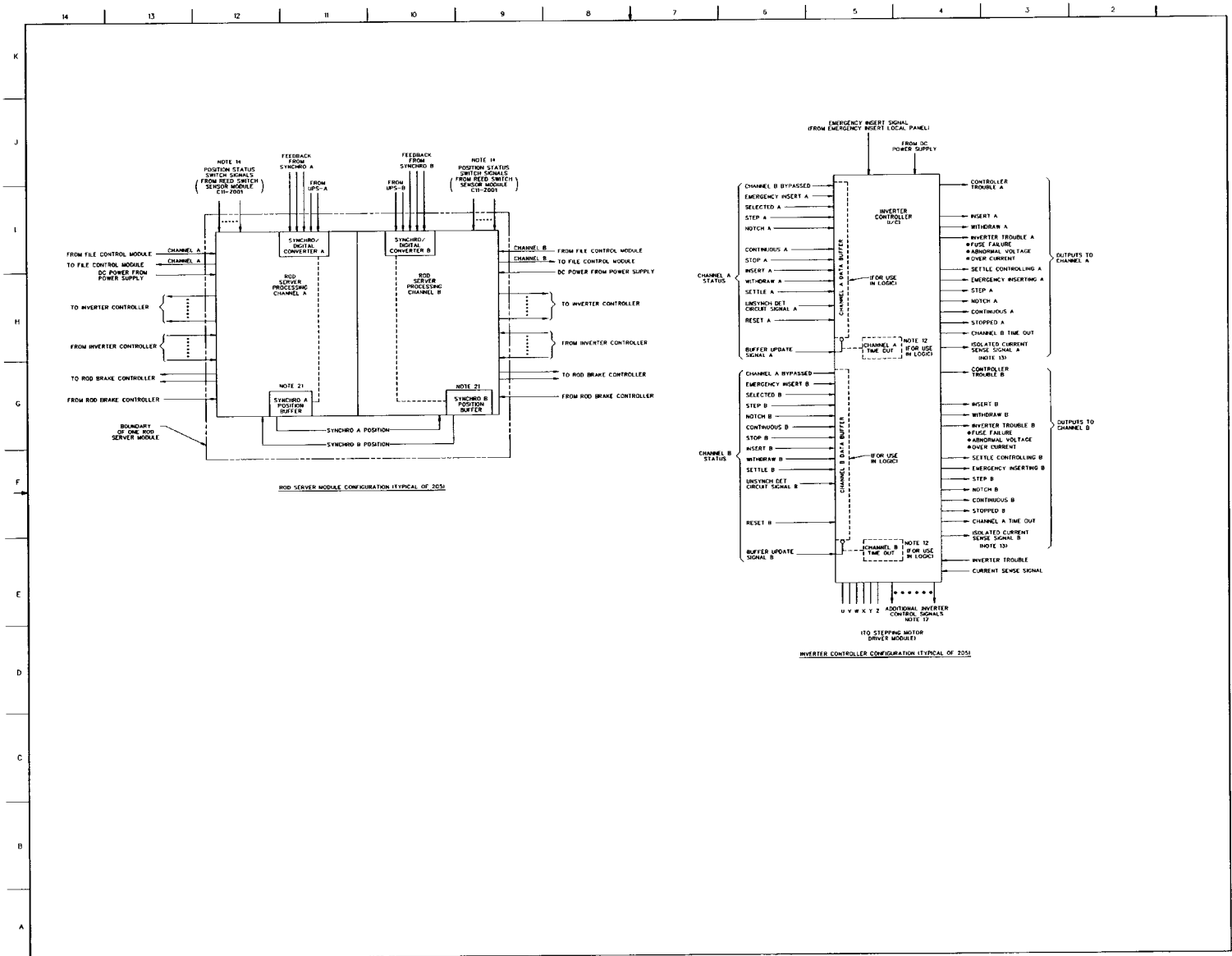


FIGURE 7.7-2 ROD CONTROL AND INFORMATION SYSTEM IED (Sheet 3 of 5)
ABWR DCD/Tier 2 Rev. 0 21-347

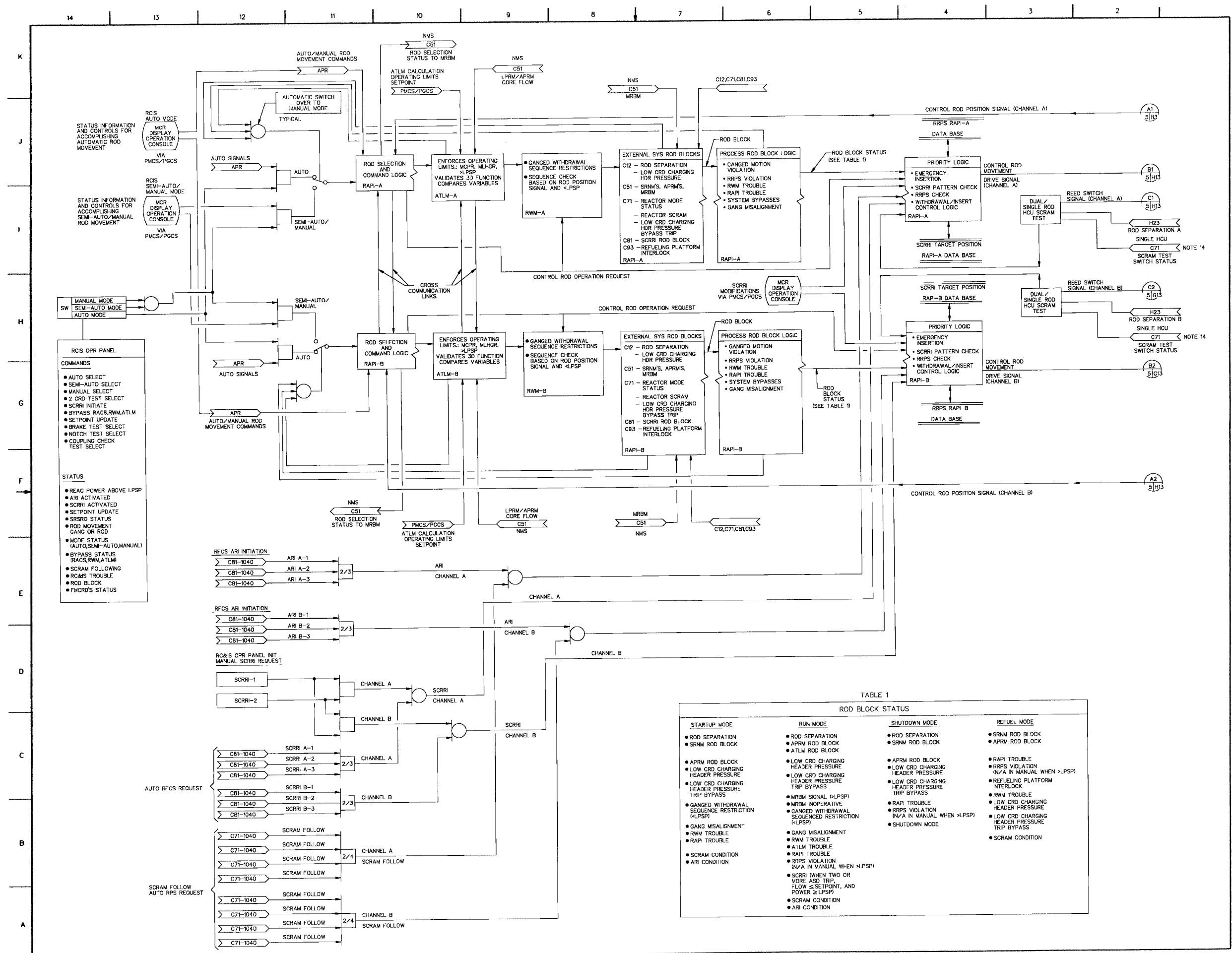
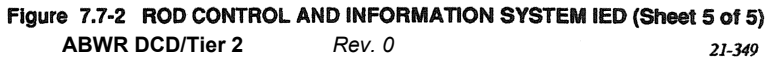
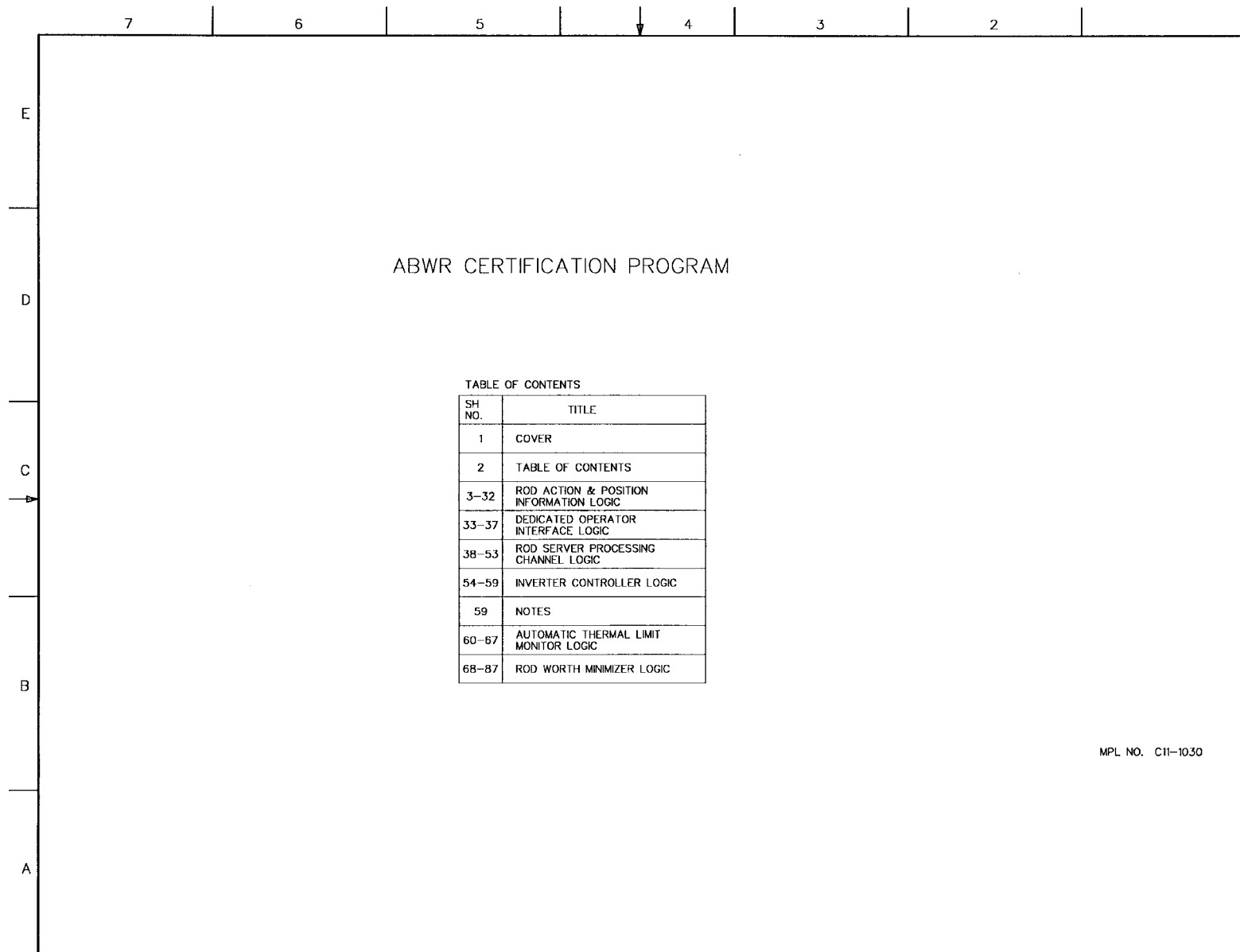


FIGURE 7.7-2 ROD CONTROL AND INFORMATION SYSTEM IED (Sheet 4 of 5)





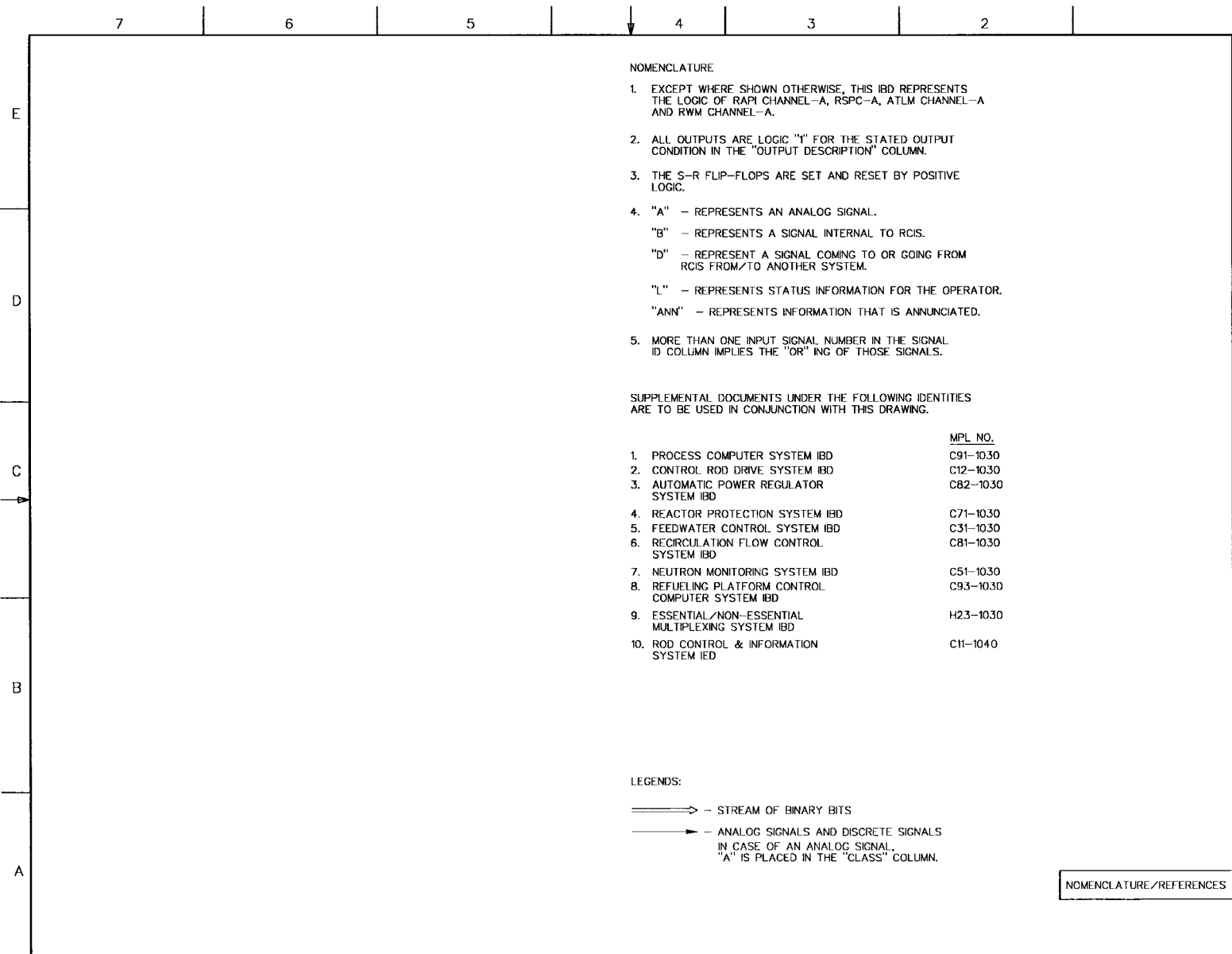


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 2 of 87)
ABWR DCD/Tier 2 Rev. 0 21-351

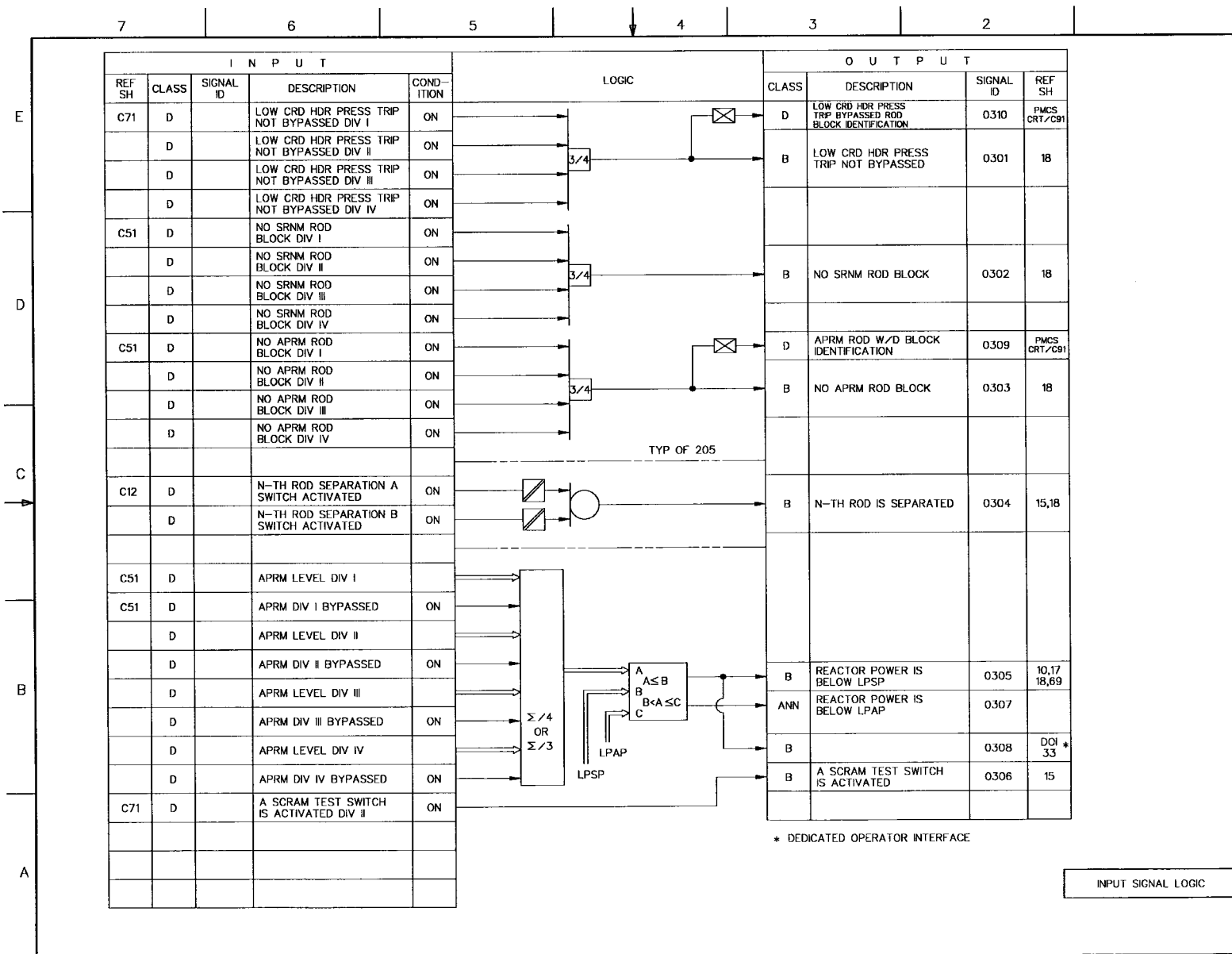


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 3 of 87)

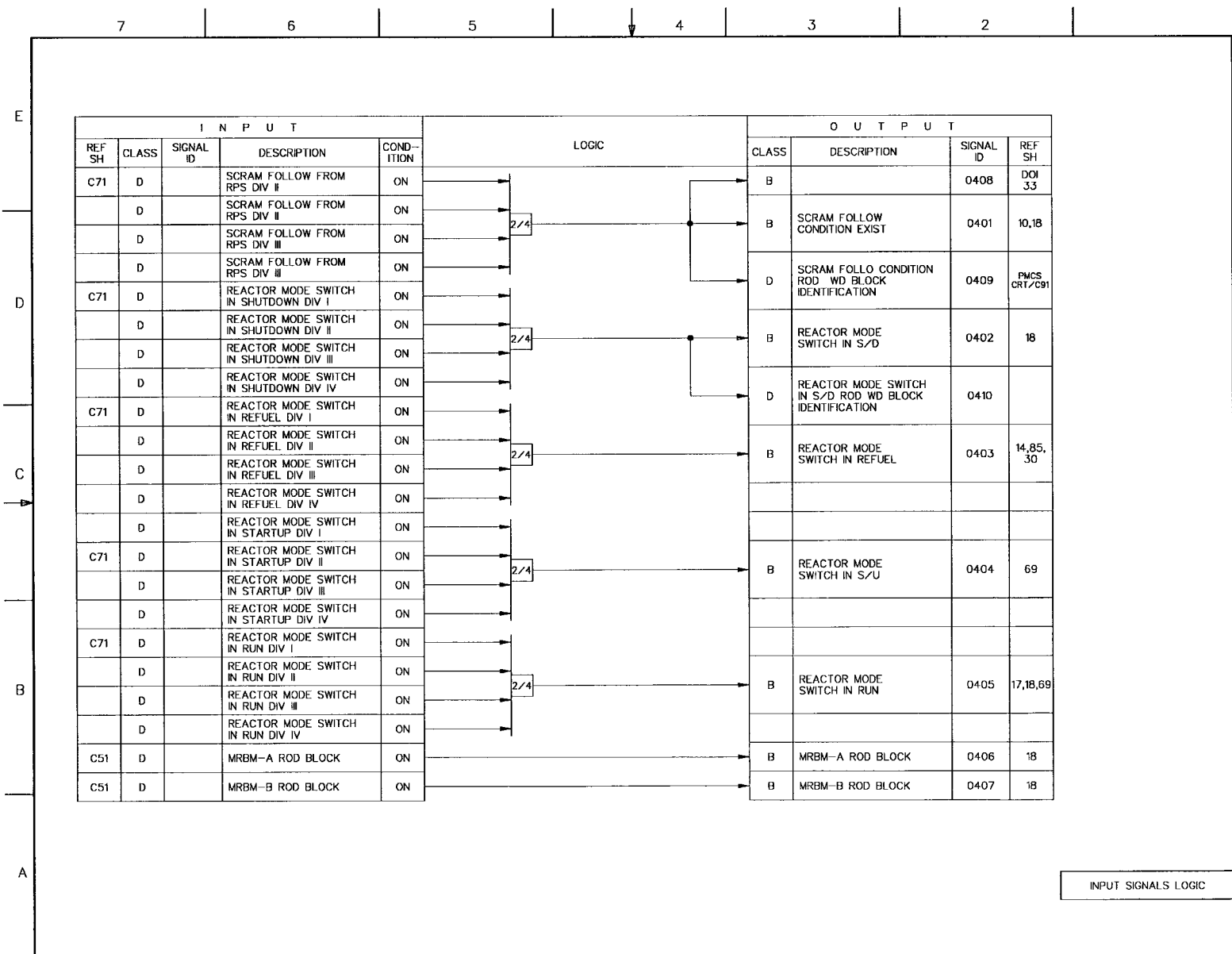


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 4 of 87)
ABWR DCD/Tier 2 Rev. 0 21-353

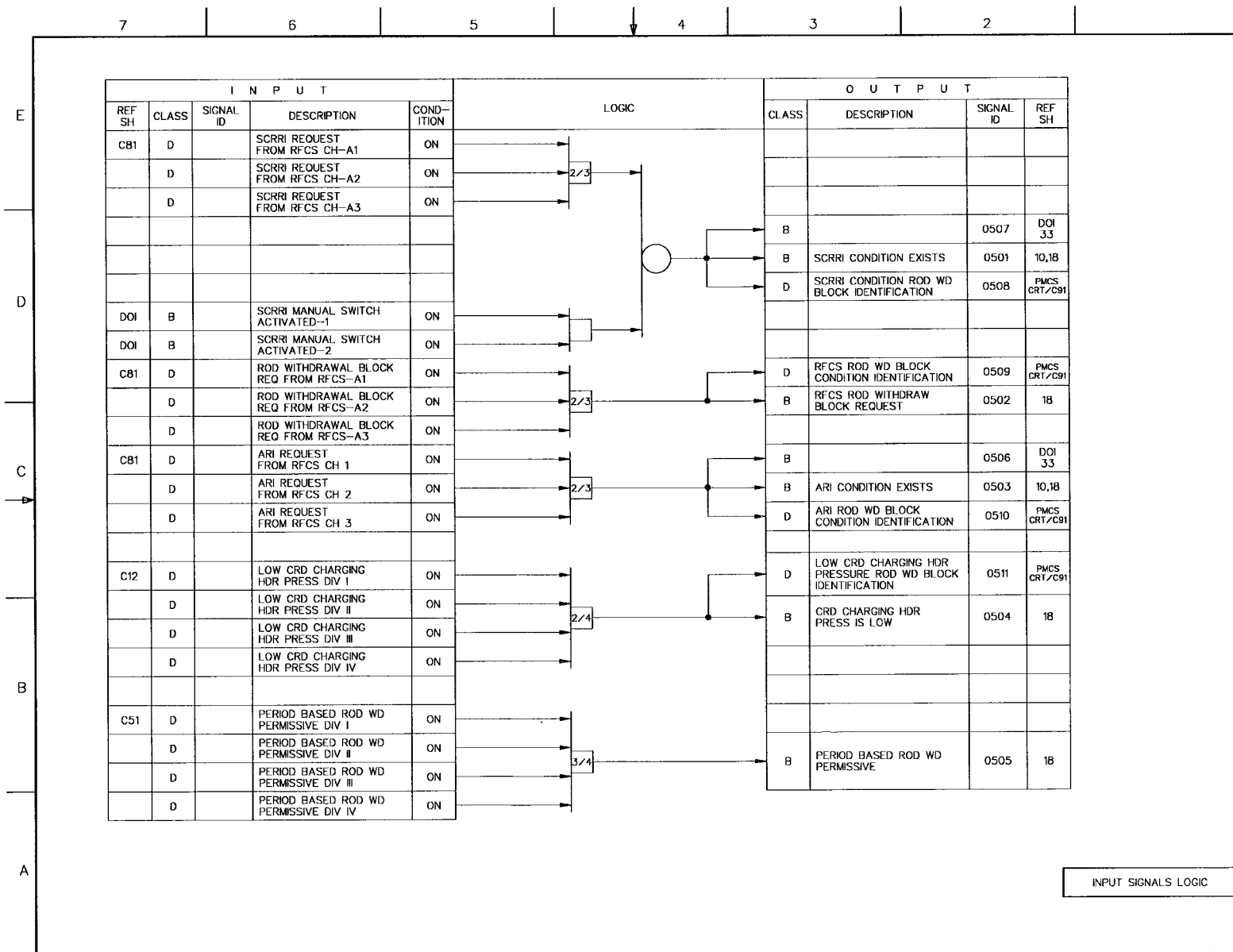


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 5 of 87)

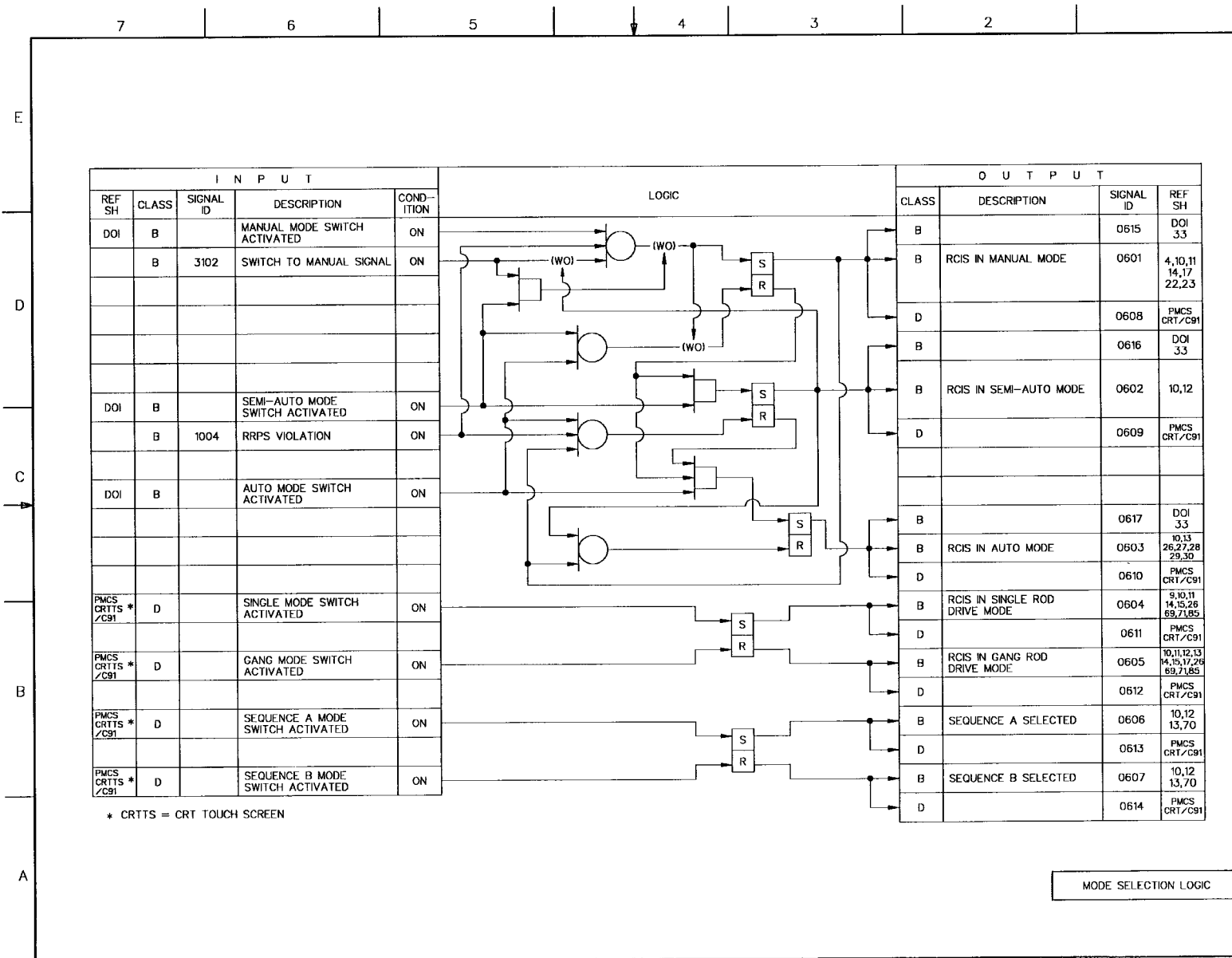


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 6 of 87)

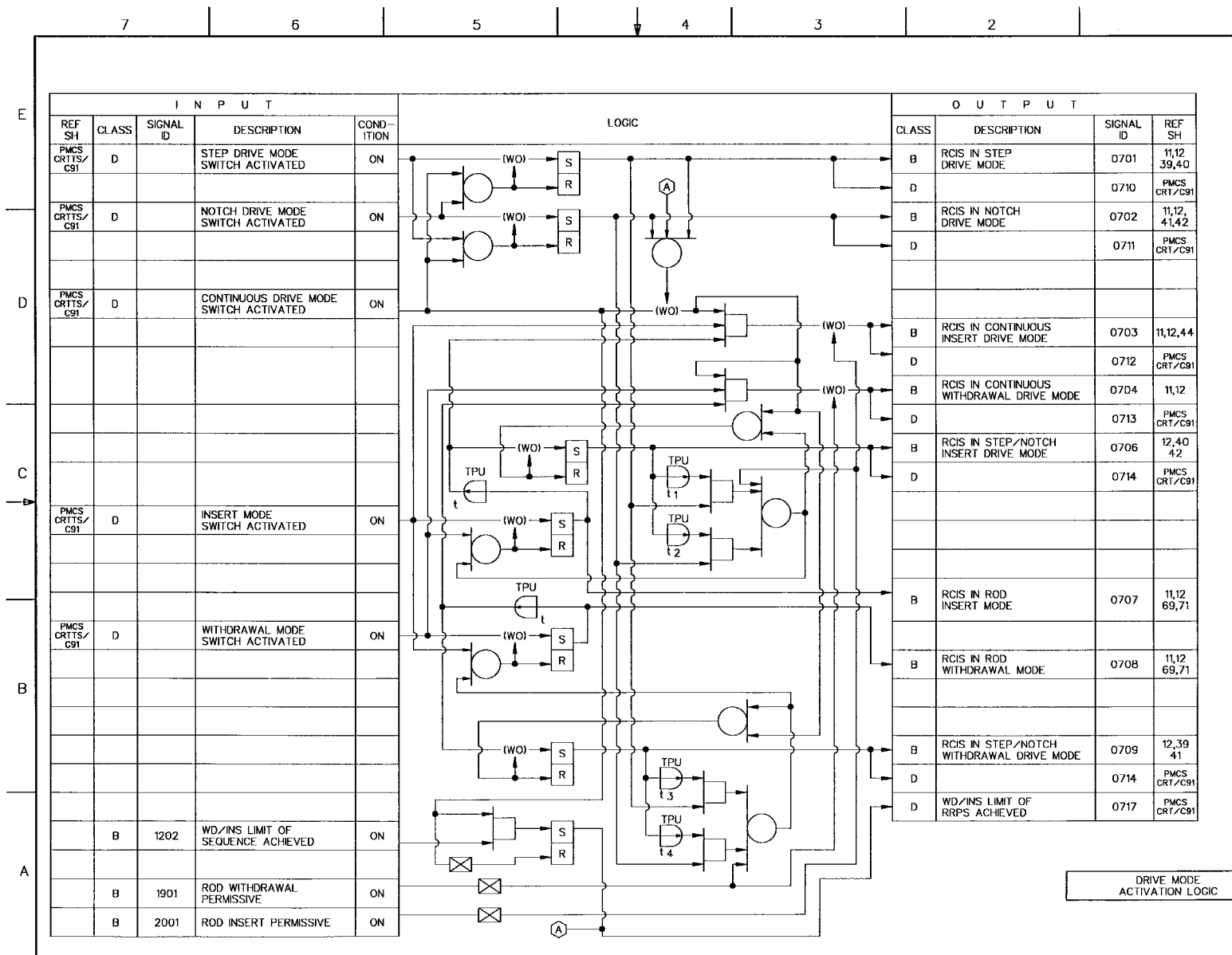


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 7 of 87)
ABWR DCD/Tier 2 Rev. 0 21-356

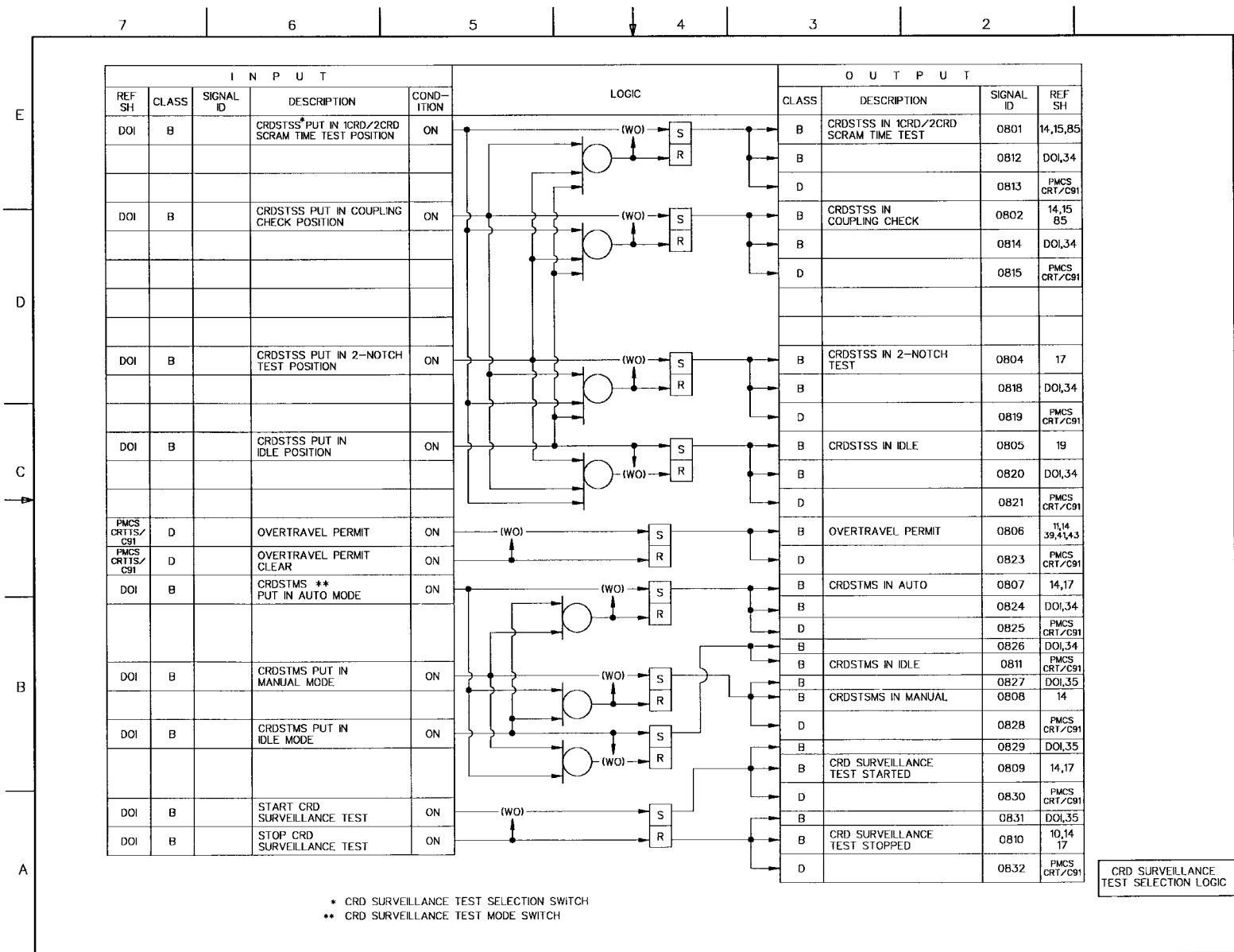
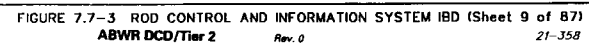


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 8 of 87)
ABWR DCD/Tier 2 Rev. 0 21-357



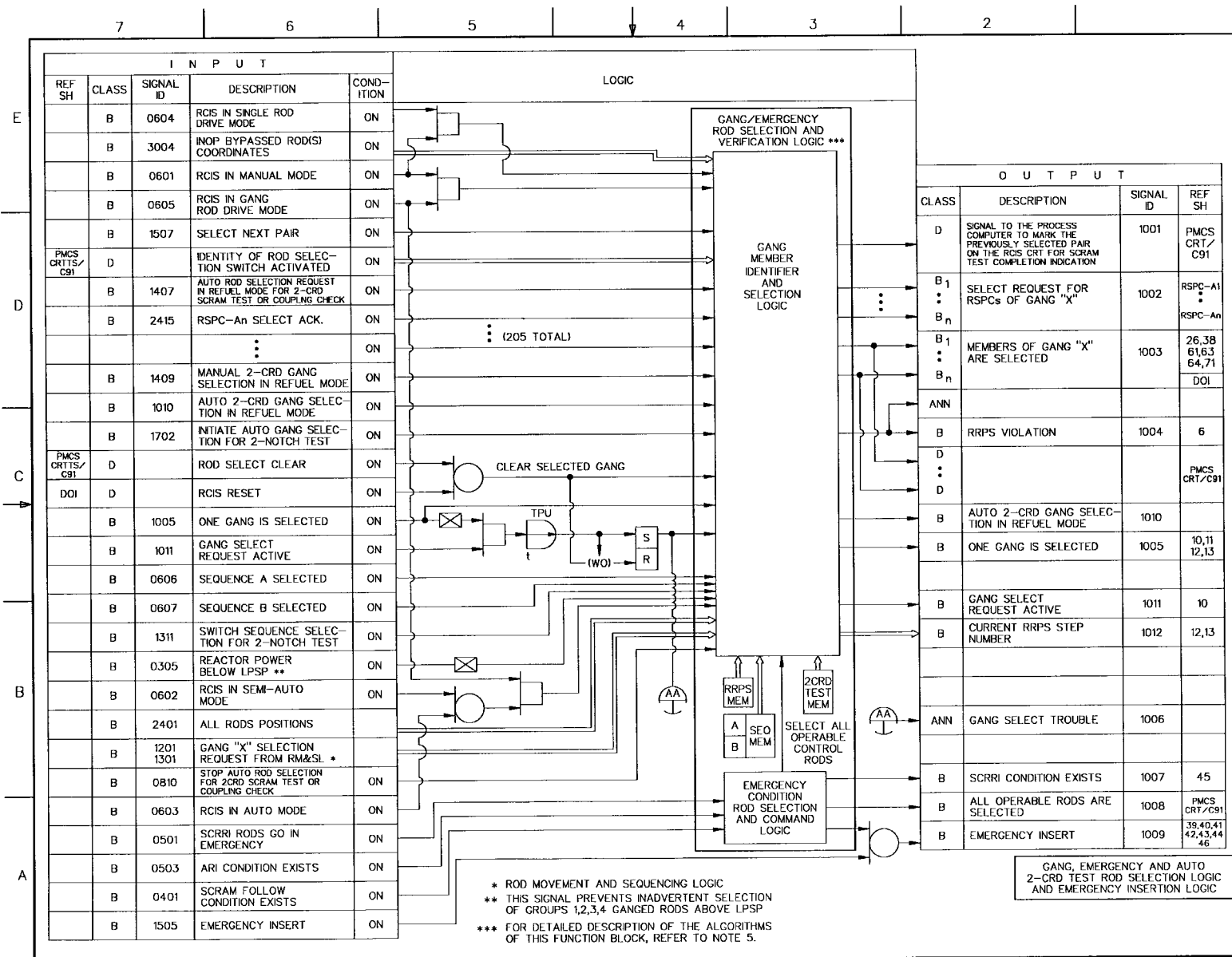


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 10 of 87)
ABWR DCD/Tier 2 Rev. 0 21-359

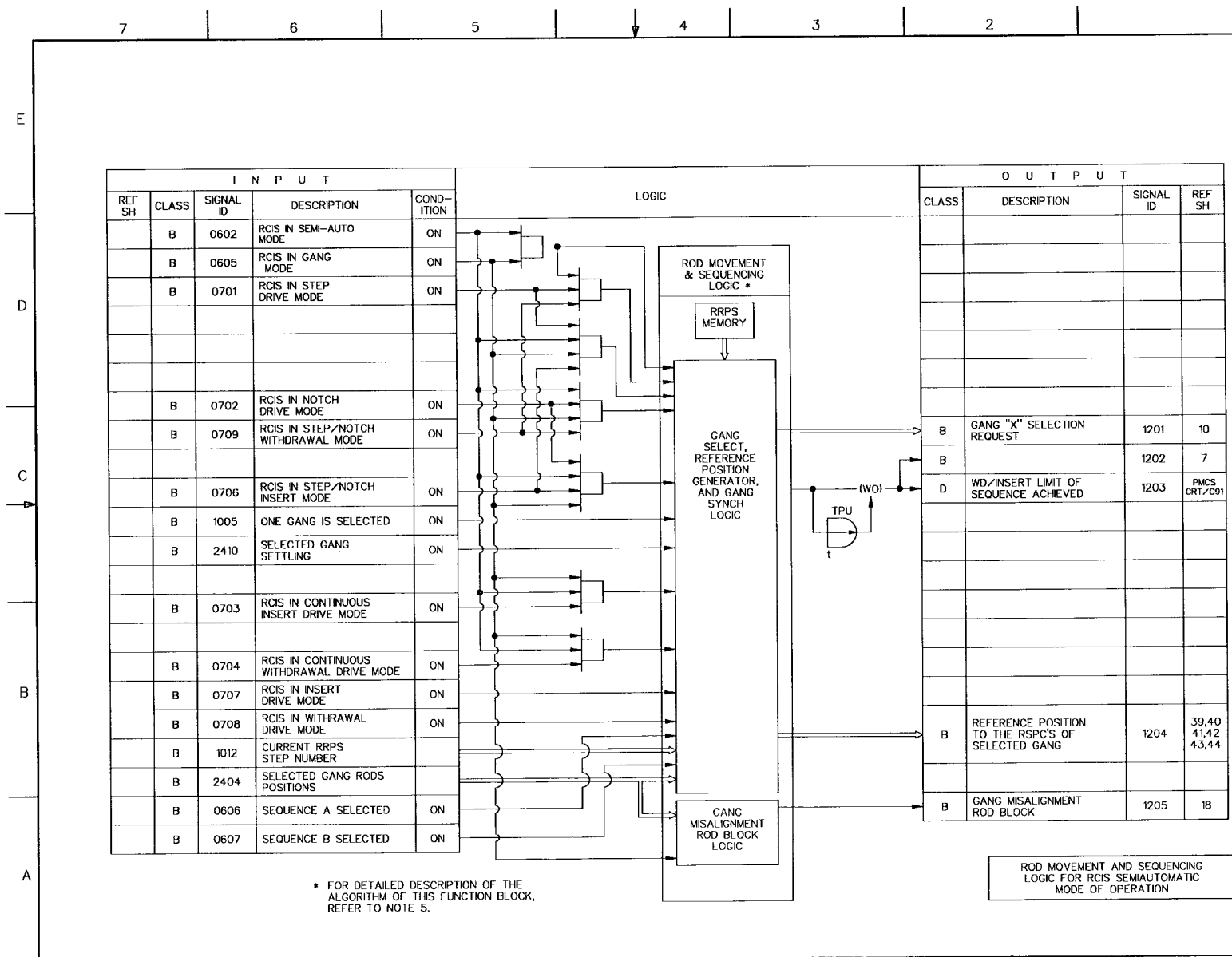
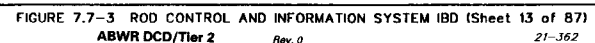


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 12 of 87)
ABWR DCD/Tier 2 Rev. 0 21-361



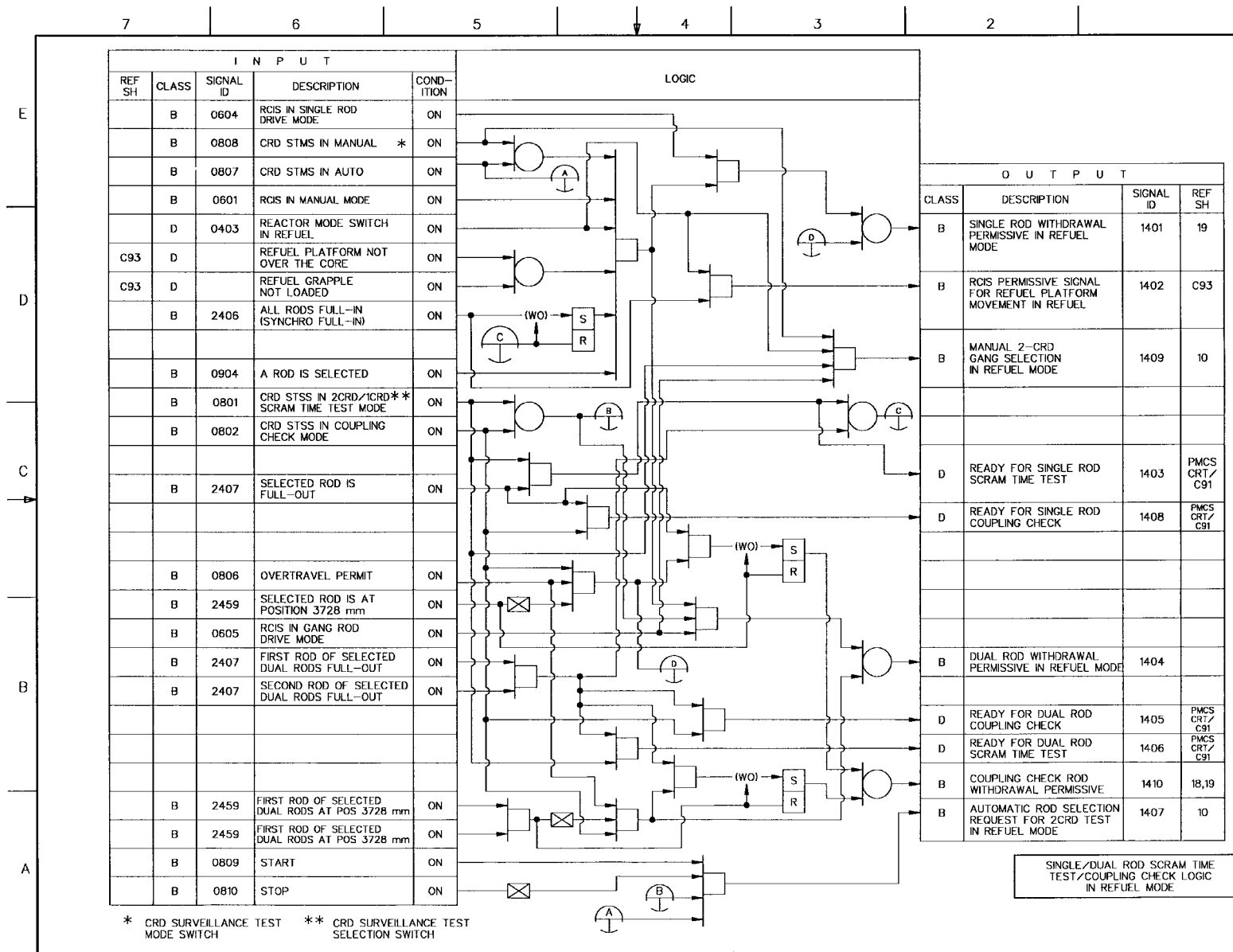


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 14 of 87)
ABWR DCD/Tier 2 Rev. 0 21-363

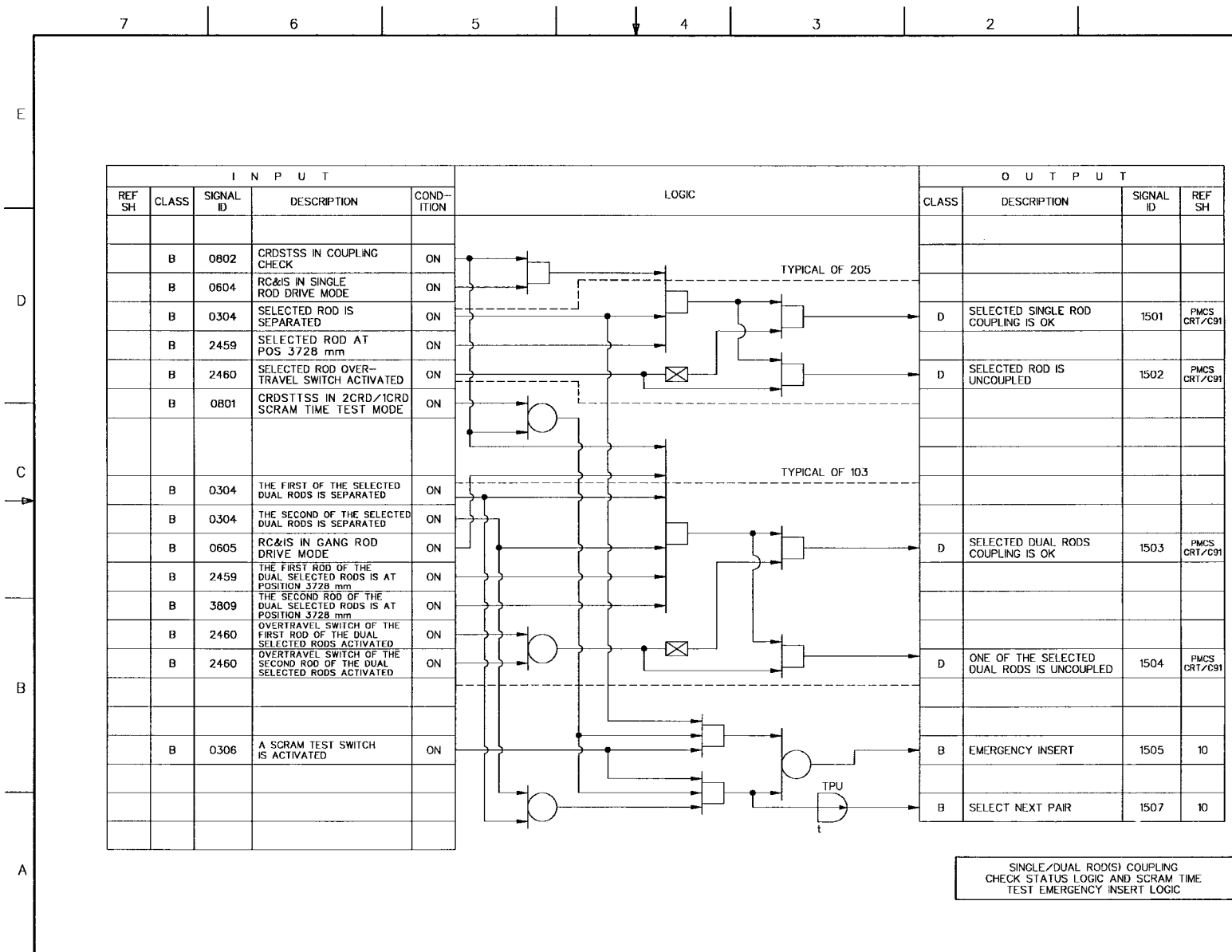
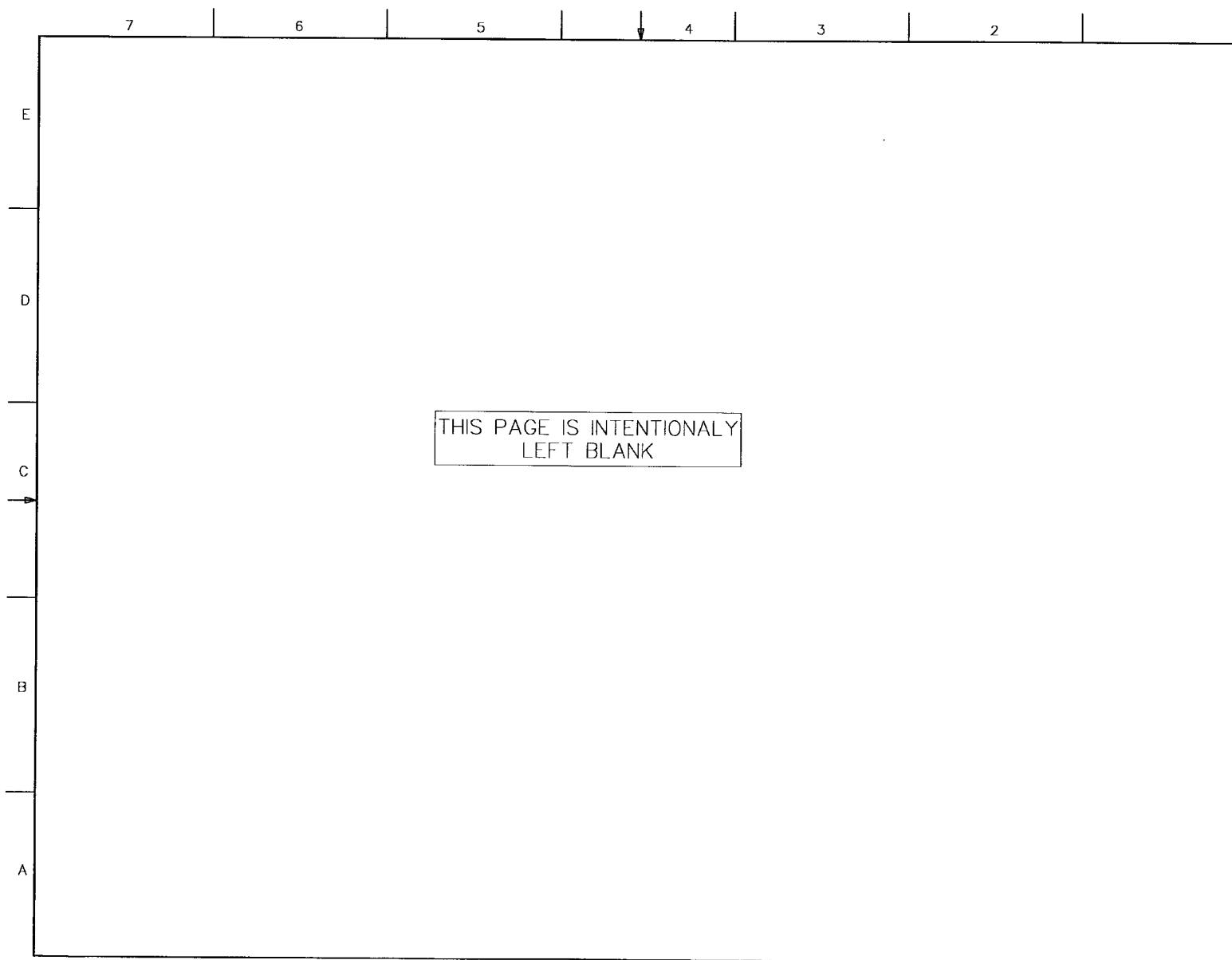


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 15 of 87)



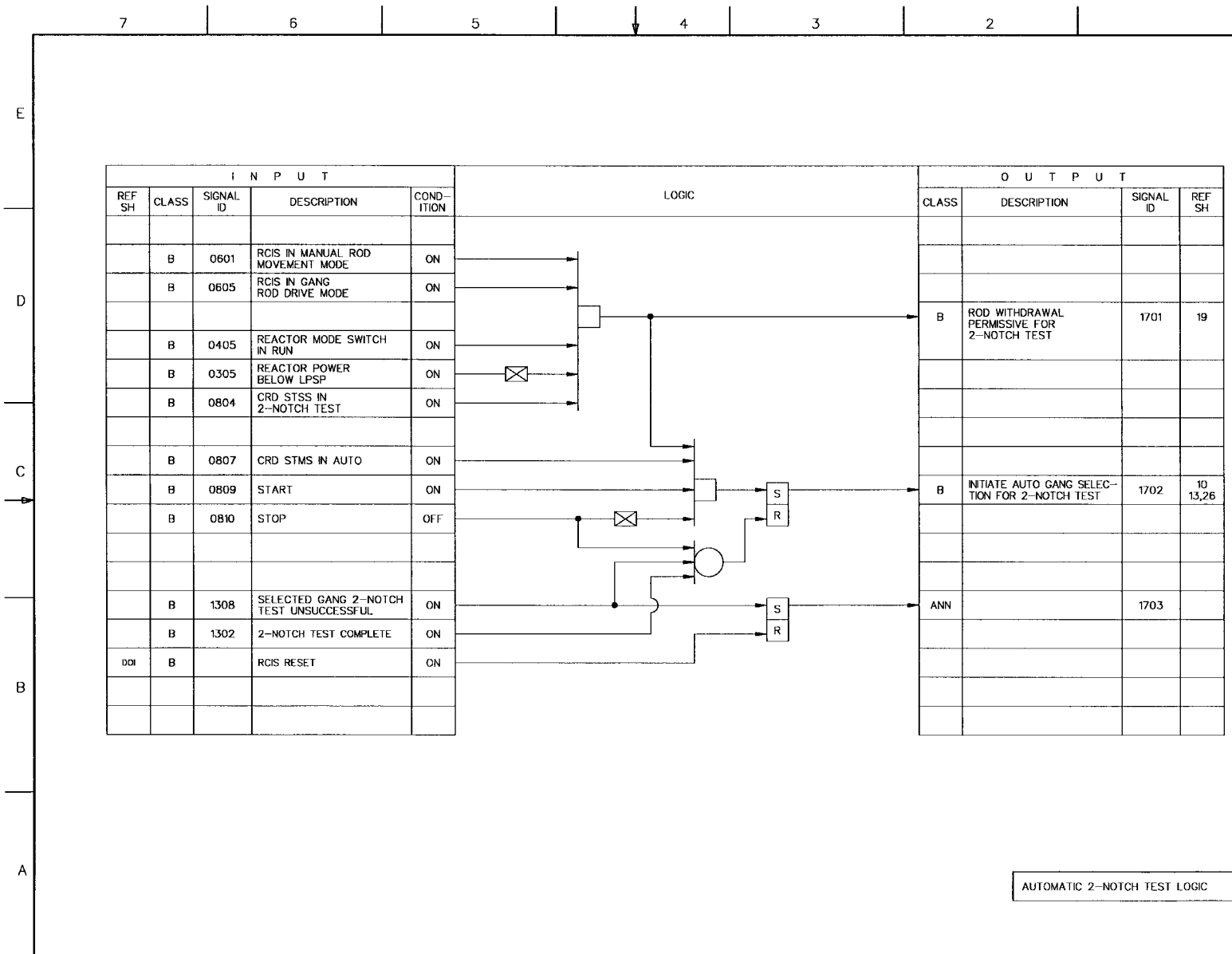


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 17 of 87)
 ABWR DCD/Tier 2 Rev. 0 21-366

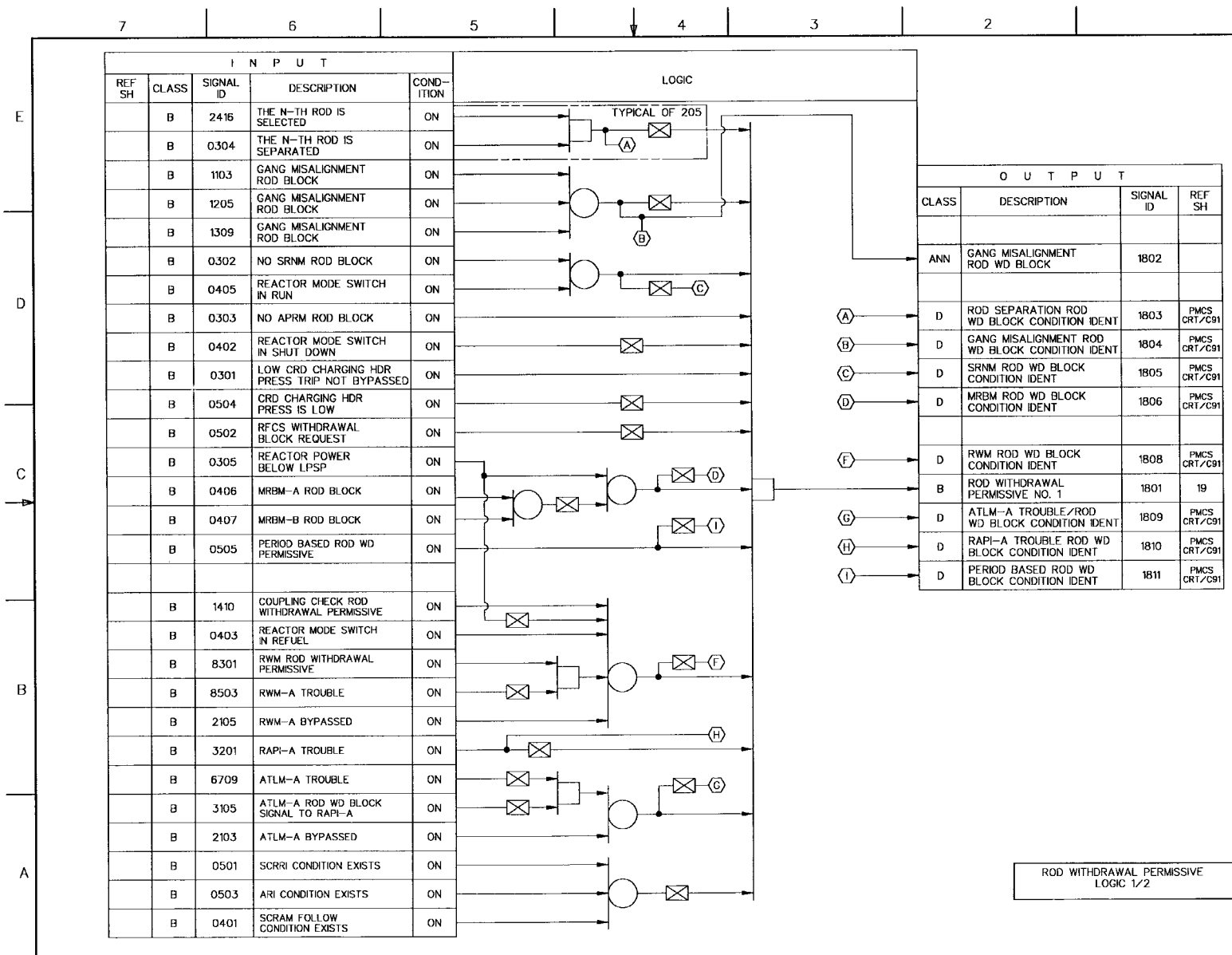
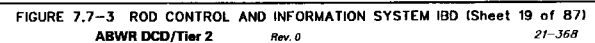


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 18 of 87)
ABWR DCD/Tier 2 Rev. 0 21-367



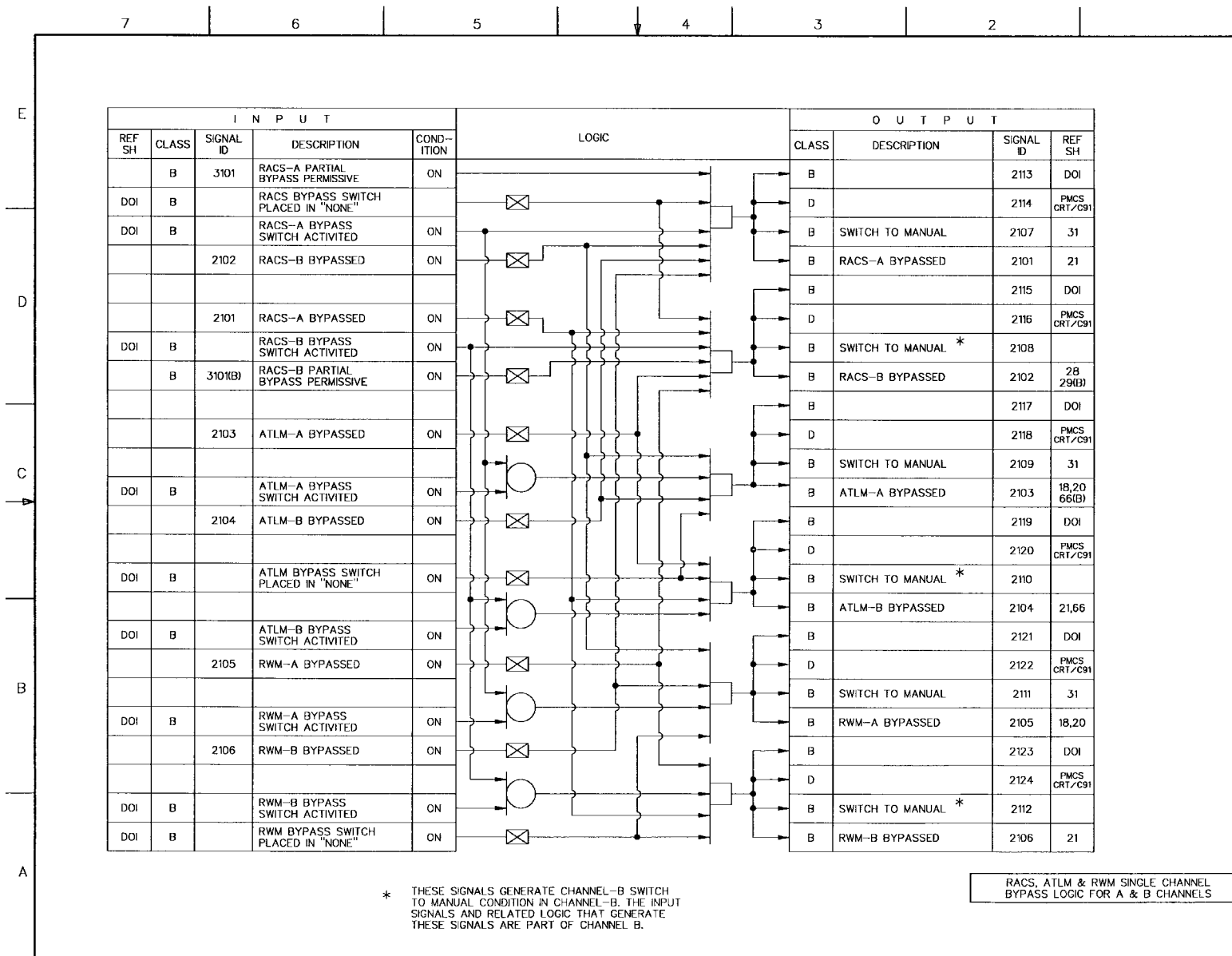
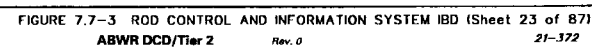


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 21 of 87)
ABWR DCD/Tier 2 Rev. 0 21-370



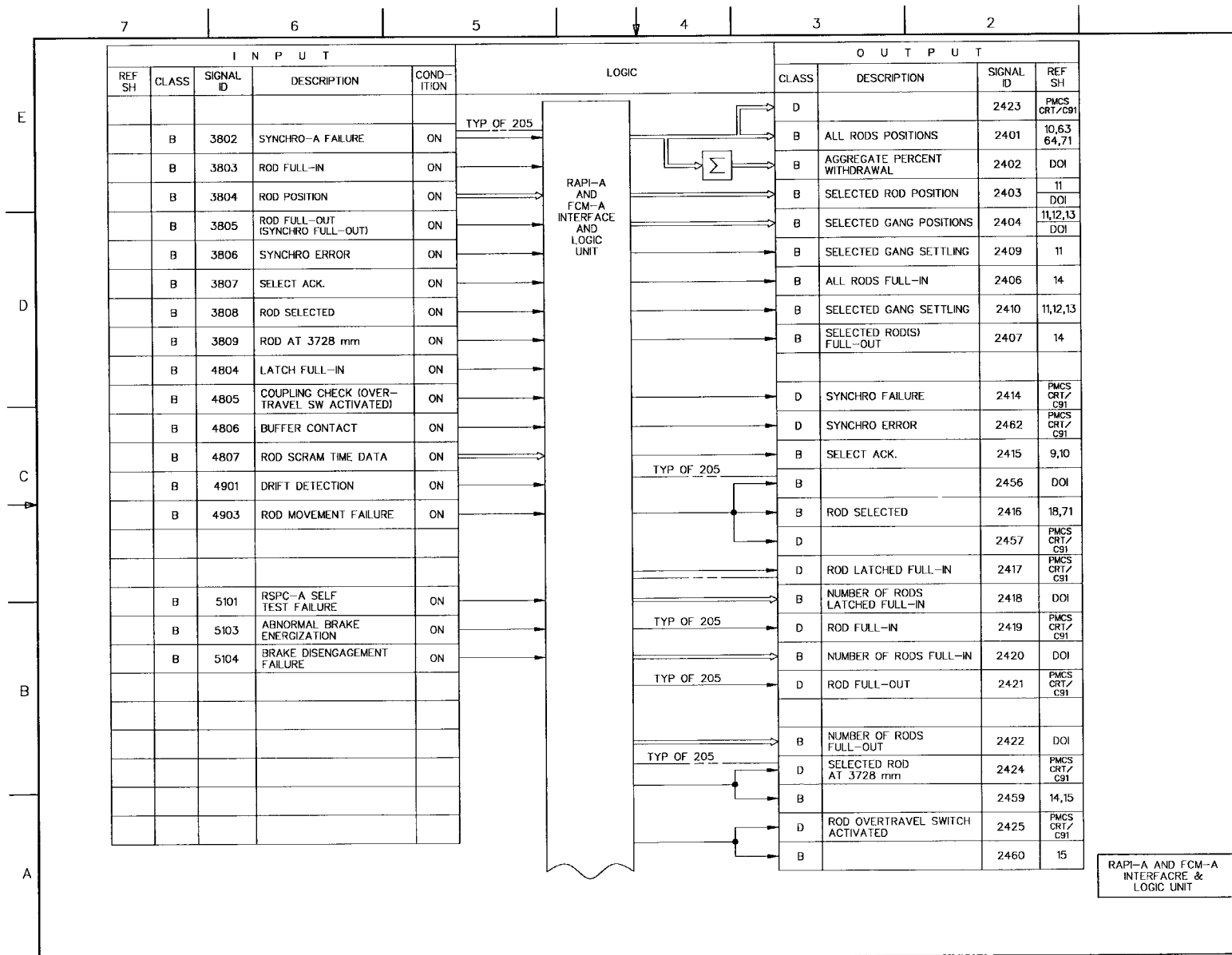


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 24 of 87)
ABWR DCD/Tier 2 Rev. 0 21-373

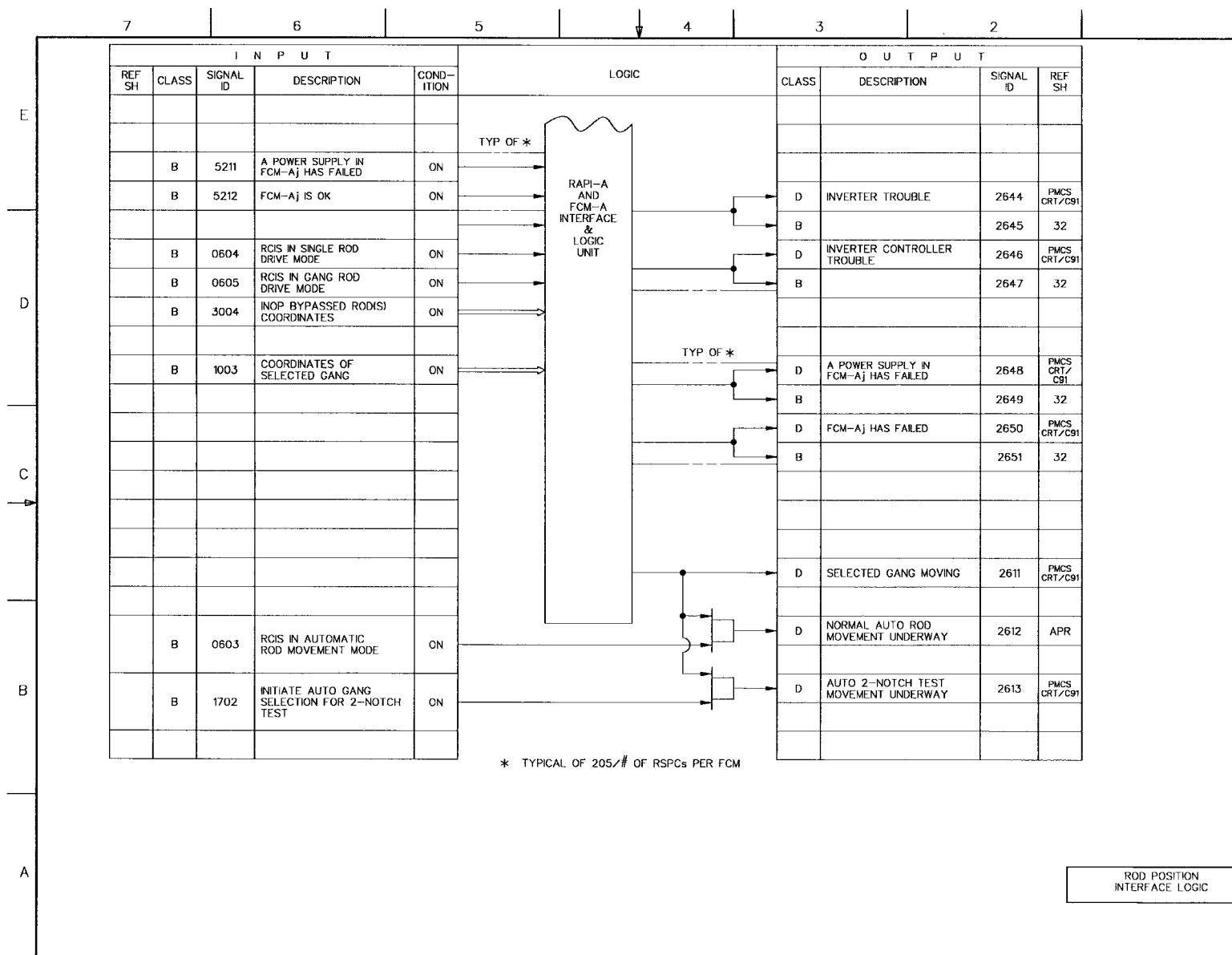
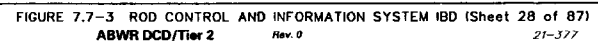
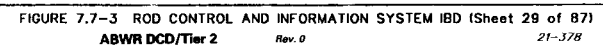
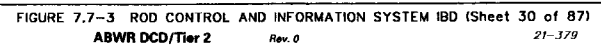


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 26 of 87)
ABWR DCD/Tier 2 Rev. 0 21-375







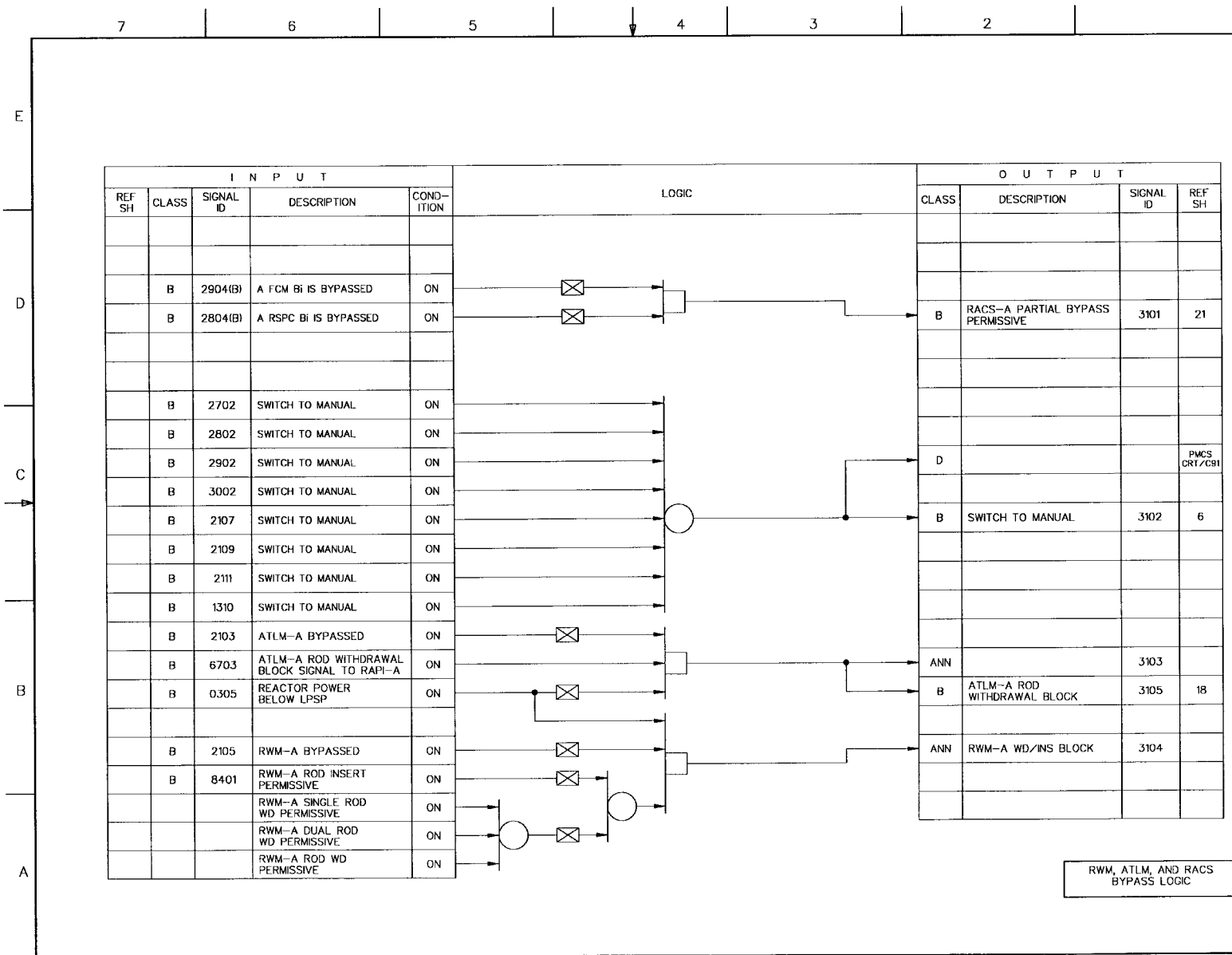
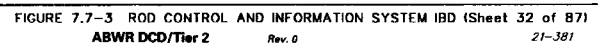


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 31 of 87)
ABWR DCD/Tier 2 Rev. 0 21-380



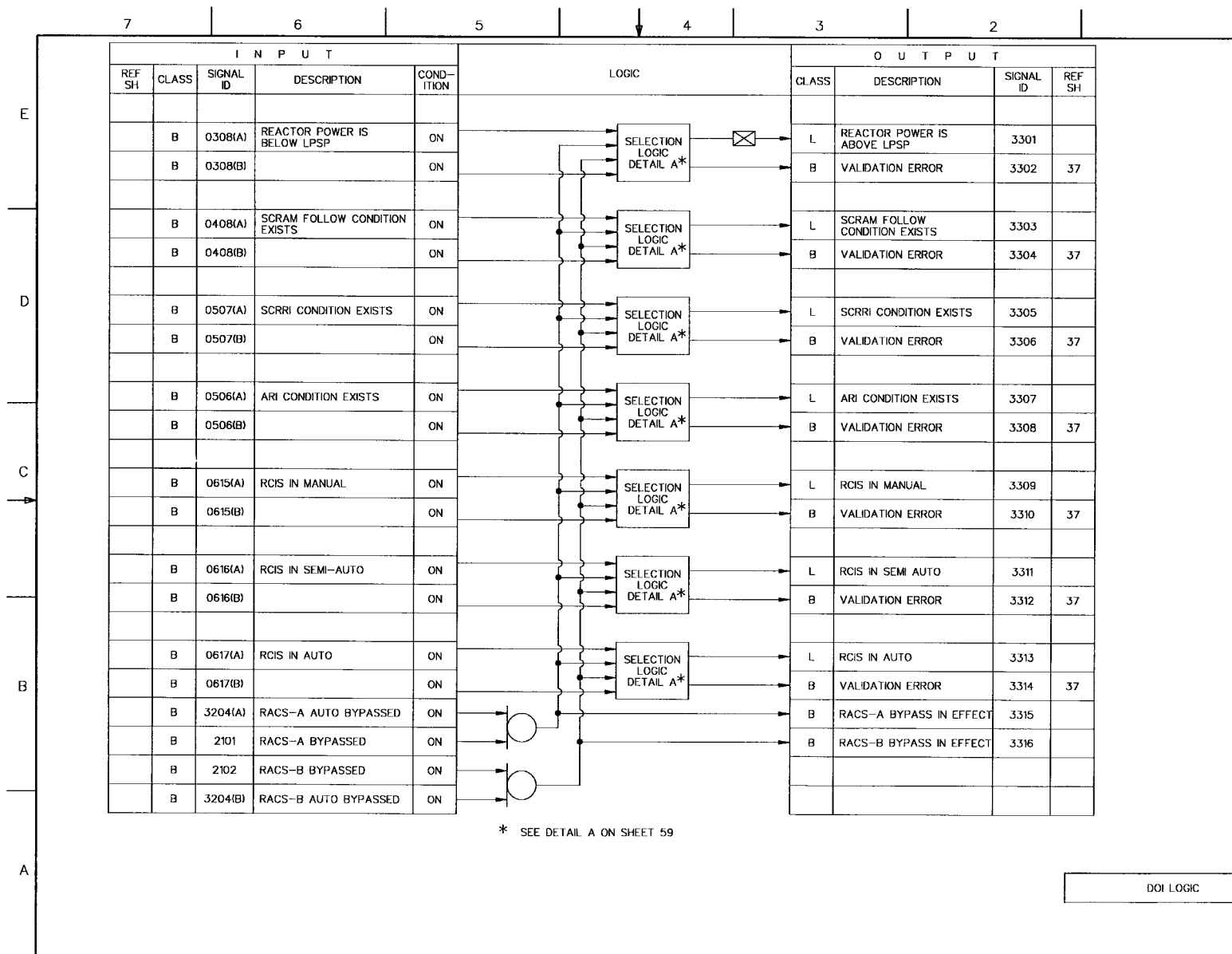


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 33 of 87)
ABWR DCD/Tier 2 Rev. 0 21-382

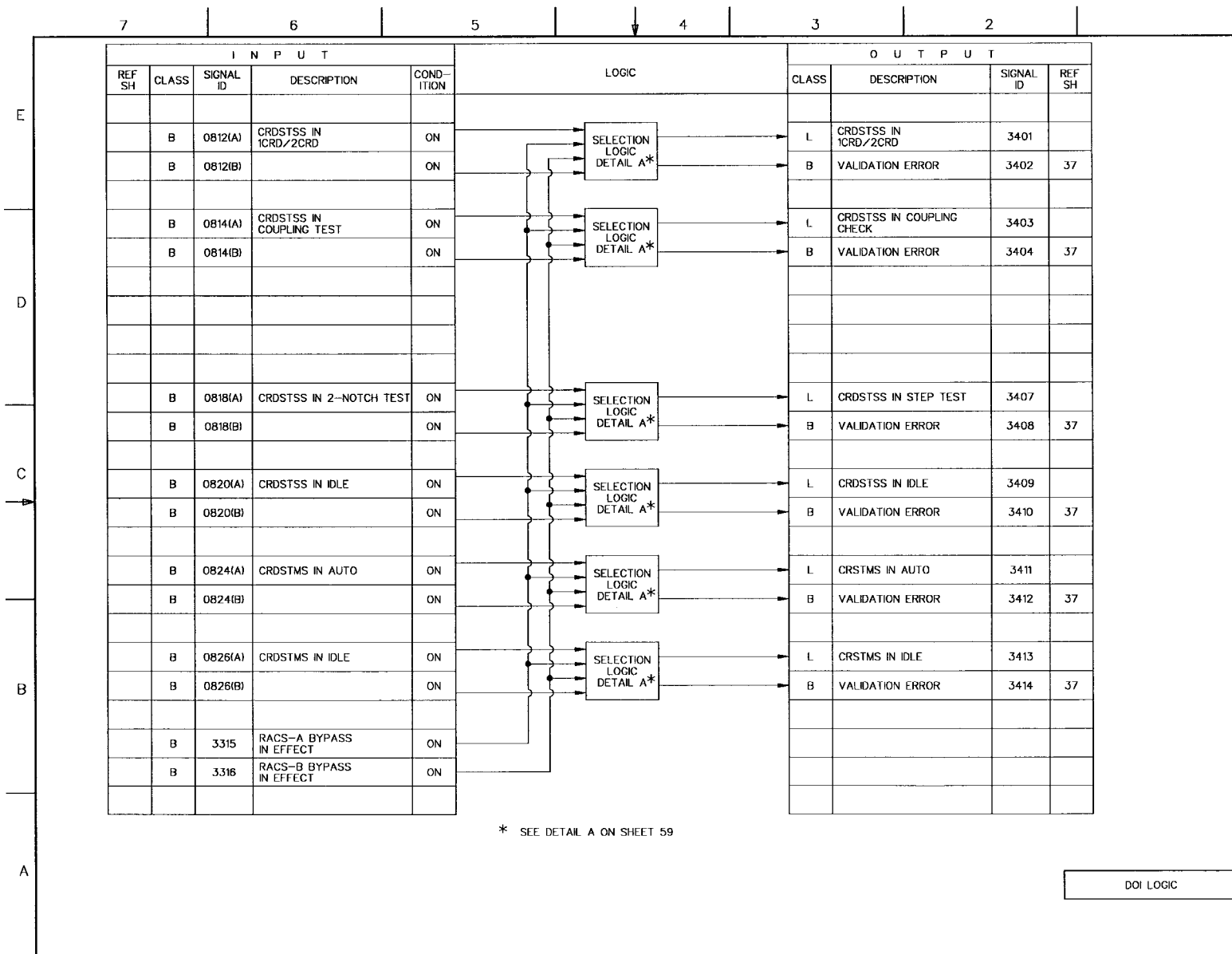


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 34 of 87)
 ABWR DCD/Tier 2 Rev. 0 21-383

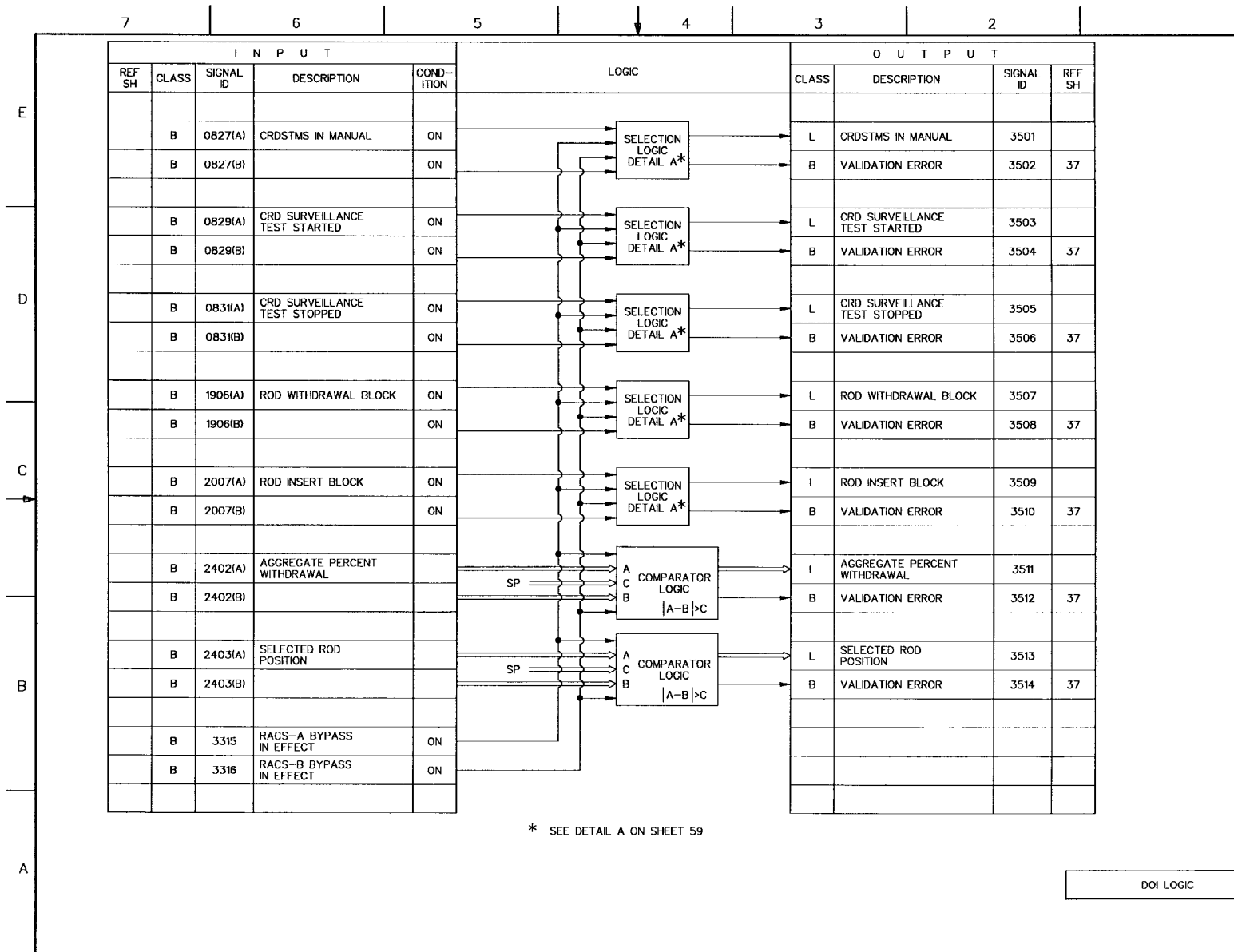


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 35 of 87)
ABWR DCD/Tier 2 Rev. 0 21-384

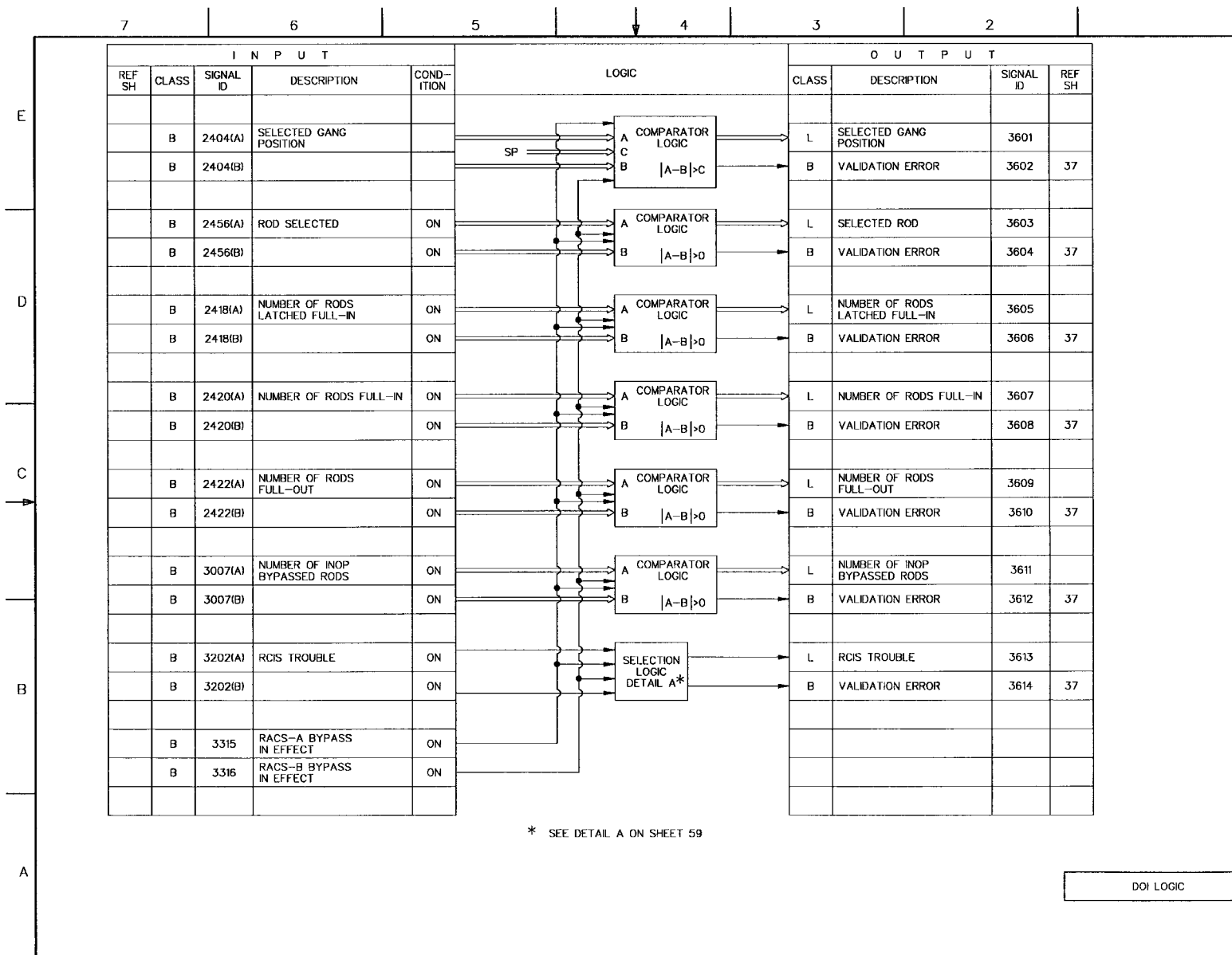


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 36 of 87)
ABWR DCD/Tier 2 Rev. 0 21-395

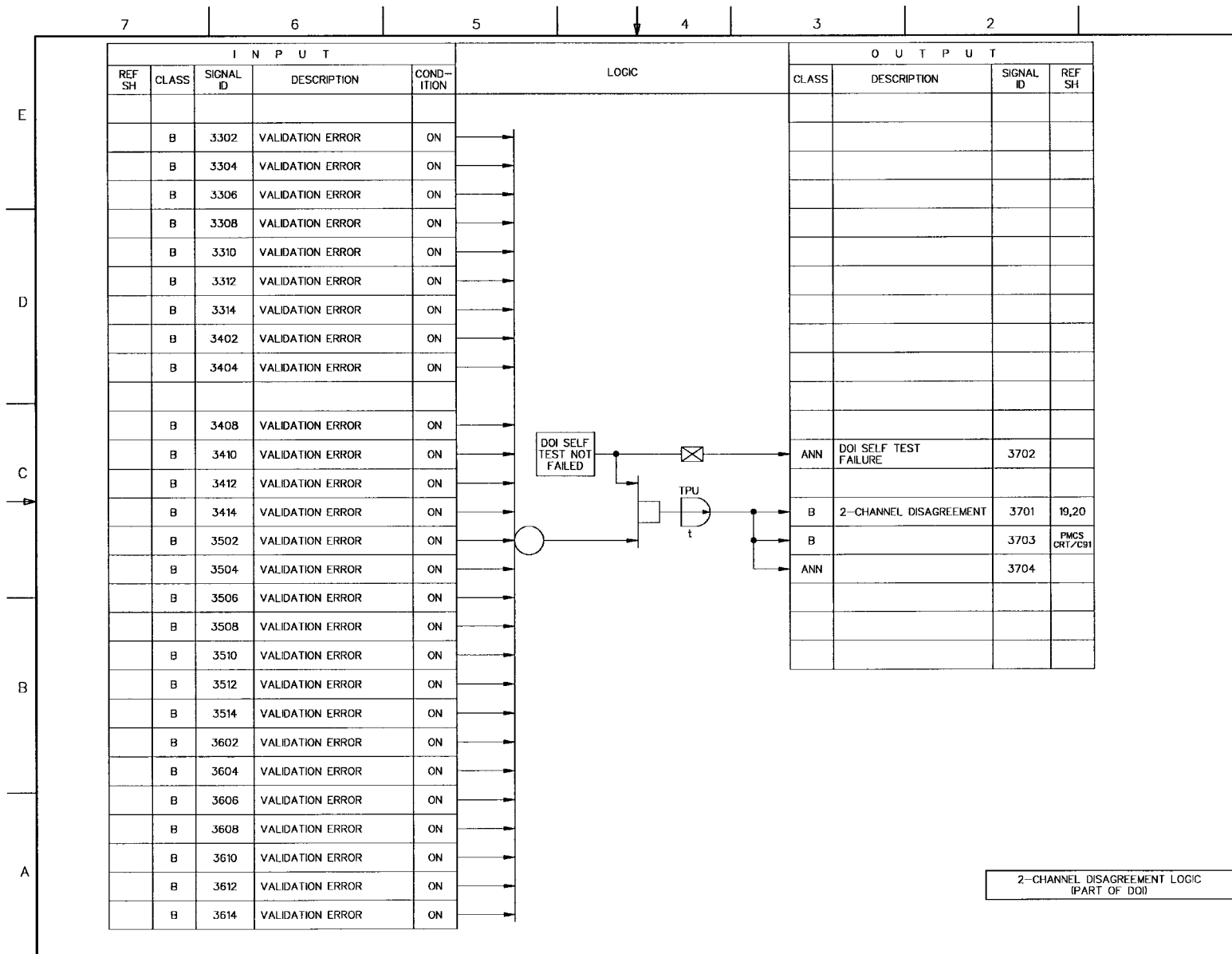


FIGURE 7.7-3 ROD CONTROL AND INFORMATION SYSTEM IBD (Sheet 37 of 87)
 ABWR DCD/Tier 2 Rev. 0 21-386

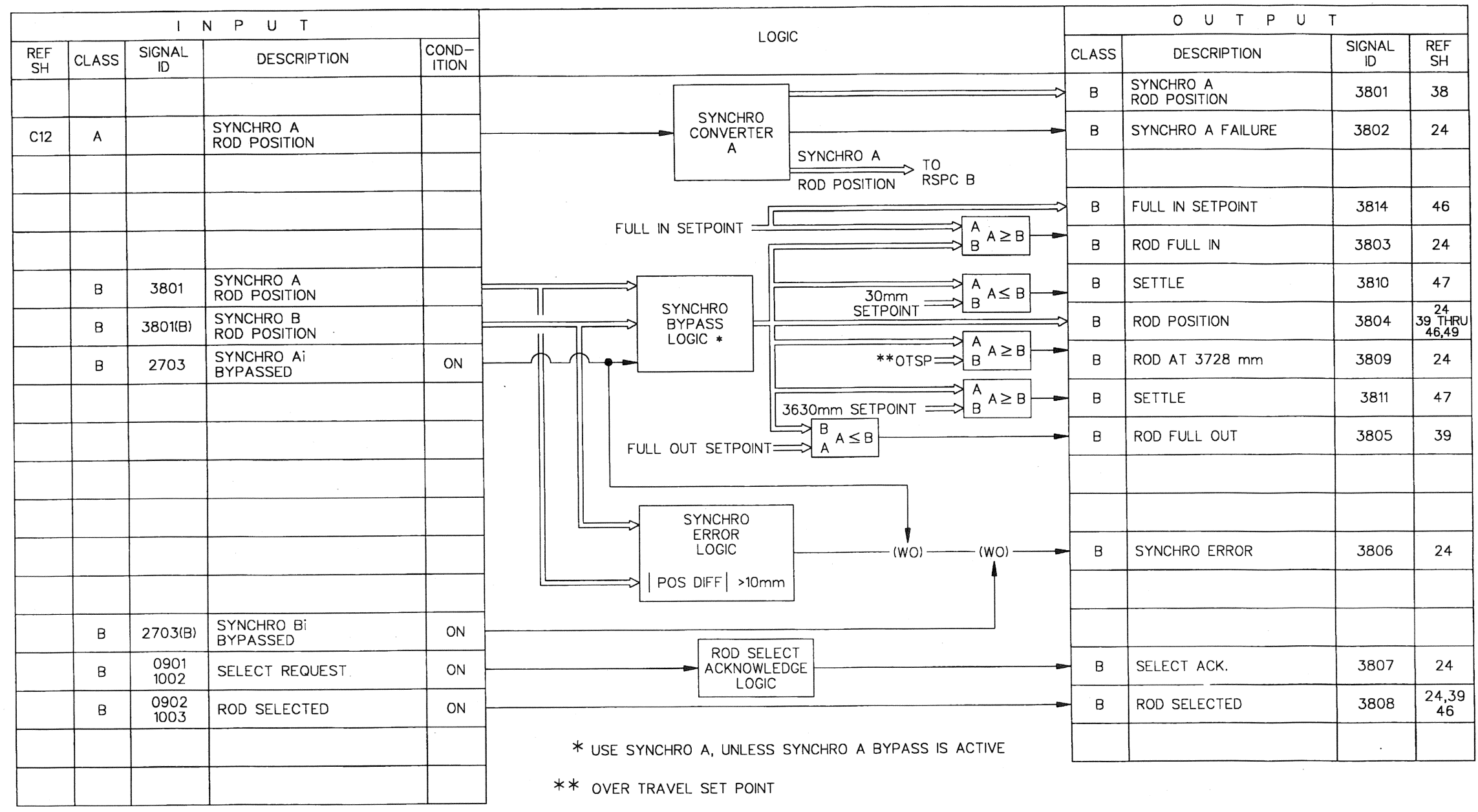
E

D

C

B

A



CHANNEL A
TYPICAL OF CHANNEL B

SYNCHRO SELECTION, ECHO,
AND ROD POSITION LOGIC

